8 Haddock in the North Sea, West of Scotland, and Skagerrak

had.27.46a20 – *Melanogrammus aeglefinus* in Subarea 4, Division 6.a, and Subdivision 20

Until 2014, haddock in Subarea 4, Division 6.a, and Subdivision 20 (referred to hereafter as Northern Shelf haddock) were assessed as two separate stocks: Subarea 4 and Subdivision 20 by WGNSSK, and Division 6.a by WGCSE. The 2014 Benchmark Workshop for Northern Haddock Stocks (WKHAD; ICES, 2014) concluded that the two notional haddock stocks should be assessed as one stock.

This stock was most recently benchmarked in early 2022 at the Benchmark Workshop on North Sea and Celtic Sea Stocks (WKNSCS; ICES, 2023). The assessment model was changed to SAM (Nielsen and Berg, 2014) which also permitted the use of stochastic forecasting methods for generating catch advice. Additionally, several updates were made to the input datasets, the most significant being moving to time varying maturity-at-age estimates and developing new delta-GAM modelled survey indices for Q1 and Q3+Q4 covering the entire stock area. Updated natural mortality estimates were provided by WGSAM (ICES, 2024) in 2024. Overall, this resulted in minor changes to stock size. Reference points were updated in response to the revision of the natural mortality inputs to the assessment. In 2025 the model used to derive the survey indices was updated and the assessment model settings were reviewed and updated (WD 1 in Annex 8). Reference points were again updated in response to these revisions.

8.1 General

8.1.1 Ecosystem aspects

Ecosystem aspects are summarized in the Stock Annex.

8.1.2 Fisheries

A general description of the fishery (along with its historical development) is presented in the Stock Annex. Most of the information presented below and in the Stock Annex pertains to the Scottish fleet, which takes the largest proportion of the haddock stock. This fleet is not just confined to the Northern Shelf area, as vessels will sometimes operate in Divisions 6.b (Rockall) and 5.b (Faroes).

Specific information on changes in the Scottish fleet during 2011–2024 was not provided to WGNSSK in 2025, and it is difficult to reach a firm conclusion on the likely effect of recent fishery changes on haddock mortality.

Currently, catches of haddock are subject to national legislation regulating discards in the UK and Norway in addition to the EU landing obligation. The EU landing obligation was implemented for the majority of fleets catching Northern Shelf haddock in January 2016 and catches of this stock have been fully under the landing obligation since 2019. It is unclear what changes in fleet dynamics and fishing behaviour this has caused. In the past, vessels have only very seldom exhausted their quota in this fishery, and previous discarding behaviour is thought to have been driven by a complicated mix of economic and other market-driven factors.

Fish from the 2019 and 2020 year classes formed the bulk of haddock catches in 2024. The entry of the large 2019 and 2020 year classes into the fishery led to a rapid rise in the discarding rate which peaked at approximately 37% in 2021. The discard rate has steadily dropped since and is 21% in 2024. By age, discarding increased substantially for age 3 in 2023 and age 4 in 2024. Haddock at age 3 are still mostly under the Minimum Conservation Reference Size (MCRS) of 30 cm and so an increase in discarding would be expected given the large size of the 2019 and 2020 year classes. However, the peak in discarding appears to follow the 2020 year class suggesting that this year class is discarded in preference over the 2019 year class. This is likely driven by the relatively smaller body sizes associated with these two year classes as observed in the catch data (see section 8.2.3) and as reported anecdotally from the industry. No specific additional information on haddock was provided by the relevant fishing industries in 2025.

8.1.3 ICES advice

ICES advice for 2025

In April 2024, ICES concluded as follows:

ICES advises that when the MSY approach is applied, total catches in 2025 should be no more than 112 435 tonnes.

ICES notes the existence of a precautionary management plan, developed and adopted by one of the relevant management authorities for this stock.

8.1.4 Management

Annual management of the fishery operates through TACs for three discrete areas; Subarea 4 (and EU Waters of 2.a), Division 3.a (EU waters), and Division 6.a. The 2024 TACs for haddock in these areas were 101 421 tonnes, 6233 tonnes and 11 301 tonnes respectively. The 2025 TACs for haddock in these areas were 95 862 tonnes, 5892 tonnes and 10 681 tonnes respectively.

The stock is managed according to advice based on the ICES MSY approach. The details of management plans relevant to North Sea haddock (Subarea 4 and Subdivision 20) and haddock in Division 6.a are provided in the Stock Annex. However, these management plans have not been evaluated by ICES for the wider Northern Shelf area. In 2018, EU-Norway requested an evaluation of multiple management strategies (ICES, 2019a) however, these are no longer applicable following the revisions to the assessment and reference points at the benchmark in 2022 (WKNSCS; ICES, 2023). In July 2018, the European Union agreed to a multiannual management plan (MAP) for demersal fisheries in the North Sea. However, the plan was not adopted by Norway or the UK and is therefore not used as the basis of advice for this shared stock. Details of the plan are given in EC 2018/973.

Fleets catching Northern Shelf haddock have previously been subject to several management schemes both aimed at reducing catches of cod. These were the Scottish Conservation Credits scheme which utilized Real Time Closures and the use of species-selective gears and the Fully documented Fisheries scheme which utilized on board monitoring cameras. Further details of these schemes are given in the Stock Annex.

The EU landing obligation was initially implemented from 1 January 2016 for directed haddock fisheries and was fully implemented in the North Sea and Northwestern Waters from 1 January 2019. Catches of haddock are also subject to UK and Norwegian national legislation regulating discards. A small number of exemptions exist for catches of haddock in ICES division 3.a. These include *de minimis* exemptions for catches of haddock from creels and some bottom trawls targeting *Nephrops* or Northern prawn. A survivability exemption exists for haddock caught using pots and fykenets.

8.2 Data available

8.2.1 Catch

In 2025, revisions of the catch data affecting discard estimates for 2021-2023 were received from UK (England). This substantially increased the discard estimates for *Nephrops* trawlers (range of ~250-500% increase). Minor revisions to the discard estimates in 2023 were also received from The Netherlands. However, overall, these revisions resulted in only a 2-7% increase in total discards for these years and a 1-3% increase in total catch due to the small overall contribution these countries make to the catch data. Changes to the numbers-at-age and mean weight-at-age were also observed to be small. A sensitivity analysis was conducted to see the effect these data revisions had on last year's assessment results (i.e., WGNSSK 2024). The differences observed were negligible.

Official landings data for each country participating in the fishery are presented in Table 8.2.1, together with the corresponding ICES estimates and the agreed international quota (listed as "total allowable catch" or TAC). International data on landings, discards, below minimum size (BMS) landings and logbook registered discards (LRD) from 2008 onwards are collated through the InterCatch system (see Section 1.2). The sampling coverage of haddock catch is usually quite high with the majority of the catch data being provided by Scotland.

Figure 8.2.1 and Tables 8.2.2 to 8.2.4 summarize the proportion of landings in the combined Northern Shelf area for which samples have been provided for 2024. While there are a large number of fleets for which landings have not been sampled, the overall contribution of these fleets to total landings is small. The proportion of landings that have been sampled in 2024 (87%) which is comparable to the sampling coverage seen last year.

Industrial bycatch (IBC) has declined considerably from the high levels observed until the late 1970s and has rarely exceed 100 tonnes in recent years. However, estimates since 2020 have been noticeably higher (>1000 tonnes). This increase has been partly driven by the larger size of recent cohorts and may also have been due to an increase in effort in the Norway pout fishery. Additionally, since 2021 there have been changes to the Danish sampling protocols for industrial bycatch. Despite this, industrial bycatch remains a minor component of the total catch, being approximately 2% of total catch in 2024.

Subarea 4 discard estimates for 2024 are derived from data submitted by Belgium, Denmark, France, Germany, the Netherlands, England, and Scotland. As Scotland is the principal haddock fishing nation in that area, Scottish discard practices dominate the overall estimates. Subdivision 20 discard estimates are derived from data submitted by Denmark, Germany, and Sweden. Division 6.a discard estimates are provided by France, Ireland, and UK (Scotland). Where landings are provided without a corresponding discard estimate, discards are raised using a discard rate inferred using simple averaging weighted by landings weight without consideration of area, quarter, country, or gear. In 2024, discard observations are available for the fleets landing the majority of haddock landings (provided for 88% of the landings; see Figure 8.2.2), and 81% of the reported discards were sampled (see Figure 8.2.3). Though these sampling levels show improvement on the lower sampling coverage seen during the COVID–19 pandemic (2020–2021), discard sampling has still not returned to pre-pandemic levels.

The collation of BMS landings and LRD in InterCatch was introduced in 2016 in accordance with the implementation of the EU landing obligation and national legislation regulating discards in the UK and Norway. Sampling information is solely provided by Scotland, however, the provision of BMS landings data and sampling information since 2016 has been sporadic. In 2024, Scotland were unable to provide reliable age estimates for BMS landings (see Figure 8.2.4). Logbook registered discard observations have not been submitted by any country for haddock in any year.

The full time-series of landings, discards, BMS landings and industrial bycatch (IBC) is presented in Table 8.2.5 and Figure 8.2.5. The total landed yield of the international fishery has been relatively stable since 2007 though has steadily risen since 2020. The total yield in 2024 is similar to that of 2023. The ICES estimates (Table 8.2.5) show that haddock discarding (as a proportion of the total catch) has fluctuated over time. The lowest discard rate since 2007 was seen in 2013 at 7.2% by weight. This may have been due in part to fleet behaviour changes related to cod avoidance measures, but also to the weak year classes since 2009 (implying that the bulk of the catch was large, mature fish that are less likely to be discarded). Since then, the discard rate has generally increased year on year, reaching a peak of 37% in 2021, from which it has steadily dropped. The discard rate in 2024 is 21% (Figure 8.2.5). The recent high discard rates have been driven mostly by the large 2019, 2020 and 2022 year classes. The recent changes in discarding are not consistent across ages (Figure 8.2.6).

It would be expected that under a landing obligation fish caught under the MCRS would be landed and recorded as BMS landings in logbooks and reported to ICES. However, low BMS values may be seen if the fish caught below MCRS are either not landed, not recorded in logbooks, not reported to ICES or a mixture of the three. BMS landings reported to ICES in 2024 are 0.37% of the total catch which is significantly lower than the discard estimate of 21% of total catch. This suggests that fish caught below MCRS are not being reported as BMS landings.

Previously, estimated discard rates could be calculated using video data from Scottish vessels carrying cameras (as part of the FDF scheme described in the Stock Annex). Neither fish ages nor weights can be measured directly using video, but a method has been developed in Scotland for estimating discard rates by measuring numbers and lengths of discarded fish and applying existing weight-length relationships to obtain a discarded weight, which can then be compared with the total landed weight (see Needle *et al.*, 2015). The lack of age information currently impedes the use of these estimates in the ICES assessment process, but work is underway in Scotland and elsewhere to address this.

8.2.2 Age compositions

Age compositions for un-sampled landings (including IBC), discards, and BMS landings are generated from the age compositions from all sampled landings, discards and BMS landings, respectively, without consideration of area, quarter, country, or gear. Total catch-at-age data are given in Table 8.2.6, while catch-at-age data for each catch component are given in Tables 8.2.7 to 8.2.10.

8.2.3 Weight-at-age

Weight-at-age for the total catch in the North Sea is given in Table 8.2.11. Weight-at-age in the total catch is a number-weighted average of weight-at-age in the human consumption landings, discards, BMS landings and industrial bycatch components. Weights-at-age in the stock are derived by applying correction factors (calculated from comparing mean weights-at-age in the catch data to survey data from NS-IBTS Q1 and UK-SCOWCGFS Q1) to the mean weights-at-age in the catch. The mean weights-at-age for the separate catch components and for the stock are given in Tables 8.2.12 to 8.2.16 and are illustrated in Figure 8.2.7 and Figure 8.2.8: this shows the declining trend in weights-at–age in recent years for ages 2+. This concurs with anecdotal reports that haddock caught have been relatively small in the last few years. This may be driving recent trends in discarding.

The mean weights-at-age in the last data year (2024) are used in the intermediate year (2025) in the assessment. However, there is some evidence of reduced growth rates for large year classes. Jaworski (2011) concluded that linear cohort-based growth models are the most appropriate method for characterizing haddock growth. Therefore, these modelled weights are used in both the intermediate year (2025) and the advice year (2026) in the short-term forecast (Section 8.6).

8.2.4 Maturity and natural mortality

The basis for estimates of maturity and natural mortality are described in the Stock Annex. Natural mortality varies with age and year as shown in Figure 8.2.9. These values shown here are derived from the latest WGSAM key run conducted in 2023 (WGSAM; ICES, 2024).

Annually varying maturity ogives are estimated from Q1 survey data following WKMOG guidelines (WKMOG; ICES, 2008) and are shown in Figure 8.2.10 and Table 8.2.18. Data on maturity in quarter 1 of the assessment year (2025) are used to derive the time-series of maturity-at-age. A GAM is then used to reduce interannual variability of the time-series and focus on long-term trends. It is these smoothed values that are used in the assessment except for the assessment year (2025) value which is omitted. This is because the GAM smoothing means the last data point in the time-series can change a substantial amount year-to-year as new data are added. Therefore, the assessment year (2025) maturity estimate is set equal to that estimated for the data year (2023). This is a similar procedure to other biological parameters.

The method for producing the maturity ogives involves a re-calculation of the entire time-series whenever new data are added. Therefore, a retrospective analysis was conducted to check the robustness of the historic values to the addition of new data. The Mohn's rho values for ages 1 to 3 were −0.2%, −11.6% and −.3% respectively, indicating a good level of robustness (Figure 8.2.11).

A new maturity scale (shown in the table below) was introduced by WKASMSF (ICES, 2018) in 2021. However, this new scale has not been used by all countries when submitting survey data. This is particularly important for maturity stage B which is split into two sub-stages, Ba and Bb. The Ba maturity code indicates first-time developers – fish which are becoming sexually mature but are unlikely to contribute to spawning that year and so are considered to be immature fish. The use of the subsets within the B stage has increased over time and stands at 84% of B stage maturity samples in 2024. This has consequences for the maturity ogive estimates as trends in maturity over time may be driven by the increased uptake of the new SMSF maturity scale rather than true trends in maturity. Most first-time spawners for haddock are seen at age 2, for which there is a decreasing trend in maturity in recent years. However, a similar decreasing trend can be seen in age 3 so it is possible that the trend at age 2 is a true trend and not an artefact of changes in data reporting.

State	Stage	Possible sub-stages
SI. Sexually immature	A. Immature	
SM. Sexually mature	B. Developing	Ba Developing but functionally immature (first-time developer)
		Bb. Developing and functionally mature
	C. Spawning	Ca. Actively spawning

State	Stage	Possible sub-stages
		Cb. Spawning capable
	D. Regressing/Regenerating	Da. Regressing
		Db. Regenerating
	E. Omitted spawning	
	F. Abnormal	

8.2.5 Catch, effort and research vessel data

The available survey data are summarized in the following table: data used to generate indices for use in the final assessment are highlighted in bold.

Area	Country	Quarter	Code	Year range	Age range
Subarea 4	Scotland	Q3	ScoGFS Aberdeen Q3	1982–1997	0+
Subarea 4	Scotland	Q3	ScoGFS Q3 GOV	1998-present	0+
Subarea 4	England	Q3	EngGFS Q3 GRT	1977–1991	0+
Subarea 4	England	Q3	EngGFS Q3 GOV	1992-present	0+
Subarea 4 and Division 3.a	International	Q1	IBTS Q1	1983-present	1+
Subarea 4 and Division 3.a	International	Q3	IBTS Q3	1991-present	0+
Subarea 6.a	Scotland	Q1	ScoGFS-WIBTS Q1	1985–2010	1+
Subarea 6.a	Scotland	Q1	UK-SCOWCGFS-Q1	2011-present	1+
Subarea 6.a	Scotland	Q4	ScoGFS-WIBTS Q4	1996–2009	0+
Subarea 6.a	Scotland	Q4	UK-SCOWCGFS-Q4	2011-present	0+
Subarea 6.a	Ireland	Q4	IGFS-WIBTS-Q4	1993–2002	0+
Subarea 6.a	Ireland	Q4	New IGFS-WIBTS-Q4	2003-present	0+

The 2022 benchmark meeting (WKNSCS; ICES, 2023) developed combined North Sea-West Coast of Scotland (NS-WC) indices for Q1 and Q3+Q4 (combined quarters) for use in the assessment using the delta-GAM modelling approach. The general methodology is described in Berg *et al*. (2014) and full methodological details are available in the report of the last benchmark (WKNSCS; ICES, 2023). However, persistent patterns in the spatial residuals were seen with this model. Therefore, in 2025 the model to derive the survey indices was updated to include annual deviations to the fixed spatial model which resulted in improvement in the spatial residuals (see Working Document 1 in Annex 8).

For the NS-WC Q1 survey indices the model is fit to ages 1–8+ and all are used in the assessment model. For the NS-WC Q3+Q4 survey indices the model is fit to ages 0–8+ all of which are used in the assessment model. A summary is given in the table below:

Modelled survey name	Input survey data	Q	Area	Years	Ages
Modelled NS-WC Q1 survey indices	ScoGFS-WIBTS-Q1; UK-SCOWCGFS-Q1; NS-IBTS-Q1	Q1	North Sea (27.4), Skagerrak (27.3.a) and West Coast of Scotland (27.6.a)	1983–2025	1–8+
Modelled NS-WC Q3+Q4 survey indices	NS-IBTS-Q3; ScoGFS-WIBTS-Q4; IGFS-WIBTS-Q4; UK-SCOWCGFS-Q4	Q3+Q4	North Sea (27.4), Skagerrak (27.3.a) and West Coast of Scotland (27.6.a)	1991–2024	0–8+

The modelled survey indices used for the calibration of the assessment are presented in Table 8.2.19. Coefficients of variation are used to weight the modelled both survey indices in the assessment model. This is to account for the increased uncertainty in the indices resulting from, for example, reduced sampling coverage. In 2022, a combination of several major storms and mechanical issues with some vessels resulted in a substantial reduction in sampling coverage of the NS-IBTS Q1 survey and the cancellation of the UK-SCOWCGFS Q1 survey.

Survey-based abundance distributions by age and year are given in Figures 8.2.12 (North Sea IBTS Q1), 8.2.13 (North Sea IBTS Q3) which clearly show the large 2019 and 2020 year classes in addition to a sizeable 2022 year class. However, the survey-based abundances indicate a very small incoming year class in 2023 and 2024. Abundance trends in survey indices are shown in Figure 8.2.14. These indicate reasonably good consistency in stock signals from the two North Sea surveys and support the perception of large year classes in 2019, 2020 and 2022.

8.3 Data analyses

The assessment has been carried out using SAM (Nielsen and Berg, 2014) as the main assessment method. The results of a SURBAR analysis are also shown for comparison with the main assessment.

8.3.1 Exploratory catch-at-age-based analyses

The catch-at-age data, in the form of log-catch curves linked by cohort (Figure 8.3.1), indicate partial recruitment to the fishery for most cohorts up to age 2 (shown by hooks towards the top of the catch curves). Gradients between consecutive values within a cohort have reduced considerably for some recent cohorts, reflecting a reduction in fishing mortality, although catch curves are considerably more variable in recent years suggesting less consistent catch data (which may reflect the lower sample size available from reduced landings, or COVID−19 impacts on sampling). Figure 8.3.2 plots the negative gradient of straight lines fitted to each cohort over the age range 2–4, which can be viewed as a rough proxy for average total mortality for ages 2–4 in the cohort. These negative gradients are also lower in most recent cohorts.

Cohort correlations in the catch-at-age matrix (plotted as log-numbers) are shown in Figure 8.3.3. These correlations show good consistency within cohorts up to the plus-group, verifying the ability of the catch-at-age data over the full time-series to track relative cohort strengths.

8.3.2 Exploratory survey-based analyses

The two survey derived indices (delta-GAM NS-WC Q1 and delta-GAM NS-WC Q3+Q4) show a high level of consistency between the indices in the plot of survey catch curves (Figure 8.3.4). Although there are indications of reduced catchability for cohorts in delta-GAM NS-WC Q1 from 2010 onwards. The plots of mean-standardized log survey indices by age and cohort (Figure 8.3.5) and the pairwise within-survey correlations (Figure 8.3.6) show that both surveys track year class strength well through the population overall.

A SURBAR run (ICES, 2010; Needle, 2015) was carried out using the same combination of tuning indices as the SAM assessment, although excluding the plus group (8+) and age 0. The summary plot from this run is given is Figure 8.3.7, which indicates good precision in estimates for total mortality, and relative estimates for biomass and recruitment. The SURBAR residual plot in Figure 8.3.8 shows that there remains an indication of some conflict (mostly positive residuals for Q1 and negative residuals for Q3+Q4).

Mean-standardizing SSB and recruitment estimates (using a common year-range for the mean) and generating SAM estimates of Z by adding F and M permits the comparison between SAM and SURBAR shown in Figure 8.3.19. SSB and recruitment estimates are very similar from the two models, although it is noticeable that the swings between high and low SURBAR SSB estimates are more pronounced than for SAM. The mean Z time-series from SURBAR are consistent mostly with those from SAM. Overall, the SURBAR assessment concurs with and support the final SAM assessment, with some relatively minor variations.

8.3.3 Final assessment

In 2025, a review of assessment model settings was conducted following the change to the survey indices used as input to the assessment (WD 1 in Annex 8). The model settings proposed by this externally reviewed process are used here. In addition to the updated indices and model settings, CVs are now used as weightings in the assessment model for both surveys whereas, previously, they were only used for the NS-WC Q1 indices.

Table 8.3.1 and Table 8.3.2 give the final SAM assessment inputs and settings, while Table 8.3.2 gives the corresponding parameter estimates from the completed run. A full description of the SAM method, inputs and settings are given in the Stock Annex, and the ICES WKNSCS (2023) report. Note that, for assessment purposes, total catch is divided into human consumption landings (referred to as "landings") and a composite of discards, BMS landings and industrial bycatch (referred to as "discards" or "discards+bycatch+BMS").

The stock summary is given in Figure 8.3.10. The spawning-stock biomass in 2025 is seen to be at the highest value in the time-series while fishing mortality is seen to be at its historical minimum. Recruitment in 2023 and 2024 are seen to be very low. However, this poor recruitment estimate follows 4-year classes that have all been larger than the recent mean (see also Figure 8.3.11).

The one-observation-ahead (Figure 8.3.12) and process error (Figure 8.3.13) residuals are relatively small and, although some patterns are seen the residuals, these are similar to those seen at the last benchmark meeting (WKNSCS; ICES, 2023). Investigation of these patterns is listed as an issue in section 8.12.

The fit to the commercial catch data, delta-GAM NS-WC Q1 indices and delta-GAM NS-WC Q3+Q4 indices are shown in figures 8.3.14–8.3.16. Overall, these indicate reasonably good fits to data.

The results of the leave-one-out analysis are shown in Figure 8.3.17. These results show similar trends when either of the survey indices are removed from the analysis indicating that the signals from each survey are consistent with each other. The leave-one-out runs are all mostly within the pointwise 95% confidence intervals of the full assessment results except for some more recent years in the SSB estimate when the delta-GAM NS-WC Q1 survey is removed.

Figure 8.3.18 summarizes the results of SAM retrospective analyses for Northern Shelf haddock. There is no significant retrospective pattern for Fbar with none of the retrospective run falls outside the pointwise 95% confidence intervals of the full time-series assessment. However, a substantial retrospective pattern can be seen for SSB. Mohn's rho values (average relative bias of retrospective estimates) were calculated for SSB, F and recruitment estimates from SAM and were −16.4%, 10.8% and −31% respectively. The Mohn's rho value for SSB falls outside of the limits specified by WKFORBIAS (ICES, 2020) and several peels fall outside of the pointwise 95% confidence intervals.

The retrospective pattern for SSB is thought to be driven by a combination of the recent rapid rise in stock biomass and higher uncertainty in the survey indices for specific cohorts. The large size of two consecutive year classes in 2019 and 2020 as well as the more recent 2022 year class are driving the rapid increase in stock size. Meanwhile, uncertainty in the survey indices is seen to be larger for the 2019 and 2020 year classes at all ages compared to other cohorts. Therefore, it seems reasonable that the retrospective pattern is due to the assessment model having difficulty estimating the size of these large cohorts. The abundance estimates for these cohorts gets substantially revised each year as each new year of data is added to the time series.

As per the WKFORBIAS guidelines, the Mohn's rho value should not be considered in isolation but other model diagnostics and fit to data should be considered. The haddock assessment is a very well fitted model with no other obvious problems. Additionally, there is no significant retrospective pattern on fishing mortality. Following the WKFORBIAS decision tree indicates that advice can be given in this case because SSB is well above MSY $B_{trigger}$ and the SSB estimate gets revised upwards with each new year of data meaning that the advice meets the precautionary principle.

Fishing mortality estimates for the final SAM assessment are presented in Table 8.3.4, the stock numbers in Table 8.3.5, and the assessment summary in Table 8.3.6. A comparison with the results of last year's assessment is shown in Figure 8.3.19.

Following the revision to the natural mortality estimates from the latest SMS key run (see section 8.2.4), reference points needed to be updated using the assessment results, before conducting the short-term forecast. See section 8.8 for more details.

8.4 Historical stock trends

The historical stock and fishery trends are presented in Figure 8.3.10.

Landings stabilized between 2005 and 2014, partly due to the limitation of interannual TAC variation to ±15% in the EU-Norway management plan for the North Sea. Discards have fluctuated in the same period due to the appearance and subsequent growth of large year classes, while industrial bycatch (IBC) is now at a very low level for haddock (see Figure 8.2.5).

Estimated fishing mortality for 2008 to 2020 fluctuates between 0.2 and 0.45 and has been decreasing since 2018. Fluctuations around the previous F(target) rate (0.3) of the management plan are an expected consequence of the lag between data collection and management action and should not be taken to indicate that the plan did not work.

The underlying mean level of recruitment has declined from the early seventies until today, and recruitment remains lower in general. Over the last two decades, recruitment has been characterized by stretches of poor recruitment (e.g. 2006–2008 and 2010–2013) with occasional modestly sized year classes (e.g. 2005, 2009, 2014) which have sustained the fishery in recent years. However, the 2019 and 2020 year classes are estimated to be the largest since the 1999 year class and are very unusual for a haddock stock in that they occur consecutively. Furthermore, the last 4 year classes (2019–2022) have all been larger than the recent mean recruitment. These recruitment events have increased the estimated SSB since 2021.

8.5 Recruitment estimates

Following the Stock Annex, recruits in the intermediate year (IY = 2025) and in the advice year (IY + 1 = 2026) were sampled, with replacement, from recent (2000–2024) recruitment estimates from SAM. This method ensures that information about the frequency of the occasional larger year classes is included in the forecast. The geometric mean of the resampled recruitments is 2 014 079 thousand.

The following table summarizes the recruitment for the short-term forecast.

Year class	Age	SAM forecast (millions)
2025	0 in 2025	2014
2026	0 in 2026	2014
2027	0 in 2027	2014

8.6 Short-term forecasts

Reference points were updated (see section 8.8) prior to conducting the short-term forecast following the revision to the survey indices (see section 8.2.5).

The short-term forecast is a stochastic forecast conducted in SAM. The inputs to the short-term forecast are presented in Table 8.6.1. The forecast is conducted for a single fleet (as opposed to two fleets: human consumption catches and industrial bycatch) and industrial bycatches are included with the projected discards. Due to limitations in the options available for setting the recruitment assumption in the first forecast year, the base year of the forecast simulation is set as the last data year (2024) to permit the resampling of recruitment to occur in all forecast years (2025–2026).

Initial stock size

The initial stock sizes are simulated from the estimated distribution at the start of last data year (including covariances).

Maturity and natural mortality

Both the natural mortality and maturity estimates used in the forecast are set equal to the mean of the final 3 data years (2022–2024).

Weights-at-age

Mean weights-at-age are forecast using the method proposed by Jaworski (2011) which was discussed by WKHAD (ICES, 2014) and modified at WKNSCS (ICES, 2023). The method is summarized in the Stock Annex and involves fitting straight lines to cohort-based weight estimates and extrapolating forward in time.

The mean weights-at-age resulting from this method for the landings are summarized in Figure 8.6.1. The weights-at-age for discards, IBC and BMS were combined using the relative contribution of each component to the total catch. These combined weights were used in the extrapolation to calculate the forecast weights and are shown in Figure 8.6.2. Where there is insufficient data (i.e., less than 3 data points) to allow for cohort-based modelling of weights-at-age a simple three-year (2022–2024) means by age are used for all forecast years. The mean weights-at-age for the total catch are derived by combining the forecast weights-at-age for the landings and discards (including IBC and BMS) using the relative contribution of each component to the total catch. Survey-derived correction factors (see Section 8.2.3 for more details) were applied to the mean weights-at-age for the catch to derive mean weight-at-age for the stock. The resulting forecast weights for the stock and all catch components are shown in Figure 8.6.3. It should be noted that the SSB value for 2025 in the forecast will be slightly different from the SSB value for 2025 from the assessment since these forecast weight-at-age are different from the mean weights-at-age used in the assessment. However, the stock numbers-at-age remain unchanged.

Fishing mortality

WKNSCS (ICES, 2023) concluded that fishing mortality estimates for the intermediate year should be a mean of the final 3 data years (2022–2024), rescaled to the final year (2024) F. When this approach results in a total catch that overshoot the TAC, a TAC constraint should be considered. A TAC constraint was not needed for the intermediate year to avoid a TAC overshoot. Therefore, $F_{status\ quo}$ (F_{2024} = 0.060) was used for the intermediate year. The combined-area human consumption TAC for 2025 is 112 435 tonnes.

The catches obtained with the fishing mortality rates discussed above are split into landings and discards (including IBC and BMS) by using the relative contribution (averaged over 2022–2024) of each component to the total catch.

Splitting catch forecasts between management units

The haddock assessment presented in this section is for the combined Northern Shelf stock, following the conclusion from ICES WKHAD (2014) that this was biologically appropriate. However, catch advice is still required for the extant management units. ICES WKHAD (2014) proposed a survey-based method for splitting forecast catch into subunits on the basis of a time-smoothed survey-based estimate of the proportion of the fishable stock in each area in each year.

However, the survey-based proportions were not accepted by ACOM (in June 2014) as the basis for advice, due to concerns over the comparability of survey catchability between the three management areas covered by the assessment area. As a consequence, the catch forecasts provided in Table 8.6.2 are provided for the full stock area only (Subarea 4, Division 6.a and Subdivision 20).

Forecast results

Results for the short-term forecasts are presented in Table 8.6.2. Assuming a F of 0.060 in 2025, SSB is expected to be 740 576 tonnes in 2025, before decreasing in 2026 to 607 965 tonnes. In this case, projected landings in 2025 would be 43 332 tonnes with associated projected unwanted catch (discards + IBC + BMS) of 11 231 tonnes.

Several alternative options for 2026 have been highlighted in Table 8.6.2. These are based on various reference points including F_{MSY}, F_{pa}, B_{pa}, B_{lim}, $B_{trigger}$ as well as F_{2025}, $F_{MSY\ upper}$, and $F_{MSY\ lower}$. Under the assumption of F_{MSY}, the 2026 total catch is forecast to be 108 301 tonnes, which corresponds (if recent discard+IBC+BMS rates remain unchanged) to landings yield of 91 316 tonnes and discards (including IBC and BMS) of 16 985 tonnes. This advised catch represents a 3.7% decrease on the TAC and advised catch for 2025. This exploitation is forecasted to lead in turn to an SSB in 2027 of 470 142 tonnes, a decrease of 23% on the SSB forecasted above for 2026.

Change in advice

The change in advice (−3.7%) is due to a change in the reference points caused by updated survey indices and assessment model settings. Although the change in survey indices, assessment model settings and an additional year of data resulted in a general upwards revision in stock size across most ages this seems to be offset by the corresponding reduction in F_{MSY} following the recalculation of reference points.

A comparison of the numbers-at-age of the current (WGNSSK 2025) and the previous (WGNSSK 2024) assessment and forecast are shown in Table 8.6.3 and Figure 8.6.4. The numbers-at-age in 2025 have been revised upwards for most ages in the current assessment compared to the assumption made in the intermediate year of the forecast at WGNSSK 2024.

Mean weight-at-age in the stock are different in each forecast year due to the cohort growth model used to derive the mean weights used in the forecast (Figure 8.6.5). However, a comparison with the mean weights at age used in last year's forecast shows that there are only minor differences. The effect of the combination of changes in numbers and mean weight-at-age between the forecasts is shown in Table 8.6.4 and Figure 8.6.6. This again shows that the size of the stock is larger in this year's forecast across most ages.

A comparison of the selectivity used in this year's forecast compared to last year's is shown in Figure 8.6.7. However, this comparison shows very little difference between the two assumptions.

A comparison of the assumed SSB, Fbar, total catch and recruitment is shown in Table 8.6.5 and Figure 8.6.8. The stock sizes in this year's forecast are slightly larger compared to last year's forecast. This year's recruitment assumption for the intermediate and advice year is similar to that made at WGNSSK 2024 as this value comes from resampling recent recruitment values. The Fbar assumption made for 2024 in this year's assessment is slightly lower than last year's forecast. This is also seen in the slightly lower total catch values for 2024. The Fbar assumption made and resulting total catch for the intermediate year (2025) this year is slightly lower the intermediate year (2024) assumption from last year. Finally, the Fbar used in the advice year this year (2025) is based on revised reference points compared to the Fbar used in the advice year (2024) in last year's forecast.

8.7 Medium-term forecasts

No specific medium-term forecasts have been carried out for this stock. Management simulations over the medium-term period were previously performed for North Sea haddock (Needle, 2008a; Needle, 2008b) and West of Scotland haddock (Needle, 2010), while management strategy evaluations for Northern Shelf haddock were conducted in 2019 in response to a request for advice on a proposed EU-Norway management plan (ICES, 2019a, b).

8.8 Biological reference points

Biological reference points were calculated at WGNSSK 2025 using EqSim following the provision of updated survey indices and the associated changes in assessment model settings. The EqSim settings and assumptions remained broadly the same as those used and detailed in the WKNSCS report (ICES, 2023). Given the recent strong trends in mean weights-at-age and selectivity only the 5 most recent years were used to derive mean values for the biological parameters and for the selectivity (Figure 8.8.1). Default values were used for sigmaSSB (0.2), sigmaFbar (0.2) since the assessment sdSSB and sdFbar were both less than 0.2. Default values were also used for other settings (cvF = 0.212, phiF = 0.423, cvSSB = 0, phiSSB =0).

The simulations were based on a recruitment time-series from 2000–2024, rather than the full time-series. This is because the recruitment during this more recent period has been poor compared to the full time-series and it would be unwise to assume that a very large recruitment is likely in the near future. There are indications of a drop in productivity, as indicated by recruits per SSB, since 2000 (Figure 8.8.2a). No autocorrelation in the recruitment time-series was observed (Figure 8.8.2b). A segmented regression stock–recruitment relationship was used in the simulations (Figure 8.8.2c). To keep some consistency with the derivation of B_{lim}, which was based on the whole time-series, a segmented regression stock–recruitment relationship with a fixed breakpoint at B_{lim} was used.

The stock–recruitment relationship indicates that haddock is a type 1 stock (Figure 8.8.2d). Using the ICES guidelines for sporadic spawners (type 1 stocks) B_{lim} was set to 138 250 tonnes (the estimated SSB for 1979, the smallest stock size to produce a good recruitment where good recruitment is defined as the 95th percentile and above), and B_{pa} was revised to 1.645*sigmaSSB x B_{lim} = 192 109 tonnes (where sigmaSSB = 0.2).

An EqSim analysis run with assessment and advice error (using default values) but without the advice rule gave a value of 0.247 for the unconstrained F_{MSY} (Figure 8.8.3). The value of F_{pa} is set equal to the F that provides a 95% probability for SSB to remain above B_{lim} in the long term ($F_{P.05}$; run with assessment and advice error and advice rule) by the ICES guidelines. The value of $F_{P.05}/F_{pa}$ was calculated as 0.167. Since the value of the unconstrained F_{MSY} (0.247) was estimated to be above $F_{P.05}/F_{pa}$, the final value of F_{MSY} was capped at $F_{P.05}/F_{pa}$ (0.167) to ensure consistency between the precautionary and MSY frameworks (see Figure 8.8.4). The F_{MSY} ranges were adjusted accordingly to be consistent with the capped value of F_{MSY} ($F_{MSY\ lower}$ = $F_{P.05\ lower}$ = 0.155; $F_{MSY\ upper}$ = $F_{P.05}$ = 0.167).

Finally, the value of MSY $B_{trigger}$ was set equal to B_{pa} (192 109 tonnes) since, although the stock has been fished at a level below the unconstrained F_{MSY} for at least 5 years, the 5th percentile of B_{MSY} (40 858 tonnes) is less than B_{pa}

The reference points from these analyses are given below:

Variable	WKNSCS 2022	WGNSSK 2024	WGNSSK 2025
B_{lim}	136 541 tonnes	141 339 tonnes	
B_{pa}	189 734 tonnes	196 402 tonnes	192 109 tonnes
MSY $B_{trigger}$	189 734 tonnes	196 402 tonnes	192 109 tonnes
$F_{P.05}/F_{pa}$	0.24	0.174	0.167
F_{MSY}	0.24	0.174	0.167
$F_{MSY\ lower}$	0.186	0.161	0.155
$F_{MSY\ upper}$	0.24	0.174	0.167

8.9 Quality of the assessment

Survey data are consistent both within and between surveys, and the catch data are internally consistent. Trends in mortality from catch data and survey indices are similar.

The assessment model has a strong retrospective pattern in SSB. This is thought to be driven by the uncertainty in survey indices on the size of the recent large year classes (2019 and 2020) coupled with the rapid increase in stock size in recent years. However, given that the retrospective pattern shows upwards revisions of SSB and that SSB is well above MSY $B_{trigger}$ the advice is considered to be precautionary.

The COVID−19 pandemic affected catch sampling in 2020 and also, to a lesser extent, in 2021. In 2020, both observer and market sampling were not possible during Q2 but continues at a reduced level in Q3 and Q4. In 2021, Scottish observer sampling was not possible during Q1 though sampling proceeded at similar levels to the latter half of 2020 for the rest of the year. This reduced coverage is not thought to have had a significant impact on the quality of catch data for Scotland (which has the main fleets catching haddock).

In 2022, a combination of several major storms and mechanical issues with some vessels resulted in a reduction in the sampling coverage across the NS-IBTS and SCOWCGFS Q1 surveys. This increased the uncertainty on the Q1 survey indices used in the assessment. Relative weightings (CVs) for the Q1 survey indices were added as an input to the assessment model to minimize the impact of this increased uncertainty on the assessment results.

8.10 Status of the stock

Fishing mortality is now estimated to be at a historically low level in 2024 and is well below the estimate of F_{MSY} (0.167). Discard rates have increased above the historical minimum observed in 2013. The 2010–2013 year classes were estimated to be weak, following the relatively strong 2009 year class, but the 2014 year class was slightly larger than the recent average and the incoming 2019 and 2020 year classes appear to be the largest since 1999 and year class sizes between 2019 and 2022 have all been notably above the recent mean. Spawning-stock biomass has recovered from levels below B_{lim} to levels well above MSY $B_{trigger}$/B_{pa} (192 109 tonnes) and is predicted to remain so in 2026.

8.11 Management considerations

The previous EU-Norway management plan for North Sea haddock, and the EU management plan for Division 6.a haddock, are not appropriate to the Northern Shelf stock, as they each relate to only a part of the full stock area. Discussions took place during 2019–20 between the EU and Norway to try to establish a new management strategy on the basis of the Northern Shelf stock, but no agreement has yet been reached, and further work would also need to include the UK. In the meantime, the principal basis for management of this haddock stock is the ICES MSY approach. The survey-based proposal for splitting catch advice into management subunits, which was proposed by WGNSSK in 2014, has not been agreed by ACOM, and the split of quota into management units remains based on historical landings. It is unlikely, therefore, to follow any future changes in stock distribution across the Northern Shelf.

Considering the Northern Shelf as a whole, fishing mortality declined significantly in the early 2000s and has fluctuated around a relatively low level since. The current estimate is below F_{MSY}. Spawning-stock biomass is estimated to have reached a historical peak in 2002 with the growth of the large 1999 year class, but declined again rapidly and is now driven strongly by occasional, moderately sized year classes. The most recent of these occurred in 2005, 2009, 2014, with two substantial year classes occurring in 2019 and 2020. Year classes between 2019 and 2022 have all been well above the recent mean..

Keeping fishing mortality close to the target MSY level would be preferable to encourage the sustainable exploitation of the recent larger year classes. Estimated discard rates remain high as large numbers of small fish from the 2022 year class enter the population, and this needs to be monitored and mitigated. In particular, discard rates remain high in certain small-mesh fisheries (such as the TR2 *Nephrops* fleets in Division 6.a). Further improvements to gear selectivity measures, allowing for the release of small fish, would be highly beneficial not only for the haddock stock, but also for the survival of juveniles of other species that occur in mixed fisheries along with haddock. Similar considerations also apply to spatial management approaches (such as real-time closures), and other measures intended to reduce unwanted bycatch and discarding of various species (such as the previous Scottish Conservation Credits scheme; see Stock Annex). Haddock was included in the EU Landing Obligation regulation from 2016, though the impacts on fishing and on the stock are yet unknown.

Haddock is a specific target for some fleets but is also caught as part of a mixed fishery catching cod, whiting and *Nephrops*. It is important to consider both the species-specific assessments of these species for effective management, as well as the latest developments in the mixed fisheries approach. This is not straightforward when stocks are managed via a series of single-species, single-area management plans that do not incorporate mixed-stocks considerations. However, a reduction in effort on one stock may lead to a reduction or an increase in effort on another and the implications of any change need to be considered carefully.

8.12 "Living issues" benchmark list

Below is a list of issues which were either left unresolved from the last benchmark or have arisen during subsequent WGNSSK meetings. A scoring system has been developed to aid Working Groups in prioritizing stocks to be put forward for benchmark (see Annex 6 for further details). The current scoring for this stock is:

1. Assessment quality	2. Opportunity to improve	3. Benchmark preparedness	4. Perceived risk	5. Changing ecosystem and ability to include impact in advice	6. Time since last benchmark	Total Score
2	2	0	2	0	3	1.9

8.12.1 Stock ID and other issues

- The stock is considered to be homogenous throughout subareas 4 and 6a, but there may be relevant substock structure.
- Explore stock ID and structure, using otolith micro-chemistry, tagging data, and the spatial range of genetic data.
- Ecosystem drivers for long term forecasts and management strategy evaluations) (MSE) need investigating.

8.12.2 Data

- SSB is used to indicate both reproductive potential and harvestable biomass, and it may be a poor proxy for both.
- Investigate indices of reproductive potential and methods to use them in management advice.
- Information on haddock abundances may be available from acoustic survey data (e.g., HERAS). Investigate potential of using indices derived from acoustic survey data.

8.12.3 Assessment

- Investigate patterns within the assessment residuals, especially those associated with the plus group. These may be addressed through using model settings not explored at the last benchmark.
- Investigate the relatively poorer fit in plus group in view of increasing relative importance of this age class. Consider increasing the age of the plus group.

8.12.4 Forecast

- SAM forecast settings for selectivity are not as recommended by model developer. Explore effect of changing to recommended forecast settings for selectivity.

8.13 References

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8.14 Tables and figures

Table 8.2.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Nominal landings (000 t) during 2014–2024, as officially reported to, and estimated by, ICES, along with WG estimates of catch components, and corresponding TACs. Landings estimates for 2023 and 2024 are preliminary. Quota uptake estimates are also given, calculated as the ICES estimates of landings divided by available quota before 2018. Quota uptake from 2018 onwards is calculated as the ICES estimates of total catch divided by available quota (following the full implementation of the EU landing obligation). Reporting of BMS landings started in 2016.

Subdivision 20											
Country	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
DE	114	103	125	56	31	30	13	23	107	180	263
DK	1764	1059	908	852	542	457	449	1846	2745	2935	4819
NL	6	4	2	20	4	4	1	11	7	4	12
NO	86	63	70	65	36	27	15	8	215	209	679
SE	219	203	110	104	140	93	56	125	271	347	592
UK	0	0	0	0	0	0	<1	0	0	0	0
BMS landings				< 1	< 1	0	<1	74	7	13	125

***Preliminary**

Subarea 4											
Country	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
BE	98	47	53	30	29	29	40	148	227	345	272
DE	677	599	554	609	348	313	370	475	692	928	904
DK	1079	1442	1244	1185	1117	1174	1680	1894	1740	3031	2421
FO	0	0	0	0	0	1	2	0	0	2	0
FR	209	100	121	140	201	188	144	218	349	452	627
IR	0	0	0	0	0	0	0	0	0	1	0
LT	0	0	0	0	0	0	130	0	0	1	0
NL	99	44	146	75	102	166	218	349	757	471	977

Subarea 4											
Country	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
NO	2743	2003	1499	2164	1428	1516	3165	2215	2544	2425	2379
SE	154	136	118	181	100	111	115	146	186	274	378
UK	29851	25905	26427	25667	26091	22044	20621	17268	25611	31228	33341
BMS landings				< 1	15	160	287	208	299	326	271

*Preliminary

Division 6.a											
Country	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
DE	0	0	0	0	0	0	0	0	0	1	0
DK	0	0	2	2	1	9	4	18	9	0	1
ES	19	9	33	28	28	64	26	24	14	48	21
FO	0	0	0	<1	0	0	0	0	0	0	0
FR	67	41	62	68	66	57	86	92	180	165	60
IE	667	768	1034	641	758	0	448	603	901	937	1003
NL	0	11	28	31	17	54	23	<1	3	0	0
NO	2	7	5	1	7	10	2	0	1	3	1
UK	3261	3052	3101	2480	3441	2755	2080	2860	2415	2421	2749
BMS landings				0	2	15	26	30	15	19	11

*Preliminary

	2014	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023***	**2024***
Official landings	41115	35596	35642	34399	34487	29102	29687	28322	38974	46406	51499
ICES landings	40589	35215	35111	33799	34441	30620	28942	26458	37401	44169	51006
ICES discards	5271	6241	7582	6960	5055	5335	8693	17305	19385	20551	14244
ICES IBC	65	21	37	8	30	186	1077	1357	430	2036	1373
ICES BMS	-	-	201	93	160	180	217	179	169	320	246
ICES total catch	45926	41477	42930	40860	39687	36320	38928	45299	57384	67076	66869
TAC 4	38284	40711	61933	33643	41767	28950	35653	42785	44924	58402	101421
TAC 3.a	2355	2504	3926	2069	2569	1780	2193	2630	2761	3589	6233
TAC 6.a	3988	4536	6462	3697	4654	3226	3973	4767	5006	6507	11301
Total TAC	44627	47751	72321	39409	48990	33956	41819	50182	52691	68498	118955

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023*	2024*
ICES quota uptake	91%	74%	49%	86%	81%	107%	93%	90%	109%	98%	56%

*Official landings are preliminary

Table 8.2.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Proportion of sampling strata for discards imported into InterCatch and proportion of discards raised from averaged discard rates for 2024.

Catch category	Raised or imported	Weight (tonnes)	Proportion
BMS landings	Imported	246	100
Discards	Imported	12110	85
Discards	Raised	2187	15
Industrial bycatch	Imported	1346	100
Landings	Imported	50179	100
Logbook registered discards	Imported	0	NA

Table 8.2.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Proportion of age distributions for landings, BMS landings and discards either imported or raised in InterCatch and either sampled or estimated for 2024.

Catch category	Raised or imported	Sampled or estimated	Weight (tonnes)	Proportion
Logbook registered discards	Imported	Estimated	0	NA
Landings	Imported	Sampled	43632	87
Landings	Imported	Estimated	6546	13
Industrial bycatch	Imported	Estimated	1346	100
Discards	Imported	Sampled	10900	76
Discards	Raised	Estimated	2187	15
Discards	Imported	Estimated	1210	8
BMS landings	Imported	Estimated	246	100

Table 8.2.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Proportion by area of distributions for landings, BMS landings and discards either imported or raised in InterCatch and either sampled or estimated for 2024.

Area	Catch category	Raised or imported	Sampled or estimated	Weight (tonnes)	Proportion
27.6.a	Logbook registered discards	Imported	Estimated		NA
27.6.a	Landings	Imported	Sampled	3649	95
27.6.a	Landings	Imported	Estimated	211	5

Area	Catch category	Raised or imported	Sampled or estimated	Weight (tonnes)	Proportion
27.6.a	Industrial bycatch	Imported	Estimated	1	100
27.6.a	Discards	Imported	Sampled	1427	97
27.6.a	Discards	Raised	Estimated	49	3
27.6.a	Discards	Imported	Estimated	1	0.06
27.6.a	BMS landings	Imported	Estimated	11	100
27.4	Logbook registered discards	Imported	Estimated	0	NA
27.4	Landings	Imported	Sampled	34645	87
27.4	Landings	Imported	Estimated	5281	13
27.4	Industrial bycatch	Imported	Estimated	1344	100
27.4	Discards	Imported	Sampled	8568	76
27.4	Discards	Raised	Estimated	1703	15
27.4	Discards	Imported	Estimated	1057	9
27.4	BMS landings	Imported	Estimated	235	100
27.3.a.20	Logbook registered discards	Imported	Estimated	0	NA
27.3.a.20	Landings	Imported	Sampled	5338	83
27.3.a.20	Landings	Imported	Estimated	1055	17
27.3.a.20	Industrial bycatch	Imported	Estimated	1	100
27.3.a.20	Discards	Raised	Estimated	906	61
27.3.a.20	Discards	Imported	Sampled	434	29
27.3.a.20	Discards	Imported	Estimated	153	10
27.3.a.20	BMS landings	Imported	Estimated	0	NA

Table 8.2.5. Haddock in Subarea 4, Division 6.a, and Subdivision 20. ICES estimates of catch components by weight (000 tonnes).

Year	Subarea 4					Subdivision 20					Division 6.a				Combined				
	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	Total	Landings	Discards	BMS landings	IBC	Total
1965	161.7	62.3		74.6	298.6	0.7				0.7	32.5	3.4		35.9	194.9	65.8	0.0	74.6	335.3
1966	225.6	73.5		46.7	345.8	0.6				0.6	29.9	0.7		30.6	256.1	74.2	0.0	46.7	376.9
1967	147.4	78.2		20.7	246.3	0.4				0.4	20.3	7.4		27.7	168.0	85.6	0.0	20.7	274.4
1968	105.4	161.8		34.2	301.4	0.4				0.4	20.5	25.3		45.8	126.3	187.1	0.0	34.2	347.6
1969	331.1	260.1		338.4	929.5	0.5				0.5	26.3	25.2		51.5	357.9	285.3	0.0	338.4	981.5
1970	524.1	101.3		179.7	805.1	0.7				0.7	34.1	6.2		40.3	558.9	107.4	0.0	179.7	846.1
1971	235.5	177.8		31.5	444.8	2				2.0	46.3	12.2		58.5	283.8	190.0	0.0	31.5	505.3
1972	193.0	128.0		29.6	350.5	2.6				2.6	41.1	16.4		57.5	236.6	144.4	0.0	29.6	410.6
1973	178.7	114.7		11.3	304.7	2.9				2.9	28.8	11.4		40.2	210.4	126.1	0.0	11.3	347.8
1974	149.6	166.4		47.5	363.5	3.5				3.5	18.0	15.4		33.3	171.0	181.8	0.0	47.5	400.3
1975	146.6	260.4		41.5	448.4	4.8				4.8	13.7	33.0		46.6	165.1	293.3	0.0	41.5	499.9
1976	165.7	154.5		48.2	368.3	7				7.0	18.8	15.3		34.1	191.4	169.8	0.0	48.2	409.4

Year	Subarea 4					Subdivision 20					Division 6.a				Combined				
	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	Total	Landings	Discards	BMS landings	IBC	Total
1977	137.3	44.4		35.0	216.7	7.8				7.8	19.3	4.4		23.7	164.4	48.7	0.0	35.0	248.2
1978	85.8	76.8		10.9	173.5	5.9				5.9	17.2	1.1		18.3	108.9	77.9	0.0	10.9	197.7
1979	83.1	41.7		16.2	141.0	4				4.0	14.8	6.5		21.3	101.9	48.2	0.0	16.2	166.3
1980	98.6	94.6		22.5	215.7	6.4				6.4	12.8	4.8		17.5	117.8	99.4	0.0	22.5	239.6
1981	129.6	60.1		17.0	206.7	6.6				6.6	18.2	7.1		25.3	154.4	67.2	0.0	17.0	238.6
1982	165.8	40.6		19.4	225.8	7.5				7.5	29.6	7.7		37.3	203.0	48.2	0.0	19.4	270.6
1983	159.3	66.0		12.9	238.2	6				6.0	29.4	3.4		32.8	194.8	69.4	0.0	12.9	277.1
1984	128.2	75.3		10.1	213.6	5.4				5.4	30.0	8.1		38.1	163.6	83.4	0.0	10.1	257.0
1985	158.6	85.2		6.0	249.8	5.6				5.6	24.4	10.7		35.1	188.5	96.0	0.0	6.0	290.5
1986	165.6	52.2		2.6	220.4	2.7				2.7	19.6	5.2		24.7	187.8	57.4	0.0	2.6	247.8
1987	108.0	59.1		4.4	171.6	3.8			1.4	5.2	27.0	11.1		38.1	138.8	70.2	0.0	5.8	214.8
1988	105.1	62.1		4.0	171.2	2.9			1.5	4.4	21.1	5.0		26.1	129.1	67.1	0.0	5.5	201.8
1989	76.2	25.7		2.4	104.2	4.1			0.4	4.5	16.7	2.5		19.2	96.9	28.2	0.0	2.8	127.9

Year	Subarea 4					Subdivision 20					Division 6.a				Combined				
	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	Total	Landings	Discards	BMS landings	IBC	Total
1990	51.5	32.6		2.6	86.6	4.1			2.0	6.1	10.1	0.8		11.0	65.7	33.4	0.0	4.6	103.7
1991	44.7	40.2		5.4	90.2	4.1			2.6	6.7	10.6	4.8		15.3	59.3	45.0	0.0	8.0	112.3
1992	70.2	47.9		10.9	129.1	4.4			4.6	9.0	11.3	3.5		14.9	86.0	51.5	0.0	15.5	153.0
1993	79.6	79.6		10.8	169.9	2			2.4	4.4	19.1	7.0		26.1	100.6	86.6	0.0	13.2	200.4
1994	80.9	65.4		3.6	149.8	1.8			2.2	4.0	14.2	5.0		19.2	96.9	70.4	0.0	5.8	173.1
1995	75.3	57.4		7.7	140.4	2.2			2.2	4.4	12.4	7.7		20.0	89.9	65.0	0.0	9.9	164.8
1996	76.0	72.5		5.0	153.5	3.1			2.9	6.0	13.5	7.8		21.3	92.6	80.3	0.0	7.9	180.8
1997	79.1	52.1		6.7	137.9	3.4			0.6	4.0	12.9	7.5		20.4	95.4	59.6	0.0	7.3	162.3
1998	77.3	45.2		5.1	127.6	3.8			0.3	4.1	14.4	7.0		21.4	95.5	52.1	0.0	5.4	153.0
1999	64.2	42.6		3.8	110.7	1.4			0.3	1.7	10.4	3.9		14.3	76.0	46.5	0.0	4.1	126.6
2000	46.1	48.8		8.1	103.0	1.5			0.6	2.1	7.0	6.3		13.2	54.5	55.1	0.0	8.7	118.3
2001	39.0	118.3		7.9	165.2	1.9			0.2	2.1	6.7	8.5		15.2	47.6	126.8	0.0	8.1	182.5
2002	54.2	45.9		3.7	103.8	4.1			0.1	4.2	7.1	9.4		16.5	65.4	55.3	0.0	3.8	124.5

Year	Subarea 4					Subdivision 20					Division 6.a				Combined				
	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	Total	Landings	Discards	BMS landings	IBC	Total
2003	40.1	23.5		1.1	64.8	1.8	0.2			2.0	5.3	4.5		9.8	47.3	28.2	0.0	1.1	76.6
2004	47.3	15.4		0.6	63.2	1.4	0.1			1.6	3.2	4.5		7.7	51.9	20.0	0.0	0.6	72.5
2005	47.6	8.4		0.2	56.2	0.8	0.2			1.0	3.1	3.8		6.9	51.5	12.4	0.0	0.2	64.1
2006	36.1	16.9		0.5	53.6	1.5	1.0		0.0	2.5	5.7	5.2		10.9	43.3	23.1	0.0	0.5	67.0
2007	29.4	27.8		0.0	57.3	1.5	0.8		0.0	2.3	3.7	4.0		7.8	34.7	32.7	0.0	0.0	67.4
2008	29.0	15.5		0.0	44.5	1.3	0.4		0.0	1.8	2.8	1.6		4.4	33.2	17.5	0.0	0.0	50.7
2009	30.1	10.0		0.0	40.1	1.5	0.4		0.0	1.9	2.8	1.8		4.6	34.4	12.2	0.0	0.0	46.6
2010	28.0	7.3		0.0	35.3	1.4	0.6		0.0	2.0	2.9	1.6		4.5	32.3	9.5	0.0	0.0	41.8
2011	26.8	10.1		0.0	36.8	2.1	1.3		0.0	3.4	1.8	1.3		3.1	30.7	12.7	0.0	0.0	43.4
2012	30.5	3.8		0.0	34.4	2.6	0.8		0.0	3.4	5.1	0.5		5.6	38.2	5.1	0.0	0.0	43.4
2013	36.9	2.1		0.0	39.0	2.2	0.2		0.2	2.5	4.8	1.0		5.8	43.8	3.3	0.0	0.2	47.3
2014	34.7	4.3		0.1	39.1	2.3	0.2		0.0	2.4	4.1	0.8		4.9	41.1	5.3	0.0	0.1	46.5
2015	30.0	4.7		0.0	34.7	1.4	0.2		0.0	1.6	4.0	1.4		5.4	35.4	6.3	0.0	0.0	41.7
2016	29.7	5.9	0.2	0.0	35.8	1.2	0.1	0.0	0.0	1.3	4.3	1.6	0.0	6.0	35.3	7.6	0.2	0.0	43.1

Year	Subarea 4					Subdivision 20					Division 6.a				Combined				
	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	IBC	Total	Landings	Discards	BMS landings	Total	Landings	Discards	BMS landings	IBC	Total
2017	29.3	5.2	0.1	0.0	34.6	1.1	0.1	0.0	0.0	1.2	3.5	1.6	0.0	5.1	33.9	7.0	0.1	0.0	41.0
2018	29.3	3.6	0.1	0.0	33.1	0.8	0.1	0.0	0.0	0.9	4.4	1.4	0.0	5.8	34.5	5.1	0.2	0.0	39.8
2019	26.6	3.4	0.2	0.2	30.3	0.6	0.1	0.0	0.0	0.7	3.6	2.1	0.0	5.7	30.8	5.6	0.2	0.2	36.7
2020	26.0	7.9	0.2	0.9	35.1	0.4	0.2	0.0	0.1	0.8	2.7	0.6	0.0	3.3	29.1	8.7	0.2	1.1	39.2
2021	21.0	12.4	0.2	1.3	34.9	2.0	0.6	0.0	0.1	2.7	3.7	4.4	0.0	8.0	26.7	17.4	0.2	1.4	45.6
2022	30.9	16.9	0.2	0.4	48.3	3.4	0.9	0.0	0.0	4.3	3.5	1.8	0.0	5.2	37.7	19.6	0.2	0.4	57.9
2023	37.2	18.1	0.3	2.0	57.6	3.7	1.4	0.0	0.1	5.2	3.5	1.2	0.0	4.7	44.5	20.7	0.3	2.0	67.5
2024	40.8	11.3	0.2	1.4	53.7	6.6	1.5	0.0	0.0	8.1	3.8	1.5	0.0	5.3	51.2	14.3	0.2	1.4	67.1

Table 8.2.6. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Numbers-at-age data (thousands) for total catch. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	650218	368560	16491	721514	36301	4954	2245	626	118	97	47	0	0	0	0	0	262
1966	1672925	1007517	26186	7536	459941	11903	1109	633	222	90	23	2	0	0	0	0	337
1967	345371	856339	108401	5814	3850	202830	2843	223	231	61	34	0	0	0	0	0	326
1968	11133	1226448	477603	22671	2303	3210	60034	1052	84	22	5	0	0	0	0	0	111

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1969	75301	20554	3736629	313593	9029	2678	2894	23704	392	32	7	0	0	0	0	0	431
1970	941790	272467	218881	2003201	60200	1350	1285	401	6539	81	13	19	0	0	0	0	6652
1971	337277	1881729	74866	50845	480381	10916	589	201	167	1767	176	3	5	0	0	0	2119
1972	255110	696714	671965	43309	23547	211817	4067	241	53	27	475	11	0	0	0	0	566
1973	79461	412305	587335	260080	6450	5689	72652	1406	140	34	234	49	5	0	0	0	462
1974	665110	1283252	187149	342628	60523	1956	1795	22380	345	57	63	4	7	4	0	0	480
1975	51796	2276937	673960	62175	112242	17691	1078	718	6168	339	70	11	0	8	0	0	6596
1976	171400	192030	1127520	225532	11538	32677	5864	228	84	1863	64	3	5	0	0	0	2019
1977	119506	263702	109480	426291	45756	4984	6757	1608	163	40	460	8	0	1	0	0	672
1978	281785	223294	130963	31141	144703	11791	1582	2322	740	122	33	275	16	2	0	0	1188
1979	844410	261156	220200	45487	7978	38097	3069	377	629	181	57	13	52	3	0	0	935
1980	374573	439674	374310	80225	11364	2040	11143	827	143	168	96	34	9	7	1	0	457
1981	645352	116229	430149	180553	17044	2225	497	3320	164	78	26	32	5	1	4	0	311
1982	275508	217834	89989	390347	49835	4275	820	551	1072	60	28	8	2	2	0	0	1172
1983	513034	148158	222772	83199	166812	20055	2365	338	255	385	93	21	4	4	0	0	763
1984	95862	483045	139887	143821	29321	56077	6238	967	127	84	185	19	5	1	1	0	423
1985	127002	161400	441785	80605	41508	7082	18393	1929	296	56	29	144	9	0	0	1	535

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1986	45703	137091	144075	328016	29497	10595	1686	4421	581	156	56	47	37	16	4	1	898
1987	10249	253236	259369	56407	92705	6214	3993	1187	2596	462	56	65	35	32	17	8	3271
1988	16679	33092	424014	96795	17161	27728	2030	874	368	1076	95	21	12	13	17	1	1603
1989	19587	51743	43162	216359	21015	4189	7671	763	285	170	469	69	8	3	2	1	1007
1990	19286	82571	78881	17811	60888	4373	1104	1839	254	100	54	13	12	1	4	2	439
1991	128703	188087	101425	24822	4706	17618	1388	684	1024	171	65	11	11	1	2	2	1287
1992	277933	166550	255051	43257	7162	1486	6376	611	337	401	149	22	6	2	0	0	918
1993	136841	302610	269220	123469	11822	1986	669	2050	215	210	188	84	4	4	0	0	706
1994	89104	91674	339428	106673	35056	3381	601	366	746	132	48	36	26	5	0	0	992
1995	200150	336460	119210	182969	33802	9237	898	161	155	151	21	8	6	2	1	0	345
1996	167032	46797	505401	73987	66245	11159	4058	1080	75	72	37	9	8	3	1	0	205
1997	36954	162449	107657	251339	18037	18288	2762	937	121	16	18	5	4	4	2	0	170
1998	21919	88387	224037	60861	128348	7110	4590	850	263	60	7	8	3	2	1	1	345
1999	90634	69455	119094	110046	28510	45221	2700	2047	438	53	8	3	3	2	0	0	507
2000	12630	397390	110381	61263	33137	7254	9935	765	367	53	13	2	1	1	0	0	438
2001	3518	95086	633162	34548	12078	5573	2094	1611	257	89	28	3	4	0	0	0	382
2002	50927	36063	99685	372036	7812	2801	1615	729	603	283	25	8	5	0	0	0	923

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2003	7082	13136	15234	48729	127241	2166	786	339	144	100	48	5	1	0	0	0	299
2004	3758	25698	24627	8958	38784	97827	1010	248	82	42	37	12	1	0	0	0	174
2005	8779	17695	24596	15085	5446	27745	61457	371	132	38	11	8	4	1	0	0	193
2006	3229	122537	30995	20657	11284	6078	16415	32978	156	56	20	7	4	1	0	0	243
2007	2046	20565	171600	16796	8187	4782	2237	6876	7254	75	8	14	3	1	0	0	7355
2008	2550	25087	35038	80164	4156	2093	1354	552	1577	2473	6	4	1	1	0	0	4061
2009	27313	11967	13754	18495	77722	1904	759	563	133	242	545	8	2	0	0	0	930
2010	2508	49728	14174	17429	11677	37295	833	379	142	78	81	162	10	0	0	3	475
2011	5024	4342	64649	12999	7402	5791	20830	450	119	70	25	12	70	0	0	0	295
2012	1377	3879	5072	66385	5431	3697	2414	8051	137	176	50	28	32	22	2	0	447
2013	1303	12258	4251	4651	68803	2216	1532	840	3919	36	7	7	2	2	2	0	3976
2014	3504	7593	20031	4690	7647	46684	1080	962	371	1694	13	6	1	1	0	2	2089
2015	3776	27610	15630	17723	1719	5013	21935	1062	434	437	785	108	0	0	0	0	1765
2016	1701	9374	61656	8846	5556	655	451	10138	253	151	9	149	9	0	0	1	573
2017	2615	12732	23207	54472	3228	1498	144	367	1442	502	6	20	3	1	0	1	1974
2018	3632	5556	24263	17121	35201	925	522	210	100	970	20	0	3	3	1	1	1099
2019	3554	17834	11844	25773	7208	21282	462	378	25	31	147	2	14	2	0	0	221

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2020	1540	45286	27157	11930	14636	3281	7953	178	164	62	61	20	0	0	0	0	307
2021	7057	30700	75338	13613	5798	8366	771	4727	45	31	2	2	0	1	0	0	81
2022	2482	4840	48356	94204	7816	2306	1191	288	682	156	0	0	0	0	0	0	838
2023	1279	8589	9823	64080	76114	3990	1250	572	123	97	24	1	6	0	0	0	251
2024	908	3937	16977	17072	78857	46896	1260	212	109	50	26	9	17	0	0	0	211

Table 8.2.7. Haddock in Subarea 4, Division 6.a and Subdivision 20. Numbers-at-age data (thousands) for landings. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0	2670	3908	396363	30232	4358	2126	620	118	97	47	0	0	0	0	0	262
1966	0	13034	6899	5332	419437	11113	1082	631	222	90	23	2	0	0	0	0	337
1967	0	55548	40030	4627	3607	198991	2821	223	231	61	34	0	0	0	0	0	326
1968	0	22108	151474	17130	2160	3176	59110	1051	84	22	5	0	0	0	0	0	111
1969	0	143	759680	175763	7965	2282	2760	23452	392	32	7	0	0	0	0	0	431
1970	0	2428	52031	1211535	53570	1184	1220	398	6539	81	13	19	0	0	0	0	6652
1971	0	35945	27011	37832	448352	10551	582	201	167	1767	176	3	5	0	0	0	2119
1972	0	13354	233966	35440	22165	210167	4054	241	53	27	475	11	0	0	0	0	566
1973	0	7277	211018	209961	6085	5459	72528	1406	140	34	234	49	5	0	0	0	462
1974	0	25699	55734	236624	53054	1868	1679	22156	345	57	63	4	7	4	0	0	480

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1975	0	28773	211495	41030	93617	17406	1073	718	6163	339	70	11	0	8	0	0	6591
1976	0	3045	246027	155162	11292	29594	5846	228	84	1863	64	3	5	0	0	0	2019
1977	0	8934	33058	278741	42737	4737	6516	1608	163	40	460	8	0	1	0	0	672
1978	0	13913	55636	26119	123655	11479	1496	2317	740	122	33	275	16	2	0	0	1187
1979	0	16077	120456	38247	7752	37353	3052	377	629	181	57	13	52	3	0	0	935
1980	0	11487	154765	67241	9978	1985	11057	820	143	166	96	34	9	7	1	0	456
1981	0	1959	174018	128102	16447	2219	494	3320	164	78	26	32	5	1	4	0	311
1982	0	7623	40161	282492	45732	3811	820	551	1072	60	28	8	2	2	0	0	1172
1983	0	7669	114118	57151	152477	19147	2201	338	255	385	93	21	4	4	0	0	763
1984	0	22842	80349	115405	27331	52226	6238	967	127	84	185	19	5	1	1	0	423
1985	0	3059	267559	75242	40846	6858	18360	1929	296	56	29	144	9	0	0	1	535
1986	0	12735	67173	287995	29371	10587	1685	4421	581	156	56	47	37	16	4	1	898
1987	0	11150	120584	46970	89772	6212	3993	1187	2596	462	56	65	35	32	17	8	3271
1988	0	2371	167090	83798	16114	27515	2030	874	344	1076	95	21	12	13	17	1	1579
1989	0	5446	17801	146467	19506	4130	7549	752	283	170	467	69	8	3	2	1	1003
1990	0	6279	46366	15680	54465	4117	1054	1761	250	100	54	13	12	1	4	2	435
1991	0	21627	57480	23058	4646	17468	1388	684	1024	171	65	11	11	1	2	2	1287

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1992	0	3544	128147	38838	7038	1483	6354	611	337	401	149	22	6	2	0	0	918
1993	0	3232	92828	102781	11570	1976	669	2028	215	210	188	84	4	4	0	0	706
1994	0	1484	75783	85391	32827	3345	600	366	746	132	48	36	26	5	0	0	992
1995	0	2410	32846	114437	31198	9038	898	161	155	151	21	8	6	2	1	0	345
1996	0	1179	84349	41653	55794	11123	4058	1080	75	72	37	9	8	3	1	0	205
1997	0	2292	26774	140099	16153	17846	2762	937	121	16	18	5	4	4	2	0	170
1998	0	2167	45449	42411	106125	6959	4579	850	263	60	7	8	3	2	1	1	345
1999	0	1340	31357	60351	26260	42494	2648	2047	438	53	8	3	3	2	0	0	507
2000	0	5508	32823	34517	27247	6927	9734	765	367	53	13	2	1	1	0	0	438
2001	0	855	75731	17938	10929	5321	2094	1609	256	89	28	3	4	0	0	0	381
2002	0	816	14893	124903	6330	2710	1615	618	603	283	25	8	5	0	0	0	923
2003	0	53	2119	16076	81868	2141	777	339	144	100	48	5	1	0	0	0	299
2004	0	495	3142	4906	23978	77262	996	239	82	42	37	12	1	0	0	0	174
2005	0	788	5777	8878	4178	22915	56760	370	131	38	11	8	4	1	0	0	192
2006	0	2129	10416	11780	8602	5209	14745	30350	149	54	20	7	3	1	0	0	234
2007	0	1146	28873	11204	7361	4684	2199	6773	7183	75	8	14	3	1	0	0	7284
2008	0	327	6945	53012	3944	2063	1285	544	1576	2468	6	4	1	1	0	0	4055

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2009	0	438	4589	9317	62331	1837	725	532	133	242	541	8	2	0	0	0	925
2010	0	1036	4695	12884	10496	35690	803	379	131	78	81	161	10	0	0	3	463
2011	0	57	15441	11014	6571	5687	20609	450	119	70	24	11	70	0	0	0	294
2012	0	209	3191	52623	4927	3685	2408	8032	137	176	50	28	32	22	2	0	447
2013	0	983	2369	4165	65909	2183	1522	825	3880	36	6	7	2	2	2	0	3935
2014	0	207	12576	3684	7471	42241	1076	948	364	1690	13	6	1	1	0	2	2077
2015	0	599	10558	16192	1631	4983	20943	1061	434	437	785	108	0	0	0	0	1764
2016	0	155	36387	8648	5532	652	443	10119	253	151	9	149	9	0	0	1	572
2017	0	148	11248	47244	3123	1482	136	366	1430	502	6	20	3	1	0	1	1962
2018	0	108	12032	15031	33277	922	521	206	100	966	20	0	3	3	1	1	1095
2019	0	187	5116	21826	6998	20451	458	375	25	31	146	2	14	2	0	0	220
2020	0	972	15876	9846	13642	2892	7555	172	158	60	59	19	0	0	0	0	296
2021	0	518	27086	10794	5139	7914	718	4456	43	29	2	1	0	1	0	0	77
2022	0	67	13560	58733	7190	2178	1152	252	668	8	0	0	0	0	0	0	676
2023	0	7	2670	21842	61486	3428	1179	544	117	90	23	1	5	0	0	0	237
2024	0	0	4711	9832	57166	39819	1195	199	102	48	26	9	17	0	0	0	201

Table 8.2.8. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Numbers-at-age data (thousands) for discards. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	5757	111654	4897	141863	3704	4	1	0	0	0	0	0	0	0	0	0	0
1966	13832	445648	12742	1197	24643	35	2	0	0	0	0	0	0	0	0	0	0
1967	46372	408281	62831	1032	219	1576	9	0	0	0	0	0	0	0	0	0	0
1968	67	741402	244976	3512	97	15	186	0	0	0	0	0	0	0	0	0	0
1969	4475	5234	1273332	39179	432	16	8	0	0	0	0	0	0	0	0	0	0
1970	68905	99125	78340	306391	2663	13	4	0	0	0	0	0	0	0	0	0	0
1971	14189	1275394	37883	9623	25648	66	2	0	0	0	0	0	0	0	0	0	0
1972	18446	444794	380988	6846	1236	1212	13	0	0	0	0	0	0	0	0	0	0
1973	38129	287558	363916	50108	354	33	123	0	0	0	0	0	0	0	0	0	0
1974	88456	982287	99148	59143	2869	6	4	0	0	0	0	0	0	0	0	0	0
1975	7479	1653310	377845	16385	13423	143	0	0	0	0	0	0	0	0	0	0	0
1976	6418	122012	698428	41183	200	137	0	0	0	0	0	0	0	0	0	0	0
1977	16364	107748	47070	79922	664	9	0	0	0	0	0	0	0	0	0	0	0
1978	1193	83683	63997	4214	19568	248	80	0	0	0	0	0	0	0	0	0	0
1979	4795	119245	82074	5734	142	365	0	0	0	0	0	0	0	0	0	0	0

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1980	258	146751	197725	4726	96	0	0	0	0	0	0	0	0	0	0	0	0
1981	442	15023	225773	47838	157	1	0	0	0	0	0	0	0	0	0	0	0
1982	505	36063	35089	94315	2293	0	0	0	0	0	0	0	0	0	0	0	0
1983	24327	76672	94323	20914	12092	905	164	0	0	0	0	0	0	0	0	0	0
1984	3275	361946	48893	23714	1623	3317	0	0	0	0	0	0	0	0	0	0	0
1985	4924	146668	156400	3624	115	1	16	0	0	0	0	0	0	0	0	0	0
1986	13007	84333	75071	39219	23	1	0	0	0	0	0	0	0	0	0	0	0
1987	1996	159860	134988	9142	2795	2	0	0	0	0	0	0	0	0	0	0	0
1988	7399	27412	244105	10535	427	10	0	0	24	0	0	0	0	0	0	0	24
1989	10673	43756	23611	67102	1048	23	35	0	2	0	2	0	0	0	0	0	4
1990	16290	69073	30530	1772	4932	28	25	0	0	0	0	0	0	0	0	0	0
1991	11794	143967	40697	1163	17	107	0	0	0	0	0	0	0	0	0	0	0
1992	36231	82605	115933	4063	97	0	6	0	0	0	0	0	0	0	0	0	0
1993	12346	191714	163172	17474	170	1	0	3	0	0	0	0	0	0	0	0	0
1994	19197	75840	254112	20271	2069	30	0	0	0	0	0	0	0	0	0	0	0
1995	2118	231490	84163	67644	2539	199	0	0	0	0	0	0	0	0	0	0	0
1996	22563	35010	413599	28996	10344	36	0	0	0	0	0	0	0	0	0	0	0

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1997	15260	114893	69948	106789	1700	425	0	0	0	0	0	0	0	0	0	0	0
1998	2936	77065	162251	15801	20732	88	11	0	0	0	0	0	0	0	0	0	0
1999	20814	57336	83205	46764	1905	2561	49	0	0	0	0	0	0	0	0	0	0
2000	8472	320463	55818	24661	5703	321	201	0	0	0	0	0	0	0	0	0	0
2001	1531	71284	521655	6483	1115	244	0	2	1	0	0	0	0	0	0	0	1
2002	1120	21358	80304	243495	978	64	0	111	0	0	0	0	0	0	0	0	0
2003	2937	7101	11014	31369	43849	13	9	0	0	0	0	0	0	0	0	0	0
2004	3758	24613	21221	3967	14548	19811	5	4	0	0	0	0	0	0	0	0	0
2005	8779	16730	18722	6181	1258	4826	4496	1	1	0	0	0	0	0	0	0	1
2006	3229	118636	19862	8636	2634	823	1596	2520	6	1	1	0	0	0	0	0	8
2007	2045	19393	142509	5585	826	97	38	103	71	0	0	0	0	0	0	0	71
2008	2550	24760	28093	27151	212	30	69	8	2	4	0	0	0	0	0	0	6
2009	27313	11529	9165	9178	15390	67	34	31	0	0	4	0	0	0	0	0	4
2010	2508	48691	9479	4545	1181	1605	30	0	11	0	0	1	0	0	0	0	12
2011	5024	4285	49207	1985	831	104	222	0	0	0	1	1	0	0	0	0	2
2012	1377	3670	1880	13735	502	10	5	15	0	0	0	0	0	0	0	0	0
2013	1303	11271	1873	470	2638	25	5	12	24	0	1	0	0	0	0	0	25

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2014	3504	7386	7435	999	164	4375	2	13	7	1	0	0	0	0	0	0	8
2015	3776	27011	5065	1522	87	27	980	0	0	0	0	0	0	0	0	0	0
2016	1700	9029	24507	188	17	2	7	8	0	0	0	0	0	0	0	0	0
2017	2580	12418	11798	7122	103	16	7	2	12	0	0	0	0	0	0	0	12
2018	3632	5440	11654	2063	1818	3	0	4	0	3	0	0	0	0	0	0	3
2019	3554	17451	6410	3595	168	705	1	0	0	0	0	0	0	0	0	0	0
2020	1540	43835	10427	1593	441	280	116	0	0	0	0	0	0	0	0	0	0
2021	7057	30091	46086	2148	396	45	16	42	0	0	0	0	0	0	0	0	0
2022	2461	4732	34343	34496	539	102	26	33	7	146	0	0	0	0	0	0	153
2023	1259	8451	6922	40600	11614	398	16	3	0	3	0	0	0	0	0	0	3
2024	893	3871	11934	6857	19812	5904	33	8	4	0	0	0	0	0	0	0	4

Table 8.2.9. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Numbers-at-age data (thousands) for BMS landings. Ages 0–7 and 8+ and years 2016–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2016	0	189	725	2	1	0	0	0	0	0	0	0	0	0	0	0	0
2017	35	166	158	95	1	0	0	0	0	0	0	0	0	0	0	0	0
2018	0	8	566	13	77	0	0	0	0	0	0	0	0	0	0	0	0
2019	0	195	287	219	0	2	0	0	0	0	0	0	0	0	0	0	0

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2020	0	443	263	124	46	1	0	0	0	0	0	0	0	0	0	0	0
2021	0	65	777	117	0	1	0	0	0	0	0	0	0	0	0	0	0
2022	21	41	297	300	5	1	0	0	0	1	0	0	0	0	0	0	1
2023	20	131	108	631	181	6	0	0	0	0	0	0	0	0	0	0	0
2024	15	67	206	118	342	102	1	0	0	0	0	0	0	0	0	0	0

Table 8.2.10. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Numbers-at-age data (thousands) for IBC. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	644461	254237	7686	183288	2365	592	118	6	0	0	0	0	0	0	0	0	0
1966	1659093	548835	6546	1007	15861	755	25	2	0	0	0	0	0	0	0	0	0
1967	298999	392510	5539	155	24	2264	12	0	0	0	0	0	0	0	0	0	0
1968	11066	462938	81153	2029	46	19	738	1	0	0	0	0	0	0	0	0	0
1969	70826	15178	1703617	98650	632	380	126	252	0	0	0	0	0	0	0	0	0
1970	872884	170914	88509	485274	3967	153	61	2	0	0	0	0	0	0	0	0	0
1971	323088	570391	9972	3390	6381	299	6	0	0	0	0	0	0	0	0	0	0
1972	236664	238566	57010	1023	146	439	0	0	0	0	0	0	0	0	0	0	0
1973	41332	117470	12402	11	11	196	0	0	0	0	0	0	0	0	0	0	0

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1974	576654	275266	32266	46862	4600	82	112	224	0	0	0	0	0	0	0	0	0
1975	44317	594854	84620	4761	5203	141	5	0	5	0	0	0	0	0	0	0	5
1976	164982	66973	183064	29188	46	2946	17	0	0	0	0	0	0	0	0	0	0
1977	103142	147019	29352	67628	2355	238	240	0	0	0	0	0	0	0	0	0	0
1978	280592	125698	11330	809	1480	64	6	5	0	0	0	0	0	0	0	0	0
1979	839615	125834	17671	1507	84	379	16	0	0	0	0	0	0	0	0	0	0
1980	374315	281436	21820	8258	1291	54	86	7	0	1	0	0	0	0	0	0	1
1981	644910	99247	30358	4613	440	6	2	0	0	0	0	0	0	0	0	0	0
1982	275002	174147	14740	13540	1810	464	0	0	0	0	0	0	0	0	0	0	0
1983	488707	63818	14331	5134	2242	3	0	0	0	0	0	0	0	0	0	0	0
1984	92587	98257	10644	4702	368	535	0	0	0	0	0	0	0	0	0	0	0
1985	122078	11672	17826	1739	547	223	17	0	0	0	0	0	0	0	0	0	0
1986	32696	40023	1831	802	103	7	0	0	0	0	0	0	0	0	0	0	0
1987	8253	82226	3797	295	138	0	0	0	0	0	0	0	0	0	0	0	0
1988	9280	3309	12819	2462	620	202	0	0	0	0	0	0	0	0	0	0	0
1989	8914	2541	1751	2789	460	37	86	10	0	0	0	0	0	0	0	0	0
1990	2996	7218	1986	359	1491	227	25	78	4	0	0	0	0	0	0	0	4

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1991	116909	22493	3248	601	43	43	0	0	0	0	0	0	0	0	0	0	0
1992	241702	80402	10971	356	27	3	17	0	0	0	0	0	0	0	0	0	0
1993	124495	107664	13220	3214	82	9	0	18	0	0	0	0	0	0	0	0	0
1994	69907	14349	9534	1011	160	7	1	0	0	0	0	0	0	0	0	0	0
1995	198032	102560	2201	888	65	0	0	0	0	0	0	0	0	0	0	0	0
1996	144469	10608	7453	3338	107	0	0	0	0	0	0	0	0	0	0	0	0
1997	21694	45264	10935	4451	184	17	0	0	0	0	0	0	0	0	0	0	0
1998	18983	9155	16337	2649	1490	63	0	0	0	0	0	0	0	0	0	0	0
1999	69820	10780	4531	2932	344	166	3	0	0	0	0	0	0	0	0	0	0
2000	4158	71419	21740	2085	186	5	0	0	0	0	0	0	0	0	0	0	0
2001	1987	22946	35776	10127	35	8	0	0	0	0	0	0	0	0	0	0	0
2002	49807	13889	4489	3638	504	27	0	0	0	0	0	0	0	0	0	0	0
2003	4145	5983	2101	1285	1524	12	0	0	0	0	0	0	0	0	0	0	0
2004	0	590	265	84	258	753	8	4	0	0	0	0	0	0	0	0	0
2005	0	176	97	26	9	5	201	1	0	0	0	0	0	0	0	0	0
2006	0	1772	716	241	47	46	74	108	1	0	0	0	0	0	0	0	1
2007	1	27	218	6	1	0	0	0	0	0	0	0	0	0	0	0	0

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2008	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
2009	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2010	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2011	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2012	0	0	2	27	3	2	1	4	0	0	0	0	0	0	0	0	0
2013	0	4	9	16	256	8	6	3	15	0	0	0	0	0	0	0	15
2014	0	0	20	6	12	68	2	2	1	3	0	0	0	0	0	0	3
2015	0	0	6	10	1	3	12	1	0	0	0	0	0	0	0	0	1
2016	0	0	38	9	6	1	0	11	0	0	0	0	0	0	0	0	1
2017	0	0	3	12	1	0	0	0	0	0	0	0	0	0	0	0	0
2018	0	0	11	13	29	1	0	0	0	1	0	0	0	0	0	0	1
2019	0	1	31	132	42	124	3	2	0	0	1	0	0	0	0	0	1
2020	0	36	591	366	508	108	281	6	6	2	2	1	0	0	0	0	11
2021	0	27	1390	554	264	406	37	229	2	2	0	0	0	0	0	0	4
2022	0	1	156	675	83	25	13	3	8	0	0	0	0	0	0	0	8
2023	0	0	123	1007	2834	158	54	25	5	4	1	0	0	0	0	0	11
2024	0	0	127	265	1537	1071	32	5	3	1	1	0	0	0	0	0	5

Table 8.2.11. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weight-at-age data (kg) for total catch. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0.010	0.070	0.227	0.370	0.655	0.846	1.170	1.190	1.479	1.714	2.175	0.000	0.000	0.000	0.000	0.000	1.691
1966	0.010	0.088	0.247	0.394	0.536	0.962	1.254	1.512	1.827	1.723	2.955	2.035	0.000	0.000	0.000	0.000	1.877
1967	0.014	0.116	0.278	0.478	0.591	0.641	1.072	1.511	1.898	2.084	2.342	0.000	0.000	0.000	0.000	0.000	1.979
1968	0.010	0.129	0.254	0.516	0.743	0.827	0.829	1.483	2.071	2.622	2.065	0.000	0.000	0.000	0.000	0.000	2.179
1969	0.012	0.064	0.217	0.410	0.817	0.905	1.029	1.074	1.808	2.772	3.259	0.000	0.000	0.000	0.000	0.000	1.903
1970	0.013	0.075	0.222	0.353	0.738	0.925	1.195	1.246	1.427	2.438	3.489	3.864	0.000	0.000	0.000	0.000	1.450
1971	0.012	0.109	0.246	0.359	0.509	0.888	1.269	1.525	1.338	1.284	1.961	4.270	3.513	0.000	0.000	0.000	1.354
1972	0.025	0.117	0.242	0.383	0.503	0.585	0.987	1.380	1.967	1.979	1.618	2.861	0.000	0.000	0.000	0.000	1.692
1973	0.043	0.118	0.239	0.369	0.578	0.611	0.648	1.044	1.378	2.658	1.603	1.988	2.123	0.000	0.000	0.000	1.660
1974	0.025	0.129	0.226	0.339	0.536	0.867	0.828	0.863	1.377	1.704	1.854	4.057	1.927	0.890	0.000	0.000	1.502
1975	0.023	0.105	0.240	0.353	0.442	0.678	1.190	1.077	1.031	1.564	2.188	2.764	0.000	3.318	0.000	0.000	1.076
1976	0.014	0.129	0.225	0.394	0.505	0.578	0.916	1.829	1.656	1.247	2.296	2.425	1.679	0.000	0.000	0.000	1.300
1977	0.020	0.111	0.238	0.339	0.586	0.612	0.787	1.160	1.715	1.971	1.490	2.067	0.000	3.898	0.000	0.000	1.584
1978	0.011	0.104	0.254	0.396	0.424	0.707	0.784	0.921	1.350	1.995	1.990	1.329	2.182	4.475	0.000	0.000	1.446
1979	0.009	0.093	0.287	0.417	0.611	0.669	0.931	1.241	1.320	1.453	2.505	1.575	1.233	1.580	0.000	0.000	1.418
1980	0.012	0.081	0.276	0.464	0.693	0.985	0.908	1.264	1.511	1.501	1.676	3.104	1.050	2.134	2.921	0.000	1.664

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1981	0.009	0.060	0.264	0.445	0.726	1.055	1.222	1.195	1.545	1.672	1.531	1.515	2.982	4.273	1.896	0.000	1.612
1982	0.010	0.074	0.286	0.423	0.759	1.109	1.415	1.578	1.466	2.136	2.122	1.877	1.886	3.179	0.000	0.000	1.522
1983	0.011	0.132	0.303	0.431	0.612	0.904	1.211	1.191	1.630	1.460	1.449	1.972	2.853	4.689	0.000	0.000	1.555
1984	0.010	0.142	0.303	0.461	0.645	0.736	1.077	1.205	1.821	2.030	1.732	1.950	2.422	2.822	4.995	0.000	1.848
1985	0.010	0.148	0.296	0.466	0.649	0.835	0.934	1.344	1.638	2.097	2.109	2.061	2.555	2.471	2.721	4.139	1.846
1986	0.023	0.123	0.261	0.406	0.600	0.848	1.195	1.098	1.524	1.356	2.178	2.366	2.498	2.993	2.778	2.894	1.653
1987	0.010	0.125	0.264	0.405	0.594	0.974	1.215	1.322	1.260	1.358	1.870	2.132	2.609	2.450	2.768	2.638	1.339
1988	0.042	0.163	0.232	0.411	0.581	0.731	1.203	1.363	1.281	0.974	1.633	2.163	2.547	3.139	3.435	2.863	1.156
1989	0.036	0.200	0.282	0.367	0.590	0.770	0.935	1.259	1.586	1.507	1.034	1.534	2.431	2.559	2.307	0.980	1.322
1990	0.040	0.187	0.313	0.422	0.506	0.795	0.995	1.179	1.495	1.898	2.519	2.259	2.188	0.562	1.852	4.731	1.768
1991	0.030	0.175	0.308	0.454	0.574	0.644	0.959	1.136	1.313	1.701	2.163	2.012	1.622	1.070	1.208	2.888	1.418
1992	0.019	0.102	0.306	0.466	0.717	0.923	0.903	1.382	1.514	1.813	2.014	2.064	2.441	1.781	0.000	0.000	1.746
1993	0.010	0.110	0.282	0.454	0.660	0.877	1.053	1.062	1.545	1.460	1.830	1.894	2.155	2.460	0.000	0.000	1.646
1994	0.018	0.121	0.247	0.435	0.599	0.846	1.240	1.274	1.289	1.573	2.060	2.070	2.834	2.403	2.523	0.000	1.439
1995	0.012	0.107	0.290	0.369	0.581	0.774	1.058	1.418	1.261	1.320	1.889	2.491	1.713	1.699	2.243	0.000	1.368
1996	0.022	0.126	0.241	0.382	0.484	0.746	0.847	0.825	1.616	1.538	1.433	1.830	2.358	2.636	3.433	0.000	1.617
1997	0.029	0.138	0.280	0.360	0.585	0.634	0.923	0.997	1.293	2.196	1.961	2.058	2.757	2.270	2.867	2.782	1.548

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1998	0.027	0.153	0.255	0.396	0.444	0.665	0.777	1.041	1.109	1.251	2.373	2.334	1.656	2.433	2.085	2.509	1.210
1999	0.025	0.166	0.250	0.356	0.477	0.510	0.735	0.798	0.826	1.305	1.533	2.478	2.086	2.698	2.904	2.220	0.914
2000	0.052	0.121	0.256	0.355	0.480	0.605	0.656	1.033	0.973	1.529	1.911	2.323	2.365	2.310	3.595	1.843	1.084
2001	0.029	0.111	0.219	0.321	0.466	0.658	0.735	0.945	1.690	1.148	1.725	2.923	1.286	2.534	1.239	3.425	1.573
2002	0.017	0.109	0.255	0.311	0.527	0.703	0.829	0.818	1.279	1.945	1.798	1.839	2.352	2.762	0.000	0.000	1.508
2003	0.024	0.082	0.221	0.327	0.400	0.681	0.758	1.110	1.281	1.612	2.022	2.219	2.506	2.606	1.981	3.092	1.535
2004	0.039	0.139	0.238	0.378	0.395	0.440	0.686	0.926	1.184	1.602	1.753	2.605	2.170	0.000	0.000	0.000	1.506
2005	0.054	0.160	0.271	0.364	0.495	0.479	0.522	0.925	1.054	1.373	1.847	2.750	2.545	2.309	3.431	0.000	1.263
2006	0.042	0.126	0.283	0.352	0.442	0.507	0.538	0.550	1.048	1.395	2.031	2.525	1.834	3.532	5.274	2.580	1.276
2007	0.042	0.159	0.227	0.407	0.478	0.538	0.657	0.700	0.745	0.902	2.272	0.971	1.712	2.348	4.244	0.000	0.749
2008	0.034	0.141	0.252	0.359	0.570	0.642	0.758	0.836	0.878	0.834	2.058	1.248	3.538	2.685	3.792	2.923	0.854
2009	0.050	0.158	0.305	0.329	0.384	0.631	0.754	0.726	1.016	1.077	0.957	1.055	0.944	3.019	2.097	0.000	0.998
2010	0.031	0.104	0.305	0.417	0.456	0.470	0.718	0.897	1.308	1.414	1.381	1.423	2.725	2.245	2.654	2.567	1.414
2011	0.040	0.157	0.265	0.449	0.533	0.531	0.543	0.722	1.002	0.912	1.693	1.892	1.621	0.000	0.000	0.000	1.221
2012	0.034	0.160	0.442	0.407	0.568	0.687	0.680	0.642	1.146	0.848	1.426	2.158	2.121	2.095	2.368	0.000	1.246
2013	0.034	0.171	0.426	0.596	0.485	0.717	0.843	0.790	0.757	1.098	1.643	2.216	2.607	1.810	2.512	0.000	0.767
2014	0.042	0.140	0.432	0.590	0.653	0.534	0.772	0.825	0.928	0.793	1.692	2.800	1.323	2.682	0.000	1.602	0.831

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2015	0.031	0.145	0.421	0.564	0.765	0.702	0.631	0.683	0.970	0.723	0.714	0.719	1.425	2.954	0.000	0.000	0.780
2016	0.048	0.161	0.363	0.638	0.765	0.874	1.022	0.737	0.798	1.083	2.622	1.122	1.286	1.978	3.312	2.835	0.998
2017	0.040	0.149	0.343	0.450	0.781	0.961	1.338	1.045	1.020	0.654	2.834	0.930	2.682	2.237	4.673	5.554	0.936
2018	0.043	0.139	0.355	0.503	0.533	1.021	1.030	1.147	1.421	0.890	1.235	1.883	2.383	3.356	2.198	4.662	0.961
2019	0.046	0.144	0.311	0.459	0.625	0.580	1.019	0.972	1.827	2.069	1.341	3.550	3.972	1.815	0.000	0.000	1.690
2020	0.041	0.125	0.371	0.501	0.580	0.838	0.613	1.640	2.340	2.318	3.309	1.625	1.257	0.000	0.000	0.000	2.481
2021	0.044	0.140	0.305	0.443	0.515	0.572	0.732	0.690	1.009	1.182	1.645	1.964	0.000	2.046	0.000	0.000	1.128
2022	0.029	0.130	0.288	0.373	0.564	0.645	0.769	0.814	0.830	0.368	3.233	2.353	0.000	0.000	0.000	0.000	0.745
2023	0.046	0.141	0.345	0.370	0.454	0.633	0.689	0.740	1.271	1.050	1.385	3.011	3.940	0.000	0.000	0.000	1.267
2024	0.044	0.145	0.339	0.427	0.389	0.452	0.656	1.018	1.191	1.480	1.804	2.233	3.903	0.000	0.000	0.000	1.598

Table 8.2.12. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weight-at-age data (kg) for landings. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0.000	0.308	0.348	0.413	0.680	0.904	1.211	1.197	1.479	1.714	2.175	0.000	0.000	0.000	0.000	0.000	1.691
1966	0.000	0.300	0.382	0.445	0.554	1.001	1.275	1.515	1.827	1.723	2.955	2.035	0.000	0.000	0.000	0.000	1.877
1967	0.000	0.260	0.399	0.530	0.610	0.646	1.077	1.511	1.898	2.084	2.342	0.000	0.000	0.000	0.000	0.000	1.979
1968	0.000	0.256	0.360	0.595	0.769	0.832	0.835	1.484	2.071	2.622	2.065	0.000	0.000	0.000	0.000	0.000	2.179

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1969	0.000	0.178	0.302	0.508	0.878	0.989	1.058	1.081	1.808	2.772	3.259	0.000	0.000	0.000	0.000	0.000	1.903
1970	0.000	0.249	0.309	0.402	0.787	0.997	1.235	1.250	1.427	2.438	3.489	3.864	0.000	0.000	0.000	0.000	1.450
1971	0.000	0.256	0.332	0.393	0.525	0.905	1.280	1.525	1.338	1.284	1.961	4.270	3.513	0.000	0.000	0.000	1.354
1972	0.000	0.243	0.325	0.415	0.518	0.587	0.989	1.380	1.967	1.979	1.618	2.861	0.000	0.000	0.000	0.000	1.692
1973	0.000	0.228	0.310	0.400	0.596	0.621	0.649	1.044	1.378	2.658	1.603	1.988	2.123	0.000	0.000	0.000	1.660
1974	0.000	0.268	0.314	0.381	0.567	0.882	0.866	0.867	1.377	1.704	1.854	4.057	1.927	0.890	0.000	0.000	1.502
1975	0.000	0.254	0.336	0.400	0.476	0.683	1.193	1.077	1.031	1.564	2.188	2.764	0.000	3.318	0.000	0.000	1.076
1976	0.000	0.243	0.331	0.452	0.509	0.601	0.917	1.829	1.656	1.247	2.296	2.425	1.679	0.000	0.000	0.000	1.300
1977	0.000	0.272	0.344	0.381	0.595	0.625	0.800	1.160	1.715	1.971	1.490	2.067	0.000	3.898	0.000	0.000	1.584
1978	0.000	0.257	0.333	0.427	0.456	0.717	0.812	0.922	1.350	1.995	1.990	1.329	2.182	4.475	0.000	0.000	1.446
1979	0.000	0.262	0.348	0.447	0.620	0.675	0.932	1.241	1.320	1.453	2.505	1.575	1.233	1.580	0.000	0.000	1.418
1980	0.000	0.274	0.347	0.501	0.706	0.992	0.907	1.261	1.511	1.499	1.676	3.104	1.050	2.134	2.921	0.000	1.664
1981	0.000	0.334	0.364	0.503	0.734	1.056	1.222	1.195	1.545	1.672	1.531	1.515	2.982	4.273	1.896	0.000	1.612
1982	0.000	0.299	0.349	0.478	0.788	1.153	1.415	1.578	1.466	2.136	2.122	1.877	1.886	3.179	0.000	0.000	1.522
1983	0.000	0.320	0.375	0.464	0.624	0.914	1.242	1.191	1.630	1.460	1.449	1.972	2.853	4.689	0.000	0.000	1.555
1984	0.000	0.280	0.350	0.493	0.666	0.764	1.077	1.205	1.821	2.030	1.732	1.951	2.422	2.822	4.995	0.000	1.848
1985	0.000	0.279	0.348	0.478	0.651	0.844	0.935	1.344	1.638	2.097	2.109	2.061	2.555	2.471	2.721	4.139	1.846

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1986	0.000	0.277	0.348	0.428	0.600	0.848	1.195	1.098	1.524	1.356	2.178	2.366	2.498	2.993	2.778	2.894	1.653
1987	0.000	0.265	0.335	0.440	0.603	0.974	1.215	1.322	1.260	1.358	1.870	2.132	2.609	2.450	2.768	2.638	1.339
1988	0.000	0.236	0.322	0.437	0.594	0.732	1.203	1.363	1.370	0.974	1.633	2.163	2.547	3.139	3.435	2.863	1.173
1989	0.000	0.319	0.356	0.413	0.602	0.769	0.934	1.256	1.579	1.507	1.025	1.534	2.431	2.559	2.307	0.980	1.316
1990	0.000	0.260	0.372	0.439	0.525	0.796	1.015	1.196	1.504	1.898	2.519	2.259	2.188	0.562	1.852	4.731	1.776
1991	0.000	0.269	0.363	0.462	0.576	0.645	0.959	1.136	1.313	1.701	2.163	2.012	1.622	1.070	1.208	2.888	1.418
1992	0.000	0.287	0.367	0.486	0.723	0.924	0.904	1.382	1.515	1.813	2.014	2.064	2.441	1.781	0.000	0.000	1.747
1993	0.000	0.293	0.372	0.484	0.666	0.878	1.053	1.067	1.545	1.460	1.830	1.894	2.155	2.460	0.000	0.000	1.646
1994	0.000	0.269	0.378	0.473	0.617	0.851	1.241	1.274	1.289	1.573	2.060	2.070	2.834	2.403	2.523	0.000	1.439
1995	0.000	0.316	0.400	0.424	0.600	0.782	1.058	1.418	1.261	1.320	1.889	2.491	1.713	1.699	2.243	0.000	1.368
1996	0.000	0.326	0.364	0.471	0.519	0.747	0.847	0.825	1.616	1.538	1.433	1.830	2.358	2.636	3.433	0.000	1.617
1997	0.000	0.344	0.410	0.418	0.615	0.641	0.923	0.997	1.293	2.196	1.961	2.058	2.757	2.270	2.867	2.782	1.548
1998	0.000	0.271	0.370	0.441	0.470	0.670	0.778	1.041	1.109	1.251	2.373	2.334	1.656	2.433	2.085	2.509	1.210
1999	0.000	0.297	0.349	0.422	0.490	0.523	0.746	0.798	0.826	1.305	1.533	2.478	2.086	2.698	2.904	2.220	0.914
2000	0.000	0.334	0.368	0.421	0.515	0.617	0.663	1.033	0.973	1.529	1.911	2.323	2.365	2.310	3.595	1.843	1.084
2001	0.000	0.379	0.352	0.448	0.483	0.675	0.735	0.946	1.695	1.148	1.725	2.923	1.286	2.534	1.239	3.425	1.576
2002	0.000	0.427	0.446	0.397	0.569	0.713	0.829	0.901	1.279	1.945	1.798	1.839	2.352	2.762	0.000	0.000	1.508

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2003	0.000	0.283	0.377	0.464	0.441	0.684	0.759	1.110	1.281	1.612	2.022	2.219	2.506	2.606	1.981	3.092	1.535
2004	0.000	0.366	0.383	0.474	0.454	0.468	0.688	0.932	1.184	1.602	1.753	2.605	2.170	0.000	0.000	0.000	1.506
2005	0.000	0.399	0.399	0.428	0.548	0.516	0.536	0.926	1.056	1.373	1.847	2.750	2.545	2.309	3.431	0.000	1.265
2006	0.000	0.392	0.386	0.418	0.493	0.546	0.574	0.583	1.093	1.431	2.109	2.643	1.926	3.592	5.292	2.709	1.326
2007	0.000	0.379	0.385	0.466	0.497	0.542	0.662	0.705	0.748	0.902	2.272	0.971	1.712	2.348	4.244	0.000	0.752
2008	0.000	0.281	0.391	0.403	0.582	0.644	0.781	0.836	0.877	0.834	2.058	1.309	3.538	2.685	3.792	2.923	0.854
2009	0.000	0.463	0.418	0.401	0.408	0.641	0.774	0.747	1.016	1.078	0.961	1.058	0.950	3.019	2.097	0.000	1.001
2010	0.000	0.275	0.471	0.456	0.472	0.476	0.732	0.897	1.394	1.414	1.381	1.424	2.725	2.245	2.654	2.567	1.441
2011	0.000	0.383	0.414	0.474	0.561	0.534	0.545	0.722	1.002	0.912	1.683	1.862	1.621	0.000	0.000	0.000	1.215
2012	0.000	0.397	0.527	0.442	0.597	0.688	0.681	0.642	1.146	0.848	1.426	2.158	2.121	2.095	2.368	0.000	1.246
2013	0.000	0.437	0.566	0.622	0.491	0.719	0.843	0.794	0.757	1.098	1.557	2.216	2.607	1.810	2.512	0.000	0.766
2014	0.302	0.308	0.511	0.658	0.659	0.554	0.772	0.828	0.936	0.793	1.692	2.800	1.323	2.682	0.000	1.602	0.832
2015	0.000	0.325	0.495	0.584	0.786	0.704	0.642	0.683	0.970	0.723	0.714	0.719	1.425	2.954	0.000	0.000	0.780
2016	0.268	0.385	0.445	0.644	0.766	0.874	1.031	0.737	0.798	1.083	2.622	1.122	1.286	1.978	3.312	2.835	0.998
2017	0.000	0.254	0.457	0.472	0.789	0.967	1.362	1.047	1.024	0.654	2.836	0.930	2.682	2.333	4.673	5.554	0.939
2018	0.000	0.418	0.470	0.526	0.543	1.023	1.030	1.158	1.422	0.889	1.235	1.883	2.383	3.356	2.198	4.662	0.961
2019	0.000	0.415	0.437	0.488	0.633	0.588	1.020	0.972	1.831	2.069	1.341	3.550	3.972	1.815	0.000	0.000	1.690

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2020	0.000	0.359	0.450	0.531	0.588	0.879	0.616	1.640	2.340	2.318	3.309	1.625	1.257	0.000	0.000	0.000	2.481
2021	0.000	0.441	0.378	0.468	0.527	0.572	0.739	0.692	1.009	1.182	1.645	1.964	0.000	2.046	0.000	0.000	1.128
2022	0.000	0.444	0.411	0.417	0.584	0.659	0.775	0.878	0.834	1.072	3.233	2.353	0.000	0.000	0.000	0.000	0.838
2023	0.000	0.449	0.516	0.460	0.471	0.664	0.691	0.741	1.271	1.061	1.385	3.011	3.940	0.000	0.000	0.000	1.274
2024	0.000	0.286	0.572	0.524	0.408	0.465	0.662	1.036	1.212	1.480	1.804	2.233	3.903	0.000	0.000	0.000	1.617

Table 8.2.13. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weight-at-age data (kg) for discards. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0.062	0.131	0.203	0.335	0.607	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1966	0.053	0.141	0.208	0.245	0.309	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1967	0.043	0.170	0.210	0.273	0.306	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1968	0.054	0.181	0.212	0.257	0.317	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1969	0.049	0.129	0.216	0.238	0.300	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1970	0.057	0.131	0.210	0.239	0.263	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1971	0.052	0.135	0.202	0.244	0.264	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1972	0.045	0.140	0.207	0.239	0.261	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1973	0.051	0.135	0.201	0.237	0.263	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1974	0.046	0.146	0.201	0.234	0.259	0.321	0.321	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1975	0.041	0.126	0.201	0.257	0.275	0.348	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1976	0.053	0.172	0.198	0.239	0.291	0.337	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1977	0.062	0.191	0.198	0.220	0.306	0.347	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1978	0.042	0.175	0.199	0.222	0.225	0.265	0.284	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1979	0.037	0.128	0.221	0.245	0.259	0.314	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1980	0.051	0.147	0.232	0.276	0.325	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1981	0.074	0.160	0.199	0.296	0.621	0.727	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1982	0.055	0.194	0.247	0.265	0.289	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1983	0.066	0.184	0.237	0.343	0.458	0.711	0.792	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1984	0.047	0.160	0.245	0.315	0.309	0.290	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1985	0.040	0.154	0.221	0.271	0.356	0.423	0.353	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1986	0.057	0.140	0.185	0.246	0.337	0.329	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1987	0.026	0.160	0.201	0.227	0.286	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1988	0.072	0.167	0.172	0.239	0.256	0.352	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1989	0.054	0.188	0.229	0.266	0.336	0.708	0.844	0.000	2.572	0.000	3.048	0.000	0.000	0.000	0.000	0.000	2.810
1990	0.047	0.189	0.229	0.248	0.264	0.290	0.333	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1991	0.059	0.179	0.238	0.341	0.464	0.480	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1992	0.043	0.136	0.246	0.282	0.345	0.000	0.592	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1993	0.028	0.139	0.237	0.287	0.355	0.369	0.000	0.430	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1994	0.042	0.130	0.212	0.273	0.310	0.304	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1995	0.044	0.132	0.250	0.276	0.356	0.384	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1996	0.047	0.133	0.218	0.279	0.297	0.335	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1997	0.060	0.159	0.250	0.286	0.322	0.374	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1998	0.075	0.159	0.232	0.293	0.317	0.391	0.428	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1999	0.047	0.182	0.217	0.273	0.308	0.304	0.227	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2000	0.049	0.129	0.245	0.278	0.316	0.355	0.292	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2001	0.049	0.115	0.206	0.300	0.301	0.300	0.000	0.411	0.416	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.416
2002	0.044	0.125	0.223	0.267	0.334	0.382	0.000	0.358	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2003	0.042	0.124	0.223	0.261	0.327	0.536	0.630	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2004	0.039	0.135	0.218	0.263	0.299	0.330	0.639	0.650	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2005	0.054	0.150	0.232	0.273	0.318	0.301	0.342	0.499	0.493	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.493
2006	0.042	0.121	0.231	0.265	0.279	0.274	0.217	0.164	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2007	0.042	0.146	0.195	0.291	0.314	0.358	0.375	0.356	0.368	0.400	0.000	0.000	0.000	0.000	0.000	0.000	0.368
2008	0.034	0.139	0.218	0.275	0.350	0.523	0.327	0.817	1.146	0.544	0.000	0.164	0.000	0.000	0.000	0.000	0.683

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2009	0.050	0.146	0.249	0.256	0.288	0.370	0.329	0.362	0.324	0.396	0.425	0.368	0.324	0.000	0.000	0.000	0.422
2010	0.031	0.101	0.223	0.307	0.316	0.338	0.333	0.393	0.277	0.000	0.000	1.263	0.000	0.000	0.000	0.000	0.350
2011	0.040	0.154	0.219	0.310	0.310	0.354	0.352	0.502	0.351	0.589	1.995	2.215	0.000	0.000	0.000	0.000	2.111
2012	0.034	0.146	0.298	0.274	0.282	0.403	0.330	0.382	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2013	0.034	0.148	0.248	0.368	0.347	0.480	0.768	0.533	0.842	0.000	2.113	0.000	0.000	0.000	0.000	0.000	0.895
2014	0.042	0.135	0.299	0.339	0.393	0.339	0.571	0.603	0.483	0.900	0.000	0.000	0.000	0.000	0.000	0.000	0.526
2015	0.031	0.141	0.266	0.351	0.362	0.410	0.406	0.565	0.579	0.592	0.585	0.000	0.000	0.000	0.000	0.000	0.586
2016	0.048	0.155	0.245	0.369	0.356	0.878	0.464	0.508	0.683	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.683
2017	0.040	0.148	0.235	0.306	0.521	0.440	0.916	0.560	0.606	0.408	2.670	0.000	0.000	1.514	0.000	0.000	0.625
2018	0.043	0.133	0.243	0.337	0.353	0.481	0.000	0.561	0.616	0.905	0.000	0.000	0.000	0.000	0.000	0.000	0.903
2019	0.046	0.141	0.212	0.293	0.298	0.359	0.570	0.470	0.844	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.844
2020	0.041	0.118	0.250	0.328	0.367	0.403	0.379	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2021	0.044	0.134	0.262	0.326	0.360	0.413	0.384	0.490	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2022	0.029	0.125	0.239	0.297	0.295	0.365	0.490	0.320	0.405	0.330	0.000	0.000	0.000	0.000	0.000	0.000	0.333
2023	0.046	0.140	0.277	0.320	0.364	0.357	0.611	0.603	0.000	0.667	0.000	0.000	0.000	0.000	0.000	0.000	0.667
2024	0.044	0.145	0.246	0.286	0.333	0.360	0.430	0.541	0.639	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.639

Table 8.2.14. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weight-at-age data (kg) for BMS landings. Ages 0–7 and 8+ and years 2016–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2016	0.068	0.239	0.213	0.386	0.389	0.000	0.000	0.481	0.000	0.991	0.000	0.000	0.000	0.000	0.000	0.000	0.991
2017	0.040	0.148	0.235	0.306	0.521	0.440	0.916	0.560	0.606	0.408	2.670	0.000	0.000	1.514	0.000	0.000	0.625
2018	0.000	0.286	0.233	0.299	0.291	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2019	0.089	0.185	0.271	0.298	0.000	0.376	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2020	0.000	0.220	0.267	0.259	0.365	0.792	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2021	0.000	0.078	0.183	0.261	0.527	0.572	0.739	0.692	1.009	1.182	1.645	1.964	0.000	2.046	0.000	0.000	1.078
2022	0.029	0.125	0.239	0.298	0.309	0.387	0.532	0.336	0.523	0.330	3.233	2.353	0.000	0.000	0.000	0.000	0.341
2023	0.046	0.140	0.277	0.320	0.364	0.357	0.611	0.603	0.000	0.667	0.000	0.000	0.000	0.000	0.000	0.000	0.667
2024	0.044	0.145	0.246	0.286	0.333	0.360	0.430	0.541	0.639	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.639

Table 8.2.15. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weight-at-age data (kg) for IBC. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.500	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1966	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.500	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1967	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1968	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.500	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1969	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.500	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1970	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.500	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1971	0.010	0.040	0.180	0.302	0.400	0.420	0.440	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1972	0.023	0.067	0.136	0.255	0.288	0.231	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1973	0.035	0.068	0.141	0.246	0.327	0.396	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1974	0.022	0.058	0.150	0.260	0.359	0.579	0.277	0.447	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1975	0.020	0.039	0.173	0.275	0.267	0.413	0.585	0.000	0.585	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.585
1976	0.012	0.046	0.181	0.304	0.473	0.360	0.725	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1977	0.013	0.042	0.184	0.307	0.490	0.352	0.442	1.234	1.315	1.319	0.000	0.000	0.000	0.000	0.000	0.000	1.317
1978	0.011	0.040	0.174	0.286	0.372	0.473	0.411	0.456	1.315	0.000	1.400	0.000	0.000	0.000	0.000	0.000	1.345
1979	0.009	0.039	0.177	0.285	0.384	0.461	0.735	1.234	1.315	0.000	1.400	0.000	0.000	0.000	0.000	0.000	1.333
1980	0.012	0.039	0.176	0.268	0.623	0.722	1.102	1.591	0.000	1.796	0.000	0.000	0.000	0.000	0.000	0.000	1.796
1981	0.009	0.040	0.176	0.371	0.467	0.858	1.200	1.234	1.315	1.319	1.400	0.000	0.000	0.000	0.000	0.000	1.346
1982	0.010	0.040	0.206	0.379	0.636	0.751	1.225	1.233	1.315	1.319	0.000	0.000	0.000	0.000	0.000	0.000	1.316
1983	0.008	0.047	0.173	0.428	0.584	1.006	1.225	1.234	1.315	1.319	0.000	0.000	0.000	0.000	0.000	0.000	1.318
1984	0.009	0.045	0.211	0.414	0.626	0.751	1.225	1.234	1.315	1.319	1.400	1.400	0.000	0.000	0.000	0.000	1.356
1985	0.009	0.043	0.186	0.371	0.550	0.563	0.565	1.234	1.315	1.319	1.400	0.000	0.000	0.000	0.000	0.000	1.319
1986	0.010	0.040	0.186	0.375	0.626	1.259	1.225	1.234	1.315	1.319	1.400	0.000	0.000	0.000	0.000	0.000	1.328

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1987	0.006	0.038	0.258	0.442	0.908	1.171	1.225	1.234	1.315	1.319	0.000	0.000	0.000	0.000	0.000	0.000	1.316
1988	0.018	0.077	0.196	0.274	0.455	0.549	1.225	1.234	1.315	1.319	1.400	0.000	0.000	0.000	0.000	0.000	1.330
1989	0.015	0.165	0.251	0.347	0.670	0.923	1.065	1.492	1.315	0.000	1.400	0.000	0.000	0.000	0.000	0.000	1.329
1990	0.005	0.104	0.229	0.506	0.609	0.842	0.829	0.796	0.956	1.319	0.000	0.000	0.000	0.000	0.000	0.000	0.956
1991	0.027	0.058	0.206	0.357	0.472	0.477	1.225	1.234	1.315	1.319	0.000	0.000	0.000	0.000	0.000	0.000	1.316
1992	0.015	0.059	0.217	0.422	0.552	0.615	0.548	1.234	0.621	0.820	0.000	0.000	0.000	0.000	0.000	0.000	0.659
1993	0.008	0.053	0.206	0.399	0.521	0.578	1.225	0.582	1.315	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.315
1994	0.011	0.055	0.155	0.435	0.595	0.698	0.490	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1995	0.012	0.045	0.193	0.285	0.387	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1996	0.018	0.077	0.136	0.162	0.264	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1997	0.007	0.076	0.149	0.309	0.419	0.601	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1998	0.020	0.075	0.166	0.291	0.351	0.453	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1999	0.018	0.064	0.177	0.304	0.416	0.309	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2000	0.058	0.070	0.113	0.176	0.370	0.203	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2001	0.014	0.086	0.133	0.110	0.353	0.431	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2002	0.016	0.064	0.178	0.283	0.374	0.431	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2003	0.012	0.031	0.056	0.231	0.326	0.339	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2004	0.000	0.116	0.183	0.255	0.276	0.446	0.539	0.840	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2005	0.000	0.107	0.187	0.239	0.268	0.287	0.598	0.619	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
2006	0.000	0.127	0.232	0.273	0.273	0.280	0.283	0.286	0.287	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.287
2007	0.035	0.141	0.192	0.290	0.315	0.370	0.427	0.342	0.368	0.400	0.000	0.000	0.000	0.000	0.000	0.000	0.368
2008	0.000	0.281	0.391	0.403	0.582	0.644	0.781	0.836	0.877	0.834	2.058	1.309	3.538	2.685	3.791	0.000	0.851
2009	0.000	0.463	0.418	0.401	0.408	0.641	0.774	0.747	1.016	1.078	0.961	1.058	0.950	3.020	2.100	0.000	0.995
2010	0.000	0.275	0.471	0.456	0.472	0.476	0.732	0.897	1.394	1.414	1.381	1.424	2.725	2.245	2.654	0.000	1.403
2011	0.000	0.383	0.414	0.474	0.561	0.534	0.545	0.722	1.002	0.912	1.683	1.862	1.621	0.000	0.000	0.000	1.178
2012	0.000	0.397	0.527	0.442	0.597	0.688	0.681	0.642	1.146	0.848	1.426	2.158	2.121	2.095	2.368	0.000	1.246
2013	0.000	0.437	0.566	0.622	0.491	0.719	0.843	0.793	0.757	1.098	1.557	2.216	2.607	1.810	2.512	0.000	0.766
2014	0.302	0.308	0.511	0.658	0.659	0.554	0.772	0.828	0.936	0.793	1.692	2.800	1.323	2.682	0.000	1.856	0.832
2015	0.000	0.325	0.495	0.584	0.786	0.704	0.642	0.683	0.970	0.723	0.714	0.719	1.425	2.954	0.000	0.000	0.780
2016	0.268	0.385	0.445	0.644	0.766	0.874	1.031	0.737	0.798	1.083	2.622	1.122	1.286	1.978	3.312	3.841	0.998
2017	0.000	0.254	0.457	0.472	0.789	0.967	1.362	1.047	1.024	0.654	2.836	0.930	2.682	2.333	4.673	0.000	0.936
2018	0.000	0.418	0.470	0.526	0.543	1.023	1.030	1.158	1.422	0.889	1.235	1.883	2.383	3.356	2.198	0.000	0.958
2019	0.000	0.415	0.437	0.488	0.633	0.588	1.020	0.972	1.831	2.069	1.341	3.550	3.972	1.815	0.000	0.000	1.691
2020	0.000	0.359	0.450	0.531	0.588	0.879	0.616	1.640	2.340	2.318	3.309	1.625	1.257	0.000	0.000	0.000	2.481

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2021	0.000	0.441	0.378	0.468	0.527	0.572	0.739	0.692	1.009	1.182	1.645	1.964	0.000	2.046	0.000	0.000	1.128
2022	0.000	0.444	0.411	0.417	0.584	0.659	0.775	0.878	0.834	1.072	3.233	2.353	0.000	0.000	0.000	0.000	0.838
2023	0.000	0.449	0.516	0.460	0.471	0.664	0.691	0.741	1.271	1.061	1.385	3.011	3.940	0.000	0.000	0.000	1.274
2024	0.000	0.286	0.573	0.525	0.408	0.466	0.664	1.037	1.212	1.480	1.804	2.233	3.903	0.000	0.000	0.000	1.616

Table 8.2.16. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weight-at-age data (kg) for the stock. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0.004	0.028	0.146	0.291	0.552	0.731	0.998	0.987	1.062	1.062	1.062	1.062	1.062	1.062	1.062	1.062	1.062
1966	0.004	0.036	0.159	0.310	0.451	0.831	1.070	1.253	1.179	1.179	1.179	1.179	1.179	1.179	1.179	1.179	1.179
1967	0.006	0.047	0.178	0.376	0.498	0.554	0.914	1.253	1.243	1.243	1.243	1.243	1.243	1.243	1.243	1.243	1.243
1968	0.004	0.052	0.163	0.406	0.626	0.715	0.707	1.229	1.368	1.368	1.368	1.368	1.368	1.368	1.368	1.368	1.368
1969	0.005	0.026	0.139	0.322	0.688	0.782	0.878	0.890	1.195	1.195	1.195	1.195	1.195	1.195	1.195	1.195	1.195
1970	0.005	0.030	0.143	0.277	0.621	0.799	1.019	1.033	0.911	0.911	0.911	0.911	0.911	0.911	0.911	0.911	0.911
1971	0.005	0.044	0.158	0.282	0.429	0.767	1.082	1.264	0.851	0.851	0.851	0.851	0.851	0.851	0.851	0.851	0.851
1972	0.010	0.047	0.155	0.301	0.424	0.505	0.842	1.144	1.063	1.063	1.063	1.063	1.063	1.063	1.063	1.063	1.063
1973	0.017	0.048	0.153	0.290	0.487	0.528	0.553	0.865	1.042	1.042	1.042	1.042	1.042	1.042	1.042	1.042	1.042
1974	0.010	0.052	0.145	0.266	0.451	0.749	0.706	0.715	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943	0.943
1975	0.009	0.042	0.154	0.277	0.372	0.586	1.015	0.893	0.676	0.676	0.676	0.676	0.676	0.676	0.676	0.676	0.676

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1976	0.006	0.052	0.144	0.310	0.425	0.499	0.781	1.516	0.816	0.816	0.816	0.816	0.816	0.816	0.816	0.816	0.816
1977	0.008	0.045	0.153	0.266	0.493	0.529	0.671	0.962	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
1978	0.004	0.042	0.163	0.311	0.357	0.611	0.669	0.764	0.908	0.908	0.908	0.908	0.908	0.908	0.908	0.908	0.908
1979	0.004	0.038	0.184	0.328	0.514	0.578	0.794	1.029	0.890	0.890	0.890	0.890	0.890	0.890	0.890	0.890	0.890
1980	0.005	0.033	0.177	0.365	0.584	0.851	0.775	1.048	1.045	1.045	1.045	1.045	1.045	1.045	1.045	1.045	1.045
1981	0.004	0.024	0.169	0.350	0.611	0.912	1.042	0.991	1.012	1.012	1.012	1.012	1.012	1.012	1.012	1.012	1.012
1982	0.004	0.030	0.184	0.332	0.639	0.958	1.207	1.308	0.956	0.956	0.956	0.956	0.956	0.956	0.956	0.956	0.956
1983	0.004	0.053	0.195	0.339	0.515	0.781	1.033	0.987	0.976	0.976	0.976	0.976	0.976	0.976	0.976	0.976	0.976
1984	0.004	0.057	0.195	0.362	0.543	0.636	0.919	0.999	1.160	1.160	1.160	1.160	1.160	1.160	1.160	1.160	1.160
1985	0.004	0.060	0.190	0.366	0.546	0.721	0.797	1.114	1.159	1.159	1.159	1.159	1.159	1.159	1.159	1.159	1.159
1986	0.009	0.050	0.168	0.319	0.505	0.733	1.019	0.910	1.038	1.038	1.038	1.038	1.038	1.038	1.038	1.038	1.038
1987	0.004	0.051	0.169	0.318	0.500	0.842	1.036	1.096	0.841	0.841	0.841	0.841	0.841	0.841	0.841	0.841	0.841
1988	0.017	0.066	0.149	0.323	0.489	0.632	1.026	1.130	0.726	0.726	0.726	0.726	0.726	0.726	0.726	0.726	0.726
1989	0.015	0.081	0.181	0.288	0.497	0.665	0.798	1.044	0.831	0.831	0.831	0.831	0.831	0.831	0.831	0.831	0.831
1990	0.016	0.076	0.201	0.332	0.426	0.687	0.849	0.977	1.110	1.110	1.110	1.110	1.110	1.110	1.110	1.110	1.110
1991	0.012	0.071	0.198	0.357	0.483	0.556	0.818	0.942	0.891	0.891	0.891	0.891	0.891	0.891	0.891	0.891	0.891
1992	0.008	0.041	0.196	0.366	0.604	0.797	0.770	1.146	1.097	1.097	1.097	1.097	1.097	1.097	1.097	1.097	1.097

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1993	0.004	0.044	0.181	0.357	0.556	0.758	0.898	0.880	1.034	1.034	1.034	1.034	1.034	1.034	1.034	1.034	1.034
1994	0.007	0.049	0.159	0.342	0.504	0.731	1.058	1.056	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903	0.903
1995	0.005	0.043	0.186	0.290	0.489	0.669	0.902	1.176	0.859	0.859	0.859	0.859	0.859	0.859	0.859	0.859	0.859
1996	0.009	0.051	0.155	0.300	0.408	0.645	0.722	0.684	1.016	1.016	1.016	1.016	1.016	1.016	1.016	1.016	1.016
1997	0.012	0.056	0.180	0.283	0.493	0.548	0.787	0.827	0.972	0.972	0.972	0.972	0.972	0.972	0.972	0.972	0.972
1998	0.011	0.062	0.164	0.311	0.374	0.575	0.663	0.863	0.760	0.760	0.760	0.760	0.760	0.760	0.760	0.760	0.760
1999	0.010	0.067	0.160	0.280	0.402	0.441	0.627	0.662	0.574	0.574	0.574	0.574	0.574	0.574	0.574	0.574	0.574
2000	0.021	0.049	0.164	0.279	0.404	0.523	0.560	0.856	0.680	0.680	0.680	0.680	0.680	0.680	0.680	0.680	0.680
2001	0.012	0.045	0.141	0.252	0.392	0.569	0.627	0.783	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988
2002	0.007	0.044	0.164	0.244	0.444	0.607	0.707	0.678	0.947	0.947	0.947	0.947	0.947	0.947	0.947	0.947	0.947
2003	0.010	0.033	0.142	0.257	0.337	0.588	0.647	0.920	0.964	0.964	0.964	0.964	0.964	0.964	0.964	0.964	0.964
2004	0.016	0.056	0.153	0.297	0.333	0.380	0.585	0.768	0.946	0.946	0.946	0.946	0.946	0.946	0.946	0.946	0.946
2005	0.022	0.065	0.174	0.286	0.417	0.414	0.445	0.767	0.793	0.793	0.793	0.793	0.793	0.793	0.793	0.793	0.793
2006	0.017	0.051	0.182	0.277	0.372	0.438	0.459	0.456	0.802	0.802	0.802	0.802	0.802	0.802	0.802	0.802	0.802
2007	0.017	0.064	0.146	0.320	0.402	0.465	0.560	0.580	0.471	0.471	0.471	0.471	0.471	0.471	0.471	0.471	0.471
2008	0.014	0.057	0.162	0.282	0.480	0.555	0.647	0.693	0.536	0.536	0.536	0.536	0.536	0.536	0.536	0.536	0.536
2009	0.020	0.064	0.196	0.259	0.323	0.545	0.643	0.602	0.627	0.627	0.627	0.627	0.627	0.627	0.627	0.627	0.627

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2010	0.013	0.042	0.196	0.328	0.384	0.406	0.612	0.744	0.888	0.888	0.888	0.888	0.888	0.888	0.888	0.888	0.888
2011	0.016	0.063	0.170	0.353	0.449	0.459	0.463	0.599	0.767	0.767	0.767	0.767	0.767	0.767	0.767	0.767	0.767
2012	0.014	0.065	0.284	0.320	0.478	0.594	0.580	0.532	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783	0.783
2013	0.014	0.069	0.273	0.468	0.408	0.619	0.719	0.655	0.481	0.481	0.481	0.481	0.481	0.481	0.481	0.481	0.481
2014	0.017	0.057	0.277	0.464	0.550	0.461	0.659	0.684	0.522	0.522	0.522	0.522	0.522	0.522	0.522	0.522	0.522
2015	0.013	0.059	0.270	0.443	0.644	0.607	0.538	0.566	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490	0.490
2016	0.019	0.065	0.233	0.501	0.644	0.755	0.872	0.611	0.627	0.627	0.627	0.627	0.627	0.627	0.627	0.627	0.627
2017	0.016	0.060	0.220	0.354	0.658	0.830	1.141	0.866	0.588	0.588	0.588	0.588	0.588	0.588	0.588	0.588	0.588
2018	0.017	0.056	0.228	0.395	0.449	0.882	0.879	0.951	0.604	0.604	0.604	0.604	0.604	0.604	0.604	0.604	0.604
2019	0.019	0.058	0.200	0.361	0.526	0.501	0.869	0.806	1.061	1.061	1.061	1.061	1.061	1.061	1.061	1.061	1.061
2020	0.017	0.051	0.238	0.394	0.488	0.724	0.523	1.360	1.558	1.558	1.558	1.558	1.558	1.558	1.558	1.558	1.558
2021	0.018	0.057	0.196	0.348	0.434	0.494	0.624	0.572	0.709	0.709	0.709	0.709	0.709	0.709	0.709	0.709	0.709
2022	0.012	0.053	0.185	0.293	0.475	0.557	0.656	0.675	0.468	0.468	0.468	0.468	0.468	0.468	0.468	0.468	0.468
2023	0.019	0.057	0.221	0.291	0.382	0.547	0.588	0.613	0.796	0.796	0.796	0.796	0.796	0.796	0.796	0.796	0.796
2024	0.018	0.059	0.218	0.336	0.328	0.391	0.560	0.844	1.004	1.004	1.004	1.004	1.004	1.004	1.004	1.004	1.004

Table 8.2.17. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Estimates of natural mortality from the most recent key run of SMS (ICES WGSAM, 2023). Ages 0-7 and 8+ and years 1972-2024 are used in the assessment. * Values since 2022 are set equal to 2022 values until the next SMS key run in completed by WGSAM.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1966	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1967	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1968	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1969	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1970	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1971	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1972	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1973	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1974	1.189	1.478	0.783	0.533	0.404	0.288	0.257	0.237	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
1975	1.193	1.457	0.760	0.518	0.394	0.284	0.254	0.235	0.216	0.216	0.216	0.216	0.216	0.216	0.216	0.216	0.216
1976	1.196	1.436	0.737	0.504	0.383	0.281	0.251	0.233	0.214	0.214	0.214	0.214	0.214	0.214	0.214	0.214	0.214
1977	1.199	1.414	0.714	0.489	0.372	0.277	0.248	0.231	0.213	0.213	0.213	0.213	0.213	0.213	0.213	0.213	0.213
1978	1.201	1.392	0.690	0.474	0.362	0.273	0.245	0.228	0.212	0.212	0.212	0.212	0.212	0.212	0.212	0.212	0.212
1979	1.202	1.369	0.667	0.460	0.352	0.270	0.242	0.227	0.211	0.211	0.211	0.211	0.211	0.211	0.211	0.211	0.211
1980	1.203	1.345	0.644	0.446	0.342	0.266	0.239	0.225	0.210	0.210	0.210	0.210	0.210	0.210	0.210	0.210	0.210

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1981	1.202	1.319	0.622	0.432	0.333	0.263	0.236	0.223	0.209	0.209	0.209	0.209	0.209	0.209	0.209	0.209	0.209
1982	1.200	1.292	0.599	0.418	0.324	0.260	0.234	0.221	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208
1983	1.196	1.264	0.577	0.405	0.316	0.257	0.232	0.220	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208
1984	1.192	1.235	0.556	0.393	0.309	0.255	0.230	0.218	0.207	0.207	0.207	0.207	0.207	0.207	0.207	0.207	0.207
1985	1.188	1.207	0.535	0.382	0.302	0.253	0.228	0.217	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206
1986	1.183	1.180	0.516	0.371	0.296	0.251	0.227	0.216	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206
1987	1.178	1.155	0.498	0.361	0.291	0.249	0.226	0.215	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1988	1.173	1.133	0.481	0.352	0.287	0.248	0.225	0.215	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1989	1.168	1.114	0.466	0.344	0.283	0.247	0.225	0.214	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1990	1.163	1.098	0.452	0.336	0.280	0.247	0.225	0.214	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1991	1.160	1.086	0.440	0.330	0.277	0.247	0.225	0.214	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1992	1.157	1.077	0.429	0.325	0.276	0.247	0.226	0.215	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1993	1.155	1.071	0.420	0.321	0.274	0.247	0.227	0.216	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205	0.205
1994	1.154	1.067	0.412	0.317	0.274	0.248	0.229	0.217	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206
1995	1.154	1.065	0.406	0.315	0.274	0.249	0.230	0.218	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206	0.206
1996	1.154	1.065	0.402	0.314	0.274	0.251	0.233	0.220	0.207	0.207	0.207	0.207	0.207	0.207	0.207	0.207	0.207
1997	1.154	1.065	0.399	0.313	0.275	0.252	0.235	0.222	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1998	1.154	1.067	0.398	0.313	0.276	0.255	0.238	0.224	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208	0.208
1999	1.153	1.070	0.398	0.314	0.278	0.257	0.240	0.226	0.209	0.209	0.209	0.209	0.209	0.209	0.209	0.209	0.209
2000	1.152	1.074	0.400	0.316	0.280	0.259	0.243	0.229	0.210	0.210	0.210	0.210	0.210	0.210	0.210	0.210	0.210
2001	1.150	1.079	0.403	0.318	0.282	0.262	0.246	0.232	0.211	0.211	0.211	0.211	0.211	0.211	0.211	0.211	0.211
2002	1.147	1.085	0.408	0.320	0.284	0.265	0.250	0.235	0.213	0.213	0.213	0.213	0.213	0.213	0.213	0.213	0.213
2003	1.142	1.092	0.413	0.322	0.286	0.267	0.252	0.238	0.214	0.214	0.214	0.214	0.214	0.214	0.214	0.214	0.214
2004	1.136	1.099	0.420	0.324	0.289	0.270	0.255	0.241	0.215	0.215	0.215	0.215	0.215	0.215	0.215	0.215	0.215
2005	1.129	1.105	0.427	0.326	0.291	0.272	0.258	0.244	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217	0.217
2006	1.121	1.111	0.434	0.328	0.293	0.274	0.260	0.248	0.218	0.218	0.218	0.218	0.218	0.218	0.218	0.218	0.218
2007	1.112	1.116	0.441	0.330	0.295	0.276	0.262	0.250	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220
2008	1.104	1.120	0.449	0.332	0.297	0.278	0.264	0.253	0.222	0.222	0.222	0.222	0.222	0.222	0.222	0.222	0.222
2009	1.096	1.123	0.456	0.334	0.298	0.279	0.265	0.256	0.223	0.223	0.223	0.223	0.223	0.223	0.223	0.223	0.223
2010	1.089	1.125	0.463	0.335	0.299	0.280	0.266	0.258	0.225	0.225	0.225	0.225	0.225	0.225	0.225	0.225	0.225
2011	1.083	1.127	0.469	0.336	0.300	0.281	0.266	0.260	0.226	0.226	0.226	0.226	0.226	0.226	0.226	0.226	0.226
2012	1.079	1.127	0.474	0.337	0.301	0.281	0.266	0.261	0.227	0.227	0.227	0.227	0.227	0.227	0.227	0.227	0.227
2013	1.075	1.128	0.479	0.338	0.302	0.281	0.266	0.262	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.228	0.228
2014	1.074	1.128	0.483	0.339	0.302	0.281	0.265	0.262	0.229	0.229	0.229	0.229	0.229	0.229	0.229	0.229	0.229

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2015	1.074	1.128	0.486	0.340	0.302	0.280	0.264	0.262	0.230	0.230	0.230	0.230	0.230	0.230	0.230	0.230	0.230
2016	1.075	1.128	0.489	0.341	0.302	0.280	0.263	0.262	0.231	0.231	0.231	0.231	0.231	0.231	0.231	0.231	0.231
2017	1.078	1.128	0.492	0.342	0.302	0.279	0.262	0.262	0.232	0.232	0.232	0.232	0.232	0.232	0.232	0.232	0.232
2018	1.082	1.128	0.494	0.343	0.302	0.278	0.261	0.262	0.232	0.232	0.232	0.232	0.232	0.232	0.232	0.232	0.232
2019	1.088	1.129	0.496	0.344	0.301	0.278	0.260	0.261	0.233	0.233	0.233	0.233	0.233	0.233	0.233	0.233	0.233
2020	1.094	1.130	0.497	0.346	0.301	0.277	0.260	0.261	0.233	0.233	0.233	0.233	0.233	0.233	0.233	0.233	0.233
2021	1.102	1.131	0.499	0.347	0.301	0.276	0.259	0.261	0.234	0.234	0.234	0.234	0.234	0.234	0.234	0.234	0.234
2022*	1.109	1.132	0.500	0.348	0.301	0.276	0.258	0.260	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235
2023*	1.109	1.132	0.500	0.348	0.301	0.276	0.258	0.260	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235
2024*	1.109	1.132	0.500	0.348	0.301	0.276	0.258	0.260	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235

Table 8.2.18. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Estimates of the proportion mature-at-age. Ages 0–7 and 8+ and years 1972–2024 are used in the assessment.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1965	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1966	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1967	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1968	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1969	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1970	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1971	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1972	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1973	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1974	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1975	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1976	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1977	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1978	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1979	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1980	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1981	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1982	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1983	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1984	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1985	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1986	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
1987	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1988	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1989	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1990	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1991	0.000	0.034	0.429	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1992	0.000	0.037	0.434	0.937	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1993	0.000	0.039	0.440	0.936	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1994	0.000	0.041	0.448	0.936	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1995	0.000	0.043	0.460	0.938	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1996	0.000	0.044	0.477	0.941	0.996	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1997	0.000	0.046	0.498	0.945	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1998	0.000	0.047	0.524	0.950	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1999	0.000	0.049	0.551	0.956	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2000	0.000	0.050	0.579	0.961	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2001	0.000	0.052	0.607	0.966	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2002	0.000	0.053	0.635	0.971	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2003	0.000	0.055	0.662	0.976	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2004	0.000	0.057	0.685	0.980	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2005	0.000	0.059	0.703	0.983	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2006	0.000	0.061	0.715	0.985	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2007	0.000	0.064	0.719	0.985	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2008	0.000	0.067	0.717	0.985	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2009	0.000	0.071	0.710	0.983	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2010	0.000	0.074	0.701	0.982	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2011	0.000	0.078	0.692	0.980	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2012	0.000	0.080	0.684	0.978	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2013	0.000	0.082	0.678	0.977	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2014	0.000	0.082	0.674	0.977	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2015	0.000	0.082	0.670	0.976	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2016	0.000	0.082	0.667	0.976	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2017	0.000	0.082	0.661	0.975	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2018	0.000	0.081	0.654	0.973	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2019	0.000	0.081	0.643	0.971	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2020	0.000	0.080	0.629	0.967	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15+	8+
2021	0.000	0.080	0.612	0.962	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2022	0.000	0.079	0.593	0.957	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2023	0.000	0.079	0.575	0.951	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
2024	0.000	0.080	0.556	0.946	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Table 8.2.19. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Data available for calibration of the assessment. Only those data used in the final assessment are shown here.

Delta-GAM NS-WC Q1								
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8+**
1983	268.33	309.12	63.878	99.035	11.9255	1.8947	0.1223	0.2262
1984	1464.42	173.62	105.016	13.871	17.7329	2.1223	0.3825	0.1102
1985	289.37	608.33	67.760	23.442	3.8472	3.7335	0.5814	0.2401
1986	644.93	231.73	264.519	19.015	6.0847	1.2451	1.4629	0.1793
1987	1018.17	265.07	47.887	41.520	2.6128	1.1100	0.2344	0.7716
1988	147.96	526.62	76.526	9.855	7.7261	0.6774	0.1806	0.5696
1989	153.53	80.31	183.911	12.495	1.6151	2.5333	0.1956	0.1182
1990	175.18	74.15	25.623	32.360	2.3714	0.4681	0.7990	0.1223
1991	548.14	97.98	19.172	5.114	4.8992	1.0193	0.0883	0.2708
1992	1189.71	288.02	22.500	1.904	1.3225	1.0168	0.1348	0.1606
1993	1658.68	550.39	147.679	11.259	0.9132	0.2828	0.6878	0.3682
1994	215.80	487.54	101.166	15.741	1.7288	0.3796	0.0374	0.4880
1995	1817.43	190.09	195.410	24.776	4.2243	0.7934	0.1781	0.2620
1996	402.72	905.75	74.400	43.020	4.5430	1.1397	0.1703	0.0697
1997	1124.43	245.36	363.578	21.141	8.6154	1.5986	0.4472	0.0870
1998	456.99	363.97	77.124	88.090	6.7449	2.6043	0.8806	0.2483
1999	157.03	213.76	104.150	25.475	20.2309	1.6709	0.9293	0.3253
2000	4621.97	118.29	54.553	18.660	5.1893	4.0239	0.5536	0.2433
2001	734.47	2367.82	71.147	10.856	3.7431	1.5502	0.7353	0.2706
2002	87.18	448.24	1048.869	14.407	4.3342	1.2492	0.7893	0.2890
2003	81.35	64.87	220.875	370.947	3.5414	1.0605	0.5949	0.4470
2004	38.91	74.93	33.144	92.202	122.9071	1.1004	0.4846	0.2821
2005	37.97	42.55	35.638	10.464	28.1241	43.4931	1.2182	0.3733
2006	425.09	42.29	23.728	9.629	3.7502	7.5804	11.5693	0.9274
2007	67.22	495.67	28.732	9.351	3.0240	1.5293	3.7664	5.3271
2008	41.46	81.54	179.511	8.448	2.3936	1.1737	0.6572	2.7676
2009	33.95	53.29	49.573	66.342	2.3094	0.8311	0.7105	1.2534

Delta-GAM NS-WC Q1								
	1	2	3	4	5	6	7	8+
2010	411.55	52.46	29.529	16.321	22.2952	1.4288	0.4445	0.6953
2011	21.85	479.12	50.532	15.363	9.9671	18.2433	0.3688	0.5843
2012	11.96	33.42	241.314	15.106	7.2669	4.8777	6.9094	0.3842
2013	36.22	14.66	21.228	86.316	5.0721	2.5503	1.3067	3.3722
2014	29.24	39.13	9.476	11.168	31.1962	1.5249	0.9886	1.7498
2015	321.79	28.47	21.147	3.176	4.5420	13.3616	0.5279	1.0119
2016	52.90	235.52	15.389	6.439	1.2986	0.8495	6.2028	0.3325
2017	117.46	74.40	130.775	6.561	1.9744	0.3791	0.4652	3.0724
2018	34.34	94.81	41.685	51.522	1.4075	0.5555	0.1041	1.2217
2019	165.16	35.47	35.739	8.330	16.4359	0.5195	0.1984	0.3288
2020	265.33	124.12	20.809	14.641	3.2059	8.9106	0.1331	0.2138
2021	465.34	582.43	78.068	7.906	7.3465	1.4790	5.3332	0.1317
2022	105.64	1011.02	779.209	17.961	6.1126	6.5209	0.6782	3.8650
2023	406.44	253.55	672.940	282.413	13.0863	3.0093	1.5834	1.5981
2024	51.99	776.91	229.649	417.401	128.5958	6.5393	0.9483	0.9611
2025	68.07	174.15	556.688	215.418	195.8066	49.0537	2.0382	0.9076

Table 8.2.19. (cont.) Haddock in Subarea 4, Division 6.a, and Subdivision 20. Data available for calibration of the assessment. Only those data used in the final assessment are shown here.

Delta-GAM NS-WC Q3+Q4									
	0	1	2	3	4	5	6	7	8+
1991	3530.84	370.23	29.78	5.649	1.501	5.2362	0.2185	0.1395	0.2778
1992	6211.21	932.96	220.90	12.170	1.809	0.5623	1.3701	0.0356	0.1203
1993	758.58	1632.92	265.55	60.772	2.376	0.9634	0.1726	0.3266	0.0472
1994	4106.83	340.12	428.03	46.923	17.736	0.7901	0.0343	0.0215	0.0904
1995	1689.57	2367.46	174.15	132.466	12.275	3.4572	0.5063	0.0673	0.1340
1996	1328.93	468.71	512.29	38.847	19.743	2.3833	0.6149	0.0630	0.0434
1997	618.43	634.13	146.95	164.851	7.149	3.8731	0.8203	0.3230	0.0347
1998	311.29	345.42	164.92	39.751	31.756	2.5330	1.6366	0.2046	0.1441
1999	10308.84	158.60	87.69	39.184	9.502	6.5225	0.7386	0.5826	0.2602

Delta-GAM NS-WC Q3+Q4									
	0	1	2	3	4	5	6	7	8+
2000	1536.33	3279.43	46.24	17.183	6.541	2.2428	1.5230	0.1408	0.1075
2001	161.54	623.89	1395.34	29.788	8.061	3.4234	1.2624	0.6147	0.0582
2002	256.51	193.93	286.04	508.096	9.976	3.5514	1.8589	0.4225	0.3280
2003	118.87	138.44	59.95	150.337	213.405	2.9645	1.3104	0.3486	0.1576
2004	143.12	79.15	72.42	23.524	60.332	82.1870	1.3876	0.4825	0.2648
2005	1165.73	58.91	44.98	20.483	10.355	20.0014	18.1897	0.2448	0.1425
2006	169.01	468.22	36.02	11.646	6.954	3.4715	5.1261	8.5310	0.1620
2007	140.12	150.14	353.44	16.037	7.515	2.3856	1.2579	1.8900	2.3198
2008	86.24	52.86	65.20	118.994	5.902	3.5491	2.0414	0.6771	1.8554
2009	722.29	42.98	43.72	26.010	43.959	1.6770	1.1243	0.5065	0.4114
2010	72.98	540.83	48.04	23.480	15.775	25.0437	1.0865	0.4300	0.4197
2011	28.31	40.80	434.28	35.466	12.877	7.5800	14.7496	0.4834	0.3185
2012	134.56	22.54	34.39	150.256	7.707	4.5120	3.4674	5.2442	0.1982
2013	73.39	33.19	17.39	12.460	66.055	2.4524	1.9569	1.0022	2.2426
2014	2432.25	45.39	33.24	5.788	5.941	33.5142	1.2628	0.6954	1.3028
2015	149.69	653.26	40.06	12.770	2.222	3.0421	14.5327	0.6142	0.6253
2016	217.11	89.68	236.87	14.185	3.899	0.7900	1.2438	5.3754	0.2952
2017	70.27	165.45	93.51	131.348	5.135	1.4561	0.3123	0.5542	1.8012
2018	200.58	45.76	69.12	30.563	53.163	1.2860	0.6591	0.0530	0.3278
2019	1728.62	334.06	30.72	29.771	10.825	21.4882	0.5749	0.0544	0.0959
2020	1651.09	1451.06	163.53	20.588	19.444	5.3813	16.7352	0.1642	0.0716
2021	158.01	928.06	802.76	73.459	10.051	7.1323	2.0974	4.3122	0.0869
2022	969.72	215.90	711.70	455.802	40.473	4.5845	3.3018	0.4700	1.7832
2023	48.73	734.92	280.11	635.107	258.576	17.9988	1.7661	0.9454	0.7722
2024	121.24	81.60	522.59	215.349	269.549	72.5549	4.2709	0.5449	0.1793

Table 8.3.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. SAM final assessment: Model inputs.

	Dataset	Year	Ages	Details
Catch data	Catch numbers	1972–2024	0–8+	Catch-at-age in numbers. Combination of landings, discards, BMS landings and industrial bycatch

	Dataset	Year	Ages	Details
	Catch weights	1972–2024	0–8+	Mean weights-at-age in the total catch.
	Landing weights	1972–2024	0–8+	Mean weight-at-age in the landings.
	Discard weights	1972–2024	0–8+	Mean weight-at-age in the discards.
	Discard numbers	1972–2024	0–8+	Numbers-at-age for discards. Used for plus group calculations.
	Landings fraction (numbers)	1972–2024	0–8+	Numbers-at-age for landings. Used to separate out the landings portion of the total catch for use in the forecast. This is converted to a landings fraction within the model code.
Biological data	Stock weights	1972–2024	0–8+	Mean weights-at-age. Produced from applying survey-derived correction factors to mean weights-at-age in the catch.
	Natural mortality	1974–2022	0–8+	Smoothed natural mortality estimates from the SMS key run produced by WGSAM (1974-2022). Since 2023 time-invariant using the value of 2022.
	Maturity	1991–2024	0–8+	Proportion mature. Smoothed, time-varying, survey-derived maturity ogives estimated from quarter 1 data. Data from the intermediate year (2025) are used for estimating the smoothed estimates, but only values up to the last data year are used in assessment.
	Proportion of natural mortality before spawning	1972–2024	0–8+	Set to zero for all ages and years.
	Proportion of fishing mortality before spawning	1972–2024	0–8+	Set to zero for all ages and years.
Tuning data	Survey: delta-GAM NS-WC Q1	1983–2025	1–8+	Delta-GAM modelled survey indices covering the North Sea and West Coast of Scotland in quarter 1.
	Survey: delta-GAM NS-WC Q3+Q4	1991–2024	0–8+	Delta-GAM modelled survey indices covering the North Sea in quarter 3 and West Coast of Scotland in quarter 4.
	Weightings: coefficient of variations for delta-GAM NS-WC Q1	1983–2025	1–8+	Coefficient of variations for the delta-GAM modelled survey indices covering the North Sea and West Coast of Scotland in quarter 1.
	Weightings: coefficient of variations for delta-GAM NS-WC Q3+Q4	1991–2024	0–8+	Coefficient of variations for the delta-GAM modelled survey indices covering the North Sea in quarter 3 and West Coast of Scotland in quarter 4.

Table 8.3.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. SAM final assessment: Model settings. Settings not listed use the default setting in SAM.

Configuration setting	Details								
Assessment age range	0-8+								
Is maximum considered a plus group	1 1 1								
Coupling of the fishing mortality states	0 1 2 3 4 5 6 7 8								
Correlation of fishing mortality across ages	AR(1)								
Coupling of the survey catchability parameters	-1	-1	-1	-1	-1	-1	-1	-1	-1
	-1	0	1	2	3	4	5	6	7
	8	9	10	11	12	13	14	15	16
Coupling of process variance parameters for F	0 0 0 0 0 0 0 0 0								
Coupling of process variance parameters for N	0 1 1 1 1 1 1 1 2								
Density dependent survey catchability	-1	-1	-1	-1	-1	-1	-1	-1	-1
	-1	0	-1	-1	-1	-1	-1	-1	-1
	1	2	-1	-1	-1	-1	-1	-1	-1
Coupling of the variance parameters for observations	0	1	2	2	2	2	2	2	2
	-1	3	4	4	5	5	5	6	6
	7	8	9	10	10	10	11	11	11
Covariance structure for each fleet	Independent (ID) for all								
Stock recruitment code	Random walk (0)								
F range	2-4								

Table 8.3.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. SAM final assessment: Parameter estimates.

	Estimate	Lower bound	Upper bound	sd(par)	log(par)
logFpar_0	0.00000	0.00000	0.00002	1.017	-13.112
logFpar_1	0.00057	0.00051	0.00062	0.049	-7.476
logFpar_2	0.00058	0.00053	0.00064	0.047	-7.453
logFpar_3	0.00040	0.00036	0.00045	0.060	-7.815
logFpar_4	0.00027	0.00023	0.00031	0.075	-8.218
logFpar_5	0.00019	0.00016	0.00024	0.103	-8.553
logFpar_6	0.00012	0.00009	0.00016	0.145	-9.007
logFpar_7	0.00006	0.00003	0.00010	0.268	-9.797
logFpar_8	0.00000	0.00000	0.00001	1.417	-14.439
logFpar_9	0.00002	0.00000	0.00008	0.753	-10.983
logFpar_10	0.00072	0.00064	0.00081	0.057	-7.239
logFpar_11	0.00058	0.00051	0.00065	0.060	-7.458
logFpar_12	0.00044	0.00038	0.00051	0.072	-7.727

	Estimate	Lower bound	Upper bound	sd(par)	log(par)
logFpar_13	0.00030	0.00025	0.00037	0.093	-8.101
logFpar_14	0.00021	0.00016	0.00028	0.131	-8.451
logFpar_15	0.00010	0.00007	0.00014	0.163	-9.173
logFpar_16	0.00003	0.00002	0.00005	0.279	-10.392
logQpow_0	1.338	1.199	1.494	0.055	0.291
logQpow_1	1.455	1.267	1.671	0.069	0.375
logQpow_2	1.274	1.163	1.395	0.045	0.242
logSdLogFsta_0	0.229	0.184	0.286	0.110	-1.472
logSdLogN_0	1.036	0.827	1.297	0.112	0.035
logSdLogN_1	0.189	0.161	0.221	0.079	-1.665
logSdLogN_2	0.425	0.331	0.545	0.125	-0.857
logSdLogObs_0	0.985	0.793	1.224	0.108	-0.015
logSdLogObs_1	0.382	0.290	0.504	0.138	-0.962
logSdLogObs_2	0.248	0.216	0.286	0.070	-1.394
logSdLogObs_3	0.452	0.347	0.590	0.133	-0.793
logSdLogObs_4	1.390	1.101	1.756	0.117	0.329
logSdLogObs_5	2.036	1.717	2.414	0.085	0.711
logSdLogObs_6	2.661	2.164	3.273	0.103	0.979
logSdLogObs_7	0.464	0.304	0.709	0.212	-0.768
logSdLogObs_8	0.976	0.611	1.560	0.234	-0.024
logSdLogObs_9	1.208	0.849	1.719	0.176	0.189
logSdLogObs_10	1.280	1.008	1.625	0.119	0.247
logSdLogObs_11	2.337	1.948	2.805	0.091	0.849
itrans_rho_0	7.331	4.303	12.490	0.266	1.992

Table 8.3.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Estimates of fishing mortality-at-age from the final SAM assessment. Estimates refer to the full year (January–December) except for age 0, for which the mortality rate given refers to the second half-year only (July–December).

	0	1	2	3	4	5	6	7	8+	Mean F(2–4)
1972	0.024	0.181	0.596	0.947	0.892	0.830	0.809	0.733	0.315	0.812
1973	0.023	0.173	0.563	0.884	0.837	0.787	0.772	0.704	0.304	0.761

	0	1	2	3	4	5	6	7	8+	Mean F(2–4)
1974	0.024	0.180	0.583	0.897	0.847	0.794	0.777	0.706	0.305	0.776
1975	0.028	0.207	0.679	1.045	0.993	0.937	0.913	0.826	0.355	0.906
1976	0.027	0.197	0.652	1.001	0.954	0.900	0.862	0.773	0.330	0.869
1977	0.026	0.189	0.634	0.970	0.934	0.868	0.816	0.727	0.307	0.846
1978	0.026	0.188	0.646	1.006	0.980	0.904	0.844	0.753	0.318	0.878
1979	0.024	0.171	0.600	0.948	0.924	0.834	0.766	0.685	0.288	0.824
1980	0.021	0.153	0.539	0.860	0.842	0.740	0.667	0.598	0.251	0.747
1981	0.017	0.123	0.433	0.691	0.678	0.588	0.523	0.473	0.199	0.601
1982	0.018	0.126	0.452	0.722	0.716	0.622	0.555	0.504	0.210	0.630
1983	0.020	0.144	0.532	0.849	0.848	0.732	0.639	0.567	0.234	0.743
1984	0.020	0.150	0.568	0.908	0.913	0.788	0.682	0.596	0.243	0.797
1985	0.019	0.143	0.559	0.888	0.897	0.770	0.666	0.581	0.236	0.781
1986	0.019	0.146	0.596	0.942	0.945	0.803	0.688	0.602	0.244	0.828
1987	0.020	0.160	0.675	1.064	1.072	0.920	0.803	0.708	0.284	0.937
1988	0.019	0.152	0.653	1.025	1.023	0.874	0.758	0.671	0.267	0.900
1989	0.019	0.152	0.661	1.017	1.006	0.854	0.740	0.658	0.259	0.895
1990	0.018	0.149	0.653	0.972	0.936	0.780	0.665	0.588	0.228	0.854
1991	0.019	0.161	0.712	1.058	1.009	0.847	0.723	0.642	0.245	0.926
1992	0.016	0.137	0.609	0.913	0.887	0.748	0.638	0.563	0.212	0.803
1993	0.016	0.135	0.605	0.912	0.895	0.753	0.626	0.541	0.199	0.804
1994	0.015	0.127	0.572	0.875	0.879	0.745	0.604	0.513	0.186	0.775
1995	0.013	0.112	0.504	0.771	0.788	0.668	0.518	0.421	0.150	0.688
1996	0.014	0.127	0.581	0.899	0.947	0.823	0.628	0.492	0.170	0.809
1997	0.013	0.114	0.514	0.776	0.807	0.697	0.512	0.382	0.131	0.699
1998	0.014	0.125	0.562	0.829	0.848	0.716	0.510	0.363	0.124	0.746
1999	0.016	0.155	0.699	1.031	1.035	0.880	0.619	0.428	0.144	0.922
2000	0.015	0.144	0.643	0.948	0.944	0.800	0.572	0.393	0.135	0.845
2001	0.009	0.085	0.362	0.527	0.527	0.458	0.335	0.235	0.084	0.472
2002	0.007	0.066	0.266	0.375	0.372	0.328	0.242	0.169	0.061	0.338

	0	1	2	3	4	5	6	7	8+	Mean F(2–4)
2003	0.005	0.046	0.180	0.246	0.241	0.213	0.153	0.105	0.038	0.222
2004	0.005	0.053	0.204	0.274	0.263	0.231	0.160	0.105	0.036	0.247
2005	0.007	0.068	0.267	0.364	0.349	0.309	0.214	0.138	0.047	0.327
2006	0.009	0.091	0.363	0.511	0.498	0.438	0.302	0.199	0.066	0.457
2007	0.008	0.081	0.323	0.466	0.464	0.412	0.287	0.198	0.068	0.418
2008	0.006	0.062	0.243	0.353	0.352	0.306	0.210	0.150	0.053	0.316
2009	0.005	0.053	0.208	0.314	0.322	0.278	0.186	0.135	0.048	0.281
2010	0.004	0.045	0.176	0.275	0.288	0.252	0.167	0.123	0.044	0.246
2011	0.004	0.037	0.148	0.238	0.255	0.226	0.151	0.115	0.042	0.214
2012	0.003	0.036	0.145	0.237	0.258	0.228	0.149	0.119	0.044	0.213
2013	0.004	0.039	0.160	0.262	0.282	0.244	0.154	0.124	0.047	0.235
2014	0.004	0.049	0.212	0.356	0.380	0.324	0.196	0.161	0.060	0.316
2015	0.005	0.053	0.240	0.421	0.453	0.389	0.234	0.197	0.073	0.371
2016	0.003	0.039	0.178	0.318	0.346	0.293	0.173	0.149	0.055	0.281
2017	0.003	0.033	0.150	0.270	0.297	0.254	0.149	0.130	0.048	0.239
2018	0.003	0.034	0.156	0.290	0.330	0.291	0.175	0.154	0.056	0.259
2019	0.003	0.031	0.143	0.275	0.325	0.295	0.177	0.153	0.055	0.247
2020	0.002	0.024	0.110	0.221	0.275	0.258	0.154	0.132	0.047	0.202
2021	0.002	0.016	0.073	0.148	0.197	0.192	0.114	0.096	0.034	0.139
2022	0.001	0.010	0.043	0.090	0.123	0.124	0.074	0.061	0.022	0.086
2023	0.001	0.008	0.033	0.069	0.099	0.101	0.060	0.049	0.017	0.067
2024	0.001	0.007	0.029	0.061	0.089	0.092	0.054	0.044	0.016	0.060

Table 8.3.5. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Estimates of stock numbers-at-age (thousands) from the final SAM assessment. Estimates refer to 1 January, except for age 0 for estimates refer to 1 July. * SAM estimated survivors and projected recruitment.

	0	1	2	3	4	5	6	7	8+
1972	12438095	10776296	2092738	75336	44625	433384	8424	516	2409
1973	30786654	3536110	2214273	528944	15702	11831	143479	2952	1841
1974	53668362	9845151	600295	680540	125791	4548	3919	50466	2077
1975	6104936	17543605	1830930	136990	186283	35413	1656	1397	21134

	0	1	2	3	4	5	6	7	8+
1976	7153850	1705984	3437266	418109	28042	49844	10759	496	8116
1977	9320369	2064539	304654	924234	87326	8302	14451	3544	3359
1978	16028953	2728043	378197	71049	245613	22170	2730	4909	4335
1979	25238635	4683860	579925	91957	16105	69739	6403	886	4023
1980	7156029	7586687	1117317	158077	21164	4618	24016	2199	2333
1981	10604318	1887939	1877248	368229	40168	5803	1791	9749	2033
1982	6510711	3018962	405993	791705	117032	13755	2260	1045	6412
1983	17538077	1740928	692753	147757	283212	41480	6322	968	3999
1984	5943853	5379313	414630	236316	46525	90009	13995	2624	2360
1985	7524586	1671698	1409781	147247	69718	14833	31115	5323	2959
1986	10510952	2208095	412663	531804	47195	21809	5571	12466	4463
1987	2017577	3493147	563241	111083	144589	12506	6883	2311	12579
1988	2101038	593603	1055331	161699	28200	40199	4105	2136	7765
1989	2687092	641592	148485	369103	38025	7613	14335	1627	4168
1990	6082830	817054	158083	46348	97374	10172	2641	5480	2585
1991	11110962	1965606	204687	42267	12433	28191	3848	1147	5984
1992	15636548	3115945	632596	61572	10200	3906	10174	1438	4904
1993	4763128	4407077	840429	248814	22321	3875	1549	4809	4651
1994	15378716	1322376	1092769	229215	65046	7518	1563	684	5807
1995	6341308	5278120	387301	393151	65693	19939	3303	673	3002
1996	7263481	1601488	1605818	153573	119144	20377	7877	1799	1592
1997	4200931	2292038	422915	632765	44951	32843	7751	3562	1623
1998	2638446	1320867	599826	160345	203528	18078	13621	3936	3827
1999	25070740	765098	331545	202793	58754	64037	7208	6787	5237
2000	6984281	8568487	207876	99758	50540	17865	20370	2980	4472
2001	1791681	2217162	3306120	95161	31864	16606	7985	8335	4218
2002	2006632	587514	676390	1602551	34484	15299	7999	5109	10342
2003	1288148	576300	133096	355130	769042	14189	6973	4490	7920
2004	1249149	394108	152955	60396	202250	428716	7664	4359	6937

	0	1	2	3	4	5	6	7	8+
2005	5562520	351680	103508	65637	31272	108716	232666	4632	5984
2006	1746746	1924526	101773	48452	30447	17837	56326	140613	7851
2007	1143305	595507	778036	49341	25689	14099	9066	33918	100899
2008	985628	344178	168367	351813	20371	12088	7982	5285	66610
2009	4980551	307364	105031	86039	198930	9693	5862	5245	23055
2010	781331	1962662	109593	63576	52329	128851	6520	3956	13910
2011	462389	233517	780153	72549	39635	34421	101463	4387	9988
2012	931910	141659	66720	423406	30813	22727	20856	70120	9156
2013	845115	273687	35862	36273	254716	14507	13074	11561	75222
2014	5972787	266540	87152	17490	24788	162381	8387	8332	42692
2015	1440062	1878263	76754	41887	8073	14476	94978	5818	24636
2016	1837199	466430	518525	33357	16930	3747	5664	64426	11582
2017	883243	663655	164432	275398	15124	7492	1810	3997	49943
2018	2608676	281904	182943	78526	162708	6121	3768	1179	20188
2019	8777976	1082452	87742	90522	34672	96682	3136	2242	5997
2020	7920260	3456293	308450	50952	57043	18546	63561	1790	5539
2021	2490941	3011515	1413423	156099	30970	34144	9477	48491	3167
2022	6083569	953277	1507330	1086853	97293	20712	19933	6031	42828
2023	1217340	2367755	495683	1247532	738974	59887	14316	13086	22796
2024	1365893	494166	1075286	428376	927285	412426	31720	7983	12988
2025*	1365893	461126	208230	866517	385344	681344	275995	21105	16564

Table 8.3.6. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Stock summary table for the final SAM model. Both estimates and the 2.5th (low) and 97.5th (high) percentiles are given.

Year	Recruitment (age 0)	Recruitment low	Recruitment high	SSB estimate	SSB low	SSB high	Fbar (2–4)	Fbar low	Fbar high	TSB estimate	TSB low	TSB high
1972	12438095	7217436	21435065	425596	338310	535401	0.812	0.655	1.006	1225941	978170	1536473
1973	30786654	18168806	52167327	392671	321645	479381	0.761	0.625	0.927	1282993	998282	1648904
1974	53668362	31602058	91142579	325268	265627	398300	0.776	0.641	0.939	1417645	1086756	1849281
1975	6104936	3601692	10347984	288595	240050	346959	0.906	0.753	1.090	1218950	929903	1597843
1976	7153850	4234685	12085332	389454	314003	483036	0.869	0.721	1.047	808782	647425	1010354
1977	9320369	5530670	15706829	317480	249194	404477	0.846	0.702	1.021	523818	431445	635968
1978	16028953	9464376	27146778	161460	132073	197386	0.878	0.727	1.060	373179	307735	452540
1979	25238635	14856330	42876585	138250	115548	165412	0.824	0.678	1.002	473972	379097	592590
1980	7156029	4242623	12070069	187069	153776	227570	0.747	0.609	0.916	581248	462243	730891
1981	10604318	6408081	17548397	301854	244851	372128	0.601	0.488	0.740	577283	467279	713183
1982	6510711	4071185	10412045	379563	305315	471867	0.630	0.518	0.766	552346	461367	661266
1983	17538077	10886712	28253171	297130	252937	349044	0.743	0.617	0.896	537237	465957	619422
1984	5943853	3736764	9454541	225983	197207	258957	0.797	0.665	0.954	597526	494456	722080
1985	7524586	4736200	11954606	251640	215037	294473	0.781	0.653	0.934	535071	453080	631898
1986	10510952	6568171	16820527	254001	216586	297878	0.828	0.689	0.994	505450	434513	587967

Year	Recruitment (age 0)	Recruitment low	Recruitment high	SSB estimate	SSB low	SSB high	Fbar (2–4)	Fbar low	Fbar high	TSB estimate	TSB low	TSB high
1987	2017577	1259857	3231013	182804	158247	211173	0.937	0.788	1.114	419799	356328	494576
1988	2101038	1315366	3355995	169183	147410	194173	0.900	0.755	1.073	335827	291031	387517
1989	2687092	1689605	4273461	153494	130391	180692	0.895	0.749	1.068	266015	229618	308182
1990	6082830	3889202	9513730	88945	76402	103546	0.854	0.711	1.025	265518	218164	323151
1991	11110962	7762009	15904835	67500	58743	77564	0.926	0.777	1.104	359746	298494	433567
1992	15636548	10920496	22389241	103766	89735	119991	0.803	0.678	0.951	423507	360128	498041
1993	4763128	3387068	6698239	183366	158909	211588	0.804	0.670	0.964	479688	416475	552496
1994	15378716	10640751	22226334	199639	173890	229200	0.775	0.645	0.931	470488	408317	542125
1995	6341308	4467590	9000869	201526	174789	232353	0.688	0.569	0.830	496530	428231	575723
1996	7263481	5126220	10291825	235768	206723	268895	0.809	0.680	0.962	512310	449388	584043
1997	4200931	3007521	5867896	263754	228645	304253	0.699	0.583	0.838	484745	428397	548503
1998	2638446	1888377	3686446	204391	180267	231743	0.746	0.625	0.892	361005	322363	404279
1999	25070740	17177615	36590759	149855	132200	169867	0.922	0.778	1.091	475676	384742	588101
2000	6984281	4902592	9949874	114198	101108	128983	0.845	0.707	1.009	675210	557270	818112
2001	1791681	1252247	2563490	348929	292924	415642	0.472	0.388	0.574	649056	551351	764075
2002	2006632	1418933	2837746	494987	420933	582070	0.338	0.271	0.421	585358	504028	679811
2003	1288148	910018	1823398	386164	327099	455896	0.222	0.176	0.280	425855	364756	497189

Year	Recruitment (age 0)	Recruitment low	Recruitment high	SSB estimate	SSB low	SSB high	Fbar (2–4)	Fbar low	Fbar high	TSB estimate	TSB low	TSB high
2004	1249149	870571	1792356	279454	235678	331362	0.247	0.198	0.308	328051	281490	382312
2005	5562520	3943347	7846539	202333	165688	247083	0.327	0.264	0.404	351900	295045	419711
2006	1746746	1238443	2463676	147848	118143	185022	0.457	0.374	0.559	275198	234194	323381
2007	1143305	798746	1636499	188811	152877	233193	0.418	0.342	0.511	276088	232954	327208
2008	985628	691479	1404905	179601	148232	217608	0.316	0.256	0.390	220920	186908	261121
2009	4980551	3464229	7160579	128774	108372	153015	0.281	0.227	0.347	253072	211358	303019
2010	781331	541551	1127277	133309	112856	157468	0.246	0.200	0.303	226616	195825	262248
2011	462389	316073	676436	208866	179142	243522	0.214	0.172	0.265	271207	234171	314100
2012	931910	631736	1374714	230990	196479	271562	0.213	0.170	0.267	261491	224887	304053
2013	845115	587432	1215833	190726	155801	233478	0.235	0.189	0.292	223540	186653	267716
2014	5972787	4102646	8695409	147435	120942	179731	0.316	0.256	0.390	270989	224532	327059
2015	1440062	1025735	2021750	121527	99937	147780	0.371	0.299	0.461	249267	213103	291568
2016	1837199	1302775	2590854	164668	138215	196183	0.281	0.225	0.351	268050	230664	311497
2017	883243	618554	1261195	173283	144092	208388	0.239	0.191	0.299	238680	204433	278664
2018	2608676	1805660	3768809	153747	130774	180756	0.259	0.207	0.323	227945	196973	263787
2019	8777976	6070886	12692195	125653	107066	147466	0.247	0.198	0.308	357362	292698	436311
2020	7920260	5514518	11375521	165233	141347	193155	0.202	0.162	0.253	489973	414243	579547

Year	Recruitment (age 0)	Recruitment low	Recruitment high	SSB estimate	SSB low	SSB high	Fbar (2–4)	Fbar low	Fbar high	TSB estimate	TSB low	TSB high
2021	2490941	1684406	3683666	301724	259645	350622	0.139	0.111	0.175	614050	531869	708929
2022	6083569	4155571	8906073	569003	488034	663405	0.086	0.068	0.108	815772	708884	938776
2023	1217340	792919	1868940	768243	662977	890223	0.067	0.053	0.084	980301	857320	1120923
2024	1365893	890755	2094475	771472	667271	891945	0.060	0.047	0.077	935037	815223	1072460
2025	2014079*			884516	740996	1055833				970135	813158	1157416

*** Recruitment in 2025 is the geometric mean of resampled recruitment estimates from 2000 to 2024.**

Table 8.6.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Short-term forecast inputs.

SAM forecast	
Starting year:	2024
Number of simulations:	10000
Initial stock size:	Simulated from the estimated distribution at the start of the final data year (including covariances)
Recruitment	Sampled with replacement from 2000 to the final year of catch data
Fbar age range:	2–4
Intermediate year:	$F_{\textit{status quo}}$ (2024). F_{intYr} = 0.060
F and M before spawning:	Taken as 0

2025								
Age	**Mat**	**NM**	**SWt**	**CWt**	**LWt**	**DWt**	**Sel**	**LF**
0	0.000	1.109	0.016	0.040	0.000	0.040	0.002	0.000
1	0.079	1.132	0.056	0.138	0.393	0.137	0.018	0.005
2	0.575	0.500	0.208	0.324	0.500	0.257	0.076	0.277
3	0.951	0.348	0.326	0.415	0.467	0.360	0.160	0.513
4	0.999	0.301	0.459	0.545	0.575	0.413	0.226	0.818
5	1.000	0.276	0.364	0.421	0.417	0.450	0.230	0.884
6	1.000	0.258	0.434	0.509	0.510	0.496	0.137	0.953
7	1.000	0.260	0.605	0.730	0.740	0.614	0.111	0.922
8+	1.000	0.235	0.635	1.011	1.020	0.928	0.040	0.903

2026								
Age	**Mat**	**NM**	**SWt**	**CWt**	**LWt**	**DWt**	**Sel**	**LF**
0	0.000	1.109	0.016	0.040	0.000	0.040	0.002	0.000
1	0.079	1.132	0.056	0.138	0.393	0.137	0.018	0.005
2	0.575	0.500	0.208	0.324	0.500	0.257	0.076	0.277
3	0.951	0.348	0.306	0.389	0.467	0.306	0.160	0.513
4	0.999	0.301	0.408	0.485	0.488	0.470	0.226	0.818
5	1.000	0.276	0.520	0.602	0.615	0.504	0.230	0.884
6	1.000	0.258	0.357	0.418	0.412	0.529	0.137	0.953
7	1.000	0.260	0.449	0.542	0.540	0.567	0.111	0.922
8+	1.000	0.235	0.541	0.862	0.859	0.891	0.040	0.903
2027								
Age	**Mat**	**NM**	**SWt**	**CWt**	**LWt**	**DWt**	**Sel**	**LF**
0	0.000	1.109	0.016	0.040	0.000	0.040	0.002	0.000
1	0.079	1.132	0.056	0.138	0.393	0.137	0.018	0.005
2	0.575	0.500	0.208	0.324	0.500	0.257	0.076	0.277
3	0.951	0.348	0.306	0.389	0.467	0.306	0.160	0.513
4	0.999	0.301	0.390	0.463	0.488	0.352	0.226	0.818
5	1.000	0.276	0.513	0.594	0.596	0.580	0.230	0.884
6	1.000	0.258	0.556	0.652	0.655	0.595	0.137	0.953
7	1.000	0.260	0.351	0.423	0.407	0.607	0.111	0.922
8+	1.000	0.235	0.420	0.668	0.659	0.756	0.040	0.903

Table 8.6.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Short-term forecast output. A number of management options are highlighted.

Basis	Total catch (2026)	Projected landings (2026)	Projected discards and IBC* (2026)	F_{total} (ages 2–4) (2026)	$F_{projected\ landings}$ (ages 2–4) (2026)	$F_{projected\ discards\ and\ IBC}$ (ages 2–4) (2026)	SSB (2027)	% SSB change**	% TAC change ***	% advice change^
ICES advice basis										
MSY approach: F_{MSY}	108301	91316	16985	0.167	0.104	0.063	470142	-23	-3.7	-3.7
Other scenarios										
$F_{MSY\ lower}$	101195	85301	15894	0.155	0.097	0.058	475992	-22	-10.0	-10.0
$F_{MSY\ upper}$	108301	91316	16985	0.167	0.104	0.063	470142	-23	-3.7	-3.7
F = 0	0	0	0	0	0	0	559222	-8.0	-100	-100
F_{pa}	108301	91316	16985	0.167	0.104	0.063	470142	-23	-3.7	-3.7
SSB (2027) = B_{lim}	541852	450688	91164	1.57	0.98	0.59	138250	-77	382	382
SSB (2027) = B_{pa} = MSY $B_{trigger}$	465741	389448	76293	1.14	0.71	0.43	192109	-68	314	314
F = F_{2024}	41299	34843	6456	0.060	0.037	0.023	525491	-13.6	-63	-63

*** Including BMS landings, assuming recent discard rate.**

**** SSB 2027 relative to SSB 2026.**

***** Total advised catch in 2026 relative to TAC in 2025: Subdivision 20 (5892 tonnes) + Subarea 4 (95862 tonnes) + Division 6.a (10681 tonnes) = 112435 tonnes.**

^ Total catch 2026 relative to the advice value 2025 (112 435 t).

Table 8.6.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Ratio of numbers-at-age from stock assessment and forecast results from WGNSSK 2024 and WGNSSK 2025.

Age	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024*	2025**
0	1.28	1.02	1.08	1.08	1.01	0.95	0.75	0.65	0.91	1.16	1.04	1.53
1	1.10	1.21	1.00	1.15	1.07	1.24	1.06	0.80	0.74	0.99	1.42	1.02
2	1.23	0.99	1.14	1.10	1.07	1.06	1.24	1.22	1.07	0.99	1.40	1.43
3	0.96	1.20	0.98	1.26	1.15	1.07	1.16	1.32	1.39	1.30	1.45	1.41
4	1.16	0.97	1.23	1.05	1.53	1.25	1.32	1.28	1.51	1.42	1.50	1.50
5	1.66	1.26	1.10	1.28	1.11	1.82	1.46	1.43	1.32	1.48	1.21	1.56
6	1.27	1.73	1.19	1.24	1.43	1.35	2.09	1.49	1.47	1.37	1.17	1.27
7	1.40	1.50	2.00	1.40	1.50	1.80	1.60	2.40	1.60	1.50	1.10	1.20
8+	4.20	3.90	4.50	4.20	4.80	5.00	5.30	4.20	4.10	3.90	1.20	1.20

*** 2024 was the intermediate year at WGNSSK 2024.**

**** 2025 was the advice year at WGNSSK 2024 and is the intermediate year at WGNSSK 2025.**

Table 8.6.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Ratio of biomass-at-age from stock assessment and forecast results from WGNSSK 2024 and WGNSSK 2025.

Age	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024*	2025**
0	1.28	1.02	1.08	1.08	1.01	0.95	0.75	0.65	0.57	1.16	0.98	1.12
1	1.10	1.21	1.00	1.15	1.07	1.24	1.06	0.75	0.77	0.96	1.47	1.00
2	1.23	0.99	1.14	1.10	1.07	1.06	1.24	1.20	1.06	0.96	1.48	1.46
3	0.96	1.20	0.98	1.26	1.15	1.07	1.16	1.31	1.39	1.29	1.46	1.47
4	1.16	0.97	1.23	1.05	1.53	1.25	1.32	1.28	1.51	1.42	1.29	1.55
5	1.66	1.26	1.10	1.28	1.11	1.82	1.46	1.43	1.31	1.48	1.10	1.40
6	1.27	1.73	1.19	1.24	1.43	1.35	2.09	1.49	1.47	1.37	1.10	1.19
7	1.40	1.50	2.00	1.40	1.50	1.80	1.60	2.40	1.60	1.50	1.50	1.20
8+	4.20	3.90	4.50	4.20	4.80	5.00	5.30	4.20	4.10	3.90	2.10	1.50

*** 2024 was the intermediate year at WGNSSK 2024**

**** 2025 was the advice year at WGNSSK 2024 and is the intermediate year at WGNSSK 2025**

Table 8.6.5. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Summary of forecast assumptions from current assessment (WGNSSK 2025) and previous assessment (WGNSSK 2024).

	Year	Current assessment (2025)	Previous assessment (2024)
Assumed recruitment	2024	1 365 893	1 314 060
	2025	2 014 079	1 314 060
Catch	2024	60470	66 309
F	2024	0.060	0.084
Target F for TAC	2025	0.060	0.174

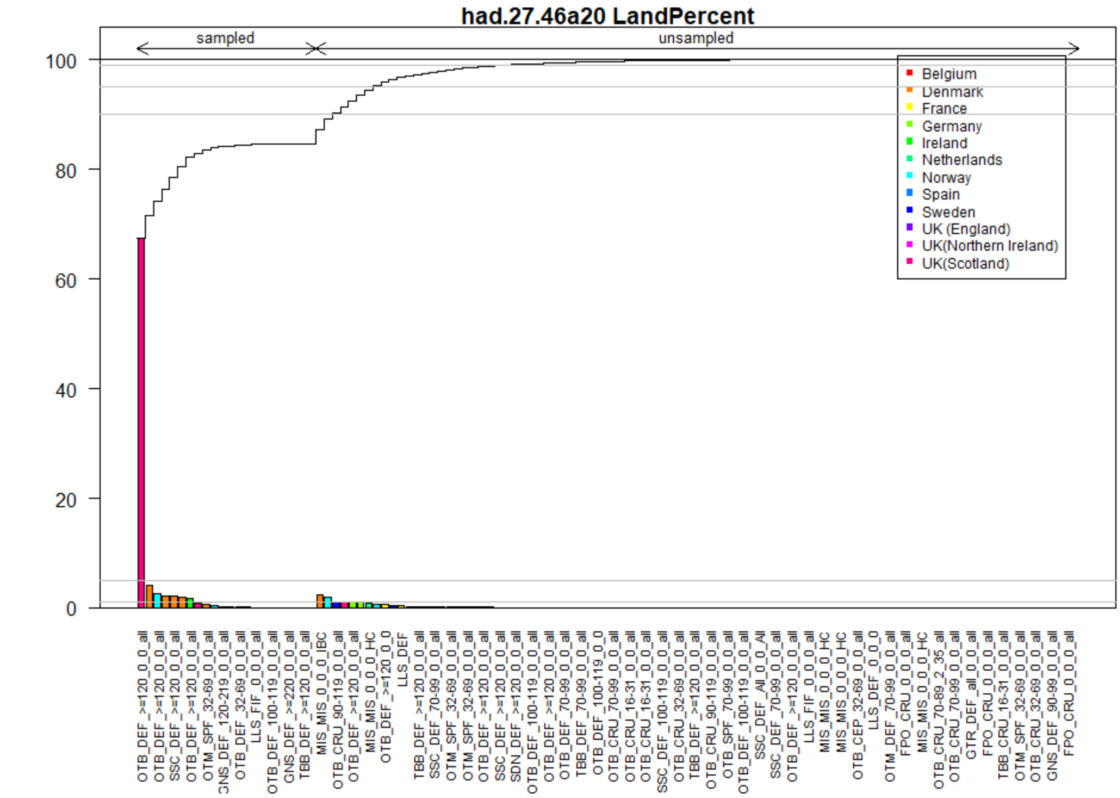


Figure 8.2.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Reported landings for each sampled and unsampled fleet in the full stock area, along with cumulative landings for fleets in descending order of yield.

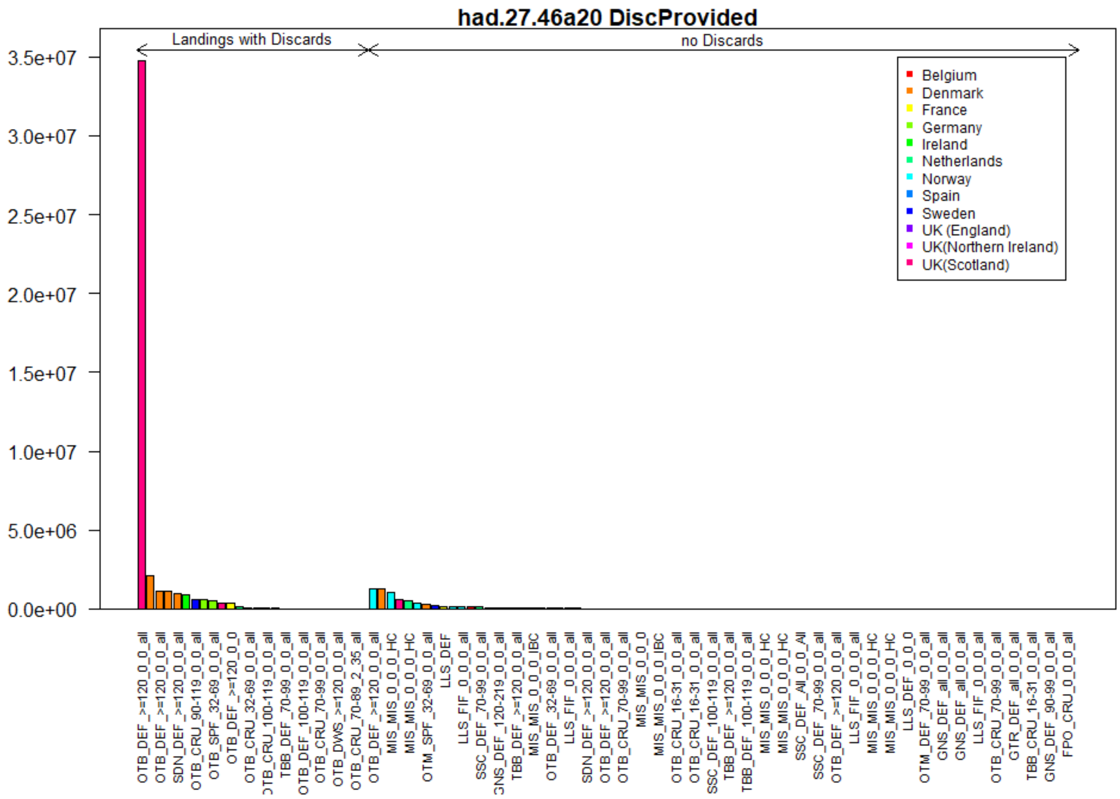


Figure 8.2.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Summary of landings for fleets with and without discard estimates.

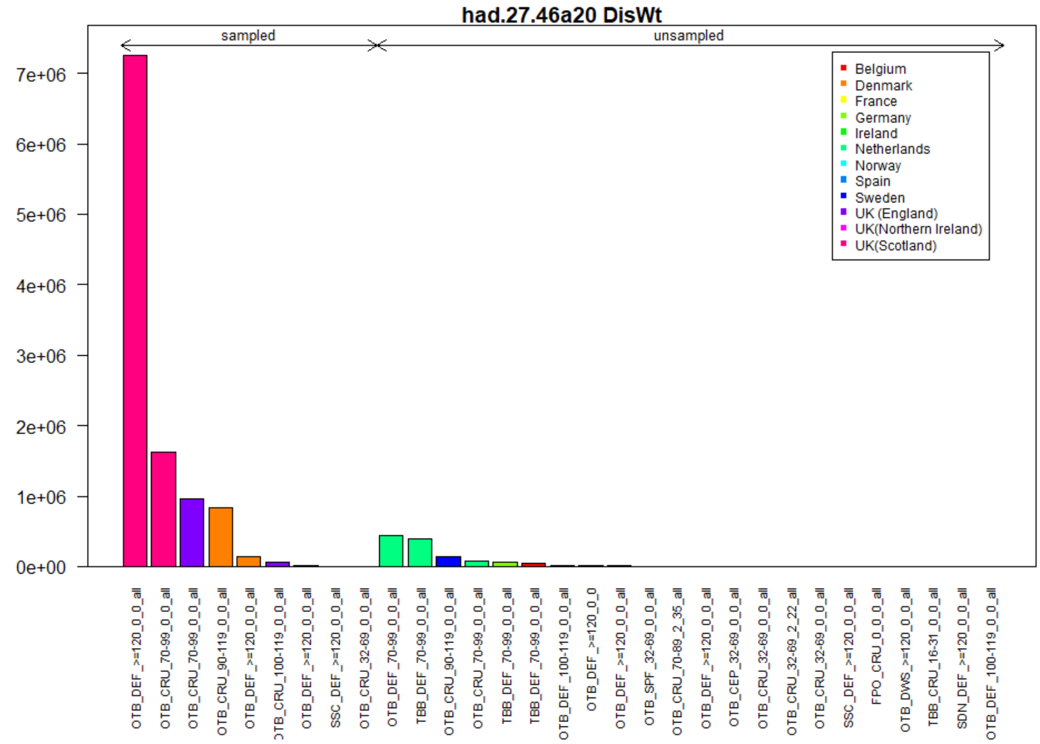


Figure 8.2.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Reported discards for each sampled and unsampled fleet in the full stock area, in descending order of yield.

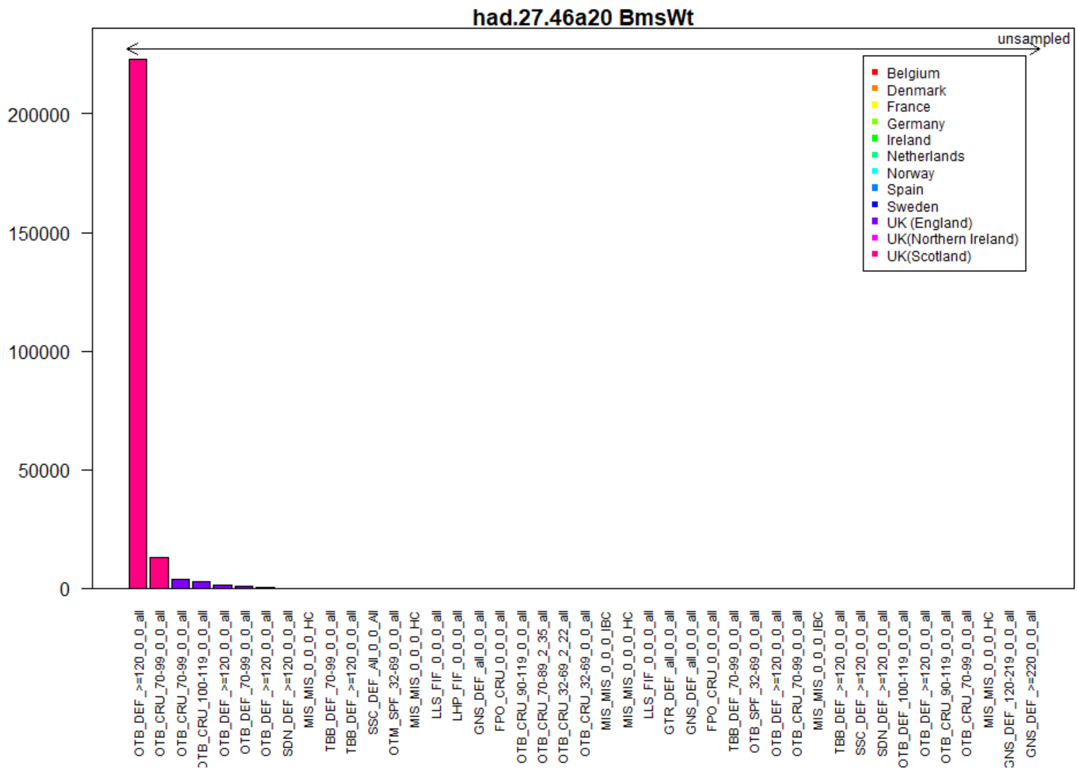


Figure 8.2.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Reported BMS landings for each sampled and unsampled fleet in the full stock area, in descending order of yield.

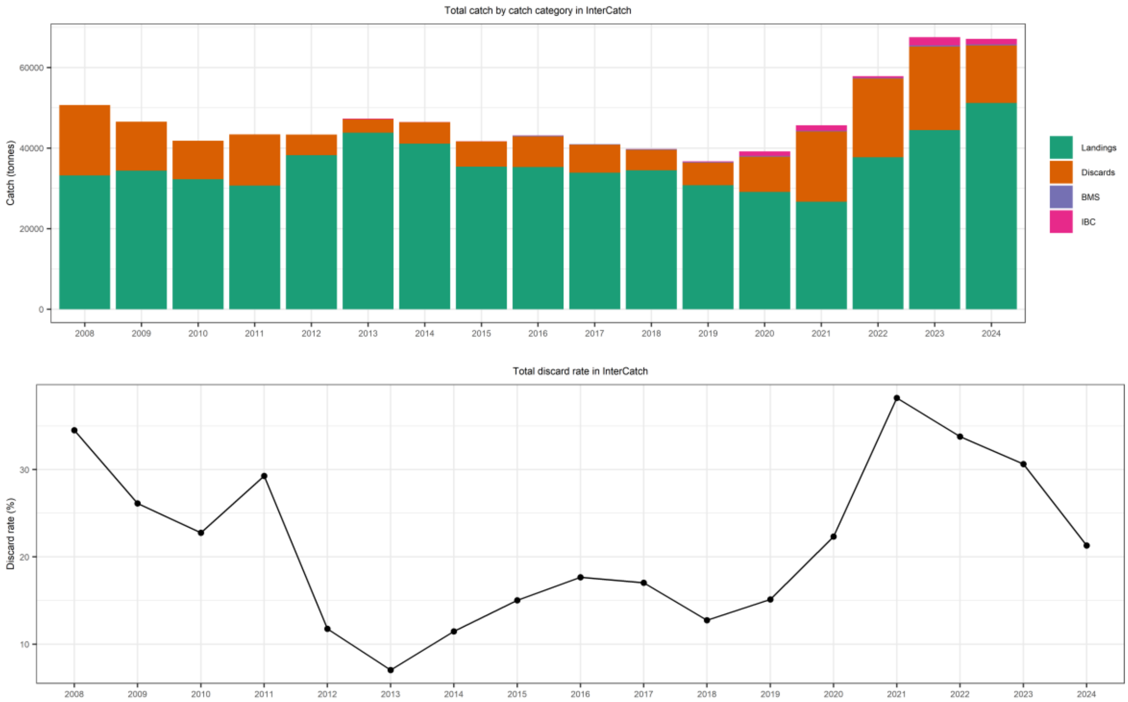


Figure 8.2.5. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Yield by catch component and discard rate.

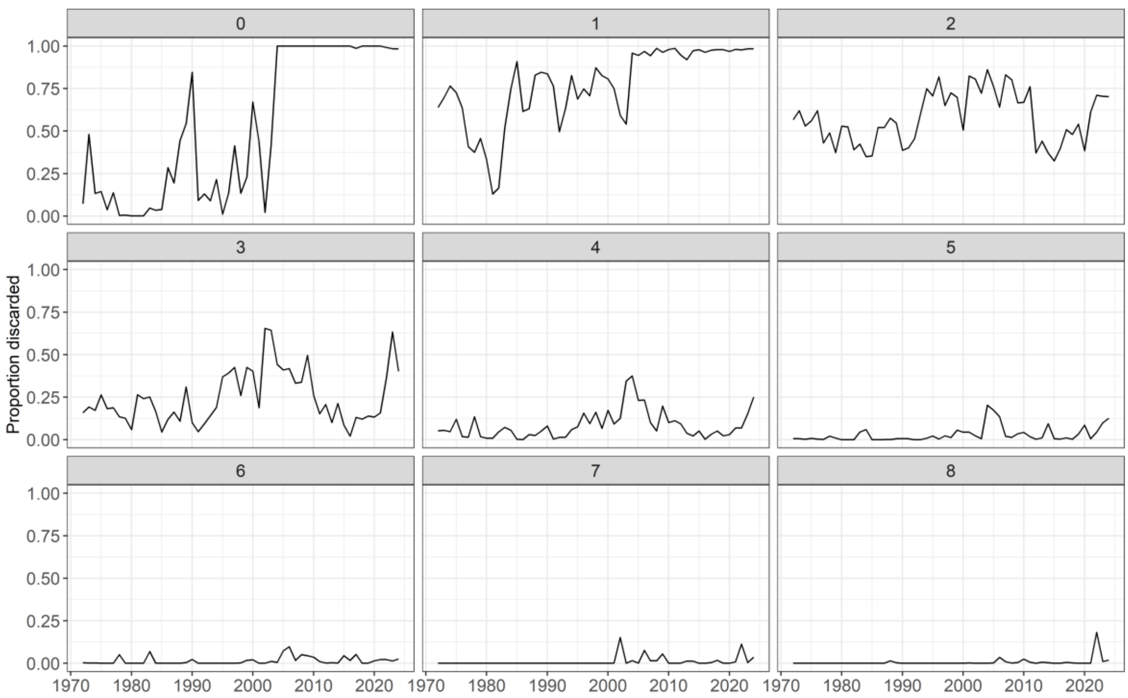


Figure 8.2.6. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Proportion of total catch discarded, by age and year.

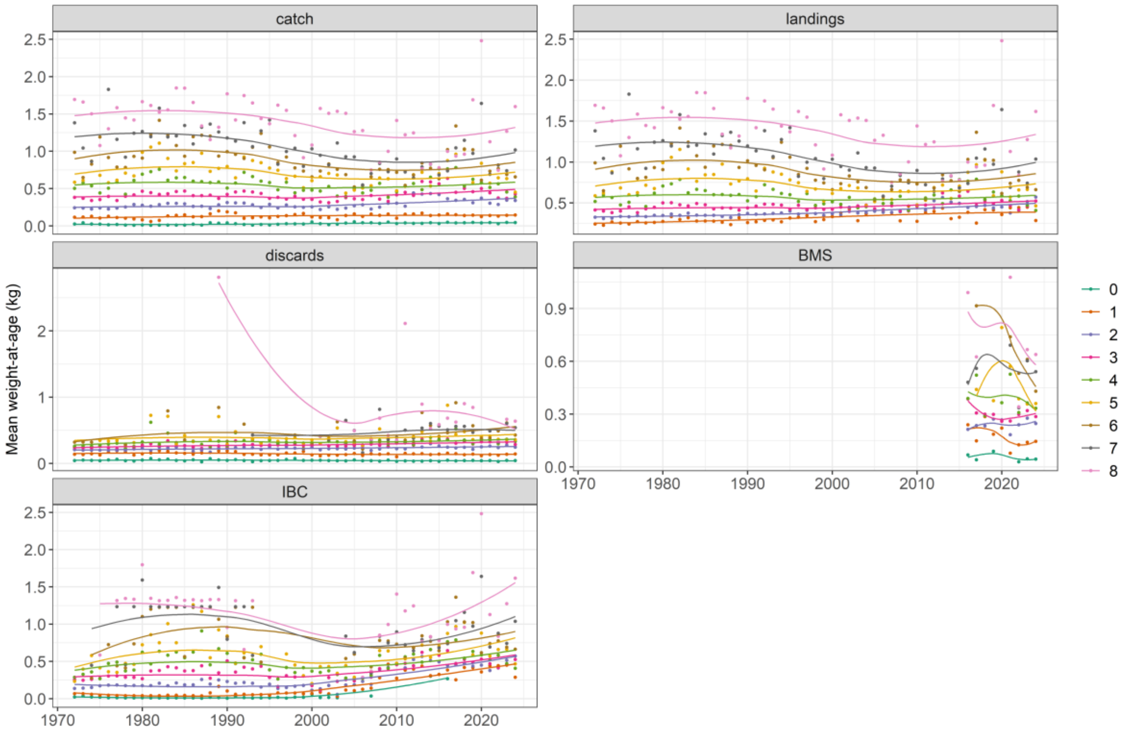


Figure 8.2.7. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weights-at-age (0–8+; kg) by catch component. Points indicate the data points. The solid lines give loess smoothers through each time-series of mean weights-at-age, to show underlying trends.

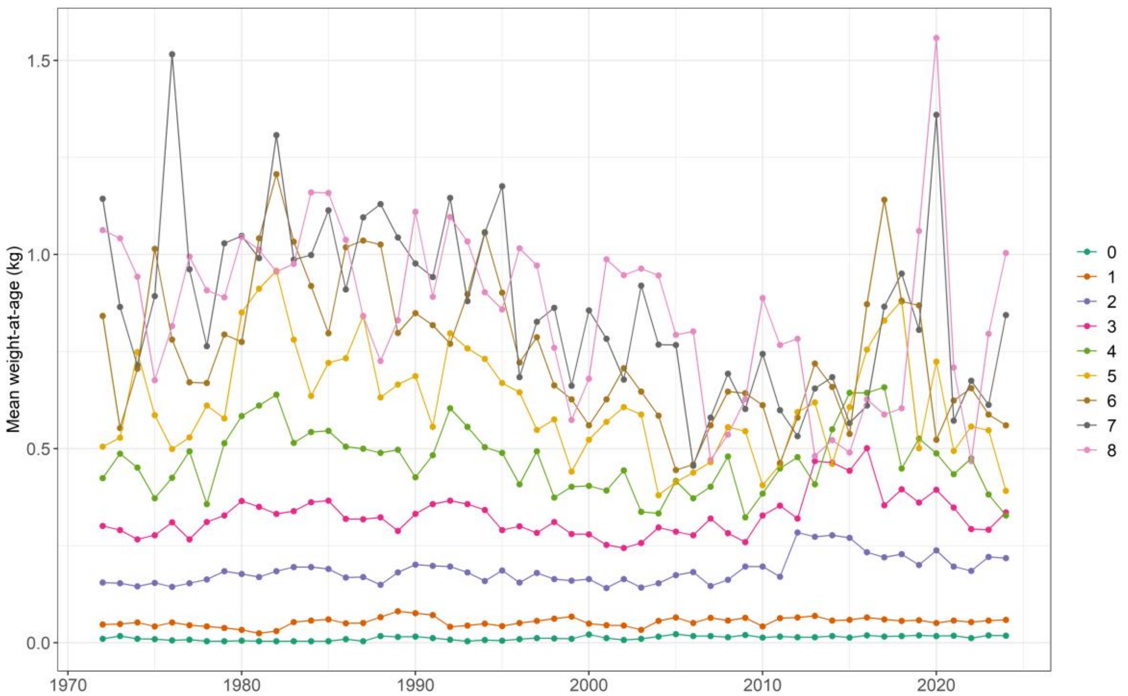


Figure 8.2.8. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weights-at-age (kg) for the stock.

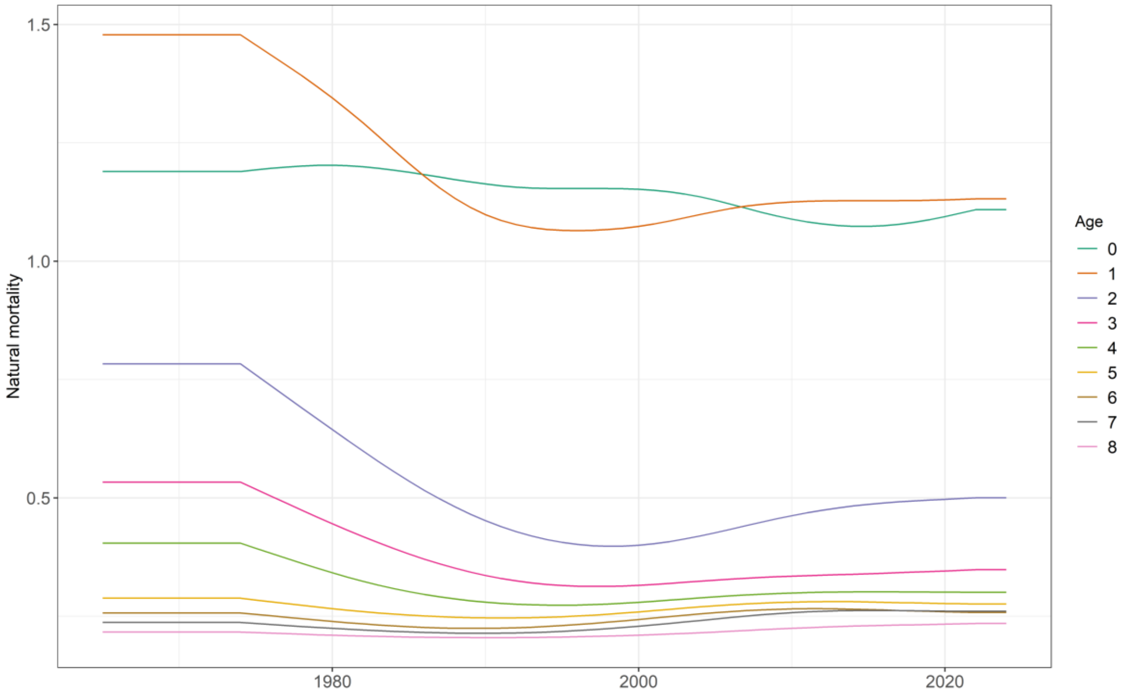


Figure 8.2.9. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Time-series of estimated natural mortality-at-age (0–8+), from ICES WGSAM (SMS 2023; ICES, 2024).

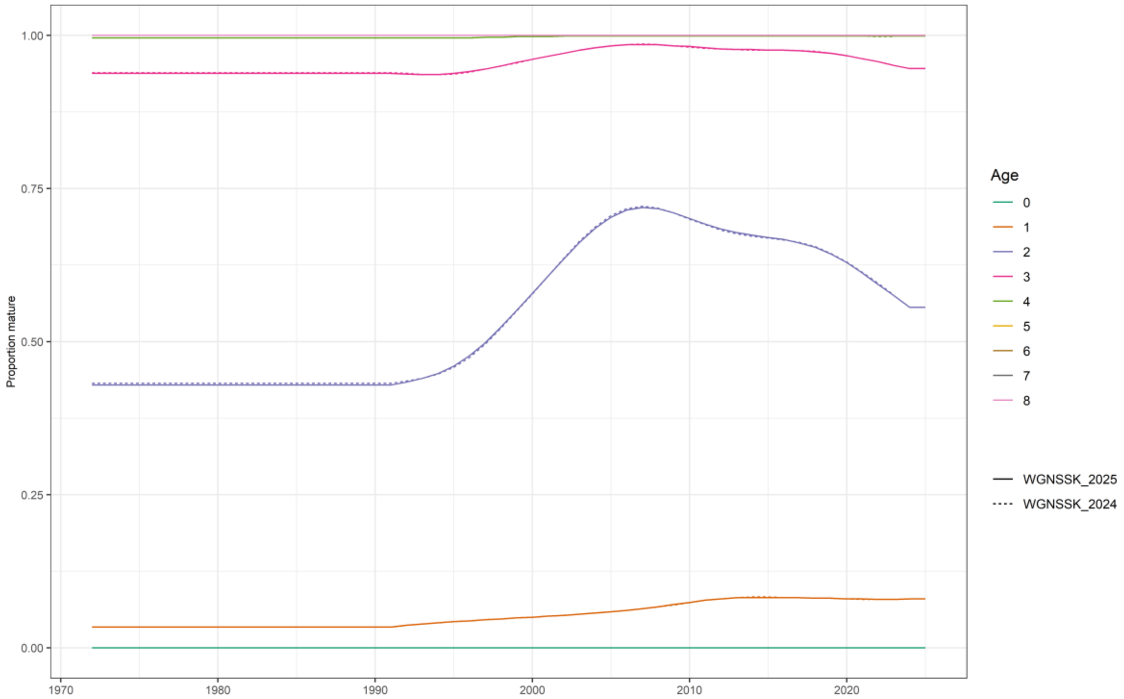


Figure 8.2.10. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Time-series of estimated maturity-at-age (0–8+) used at WGNSSK 2025 compared to those used at WGNSSK 2024.

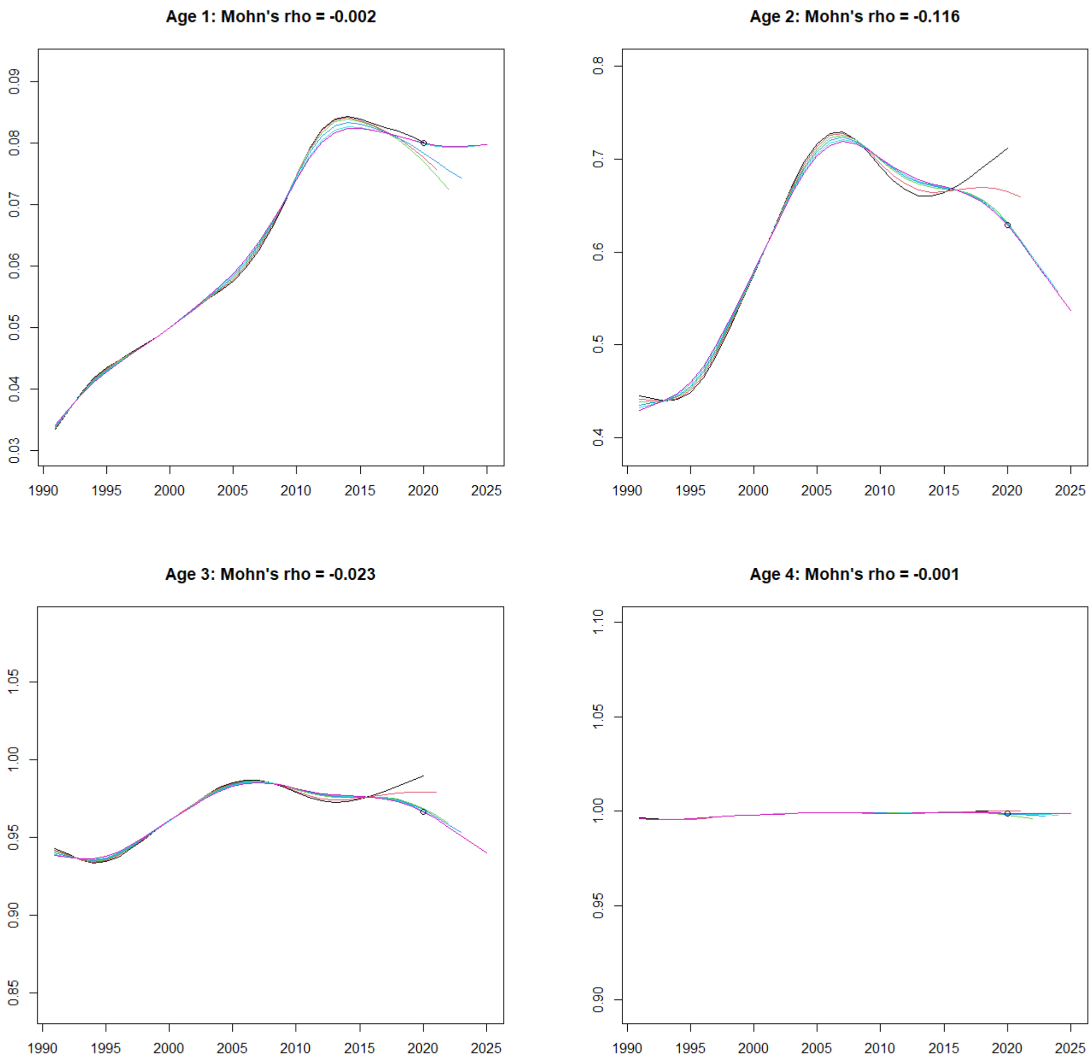


Figure 8.2.11. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Retrospective analysis of maturity time-series for ages 1–4.

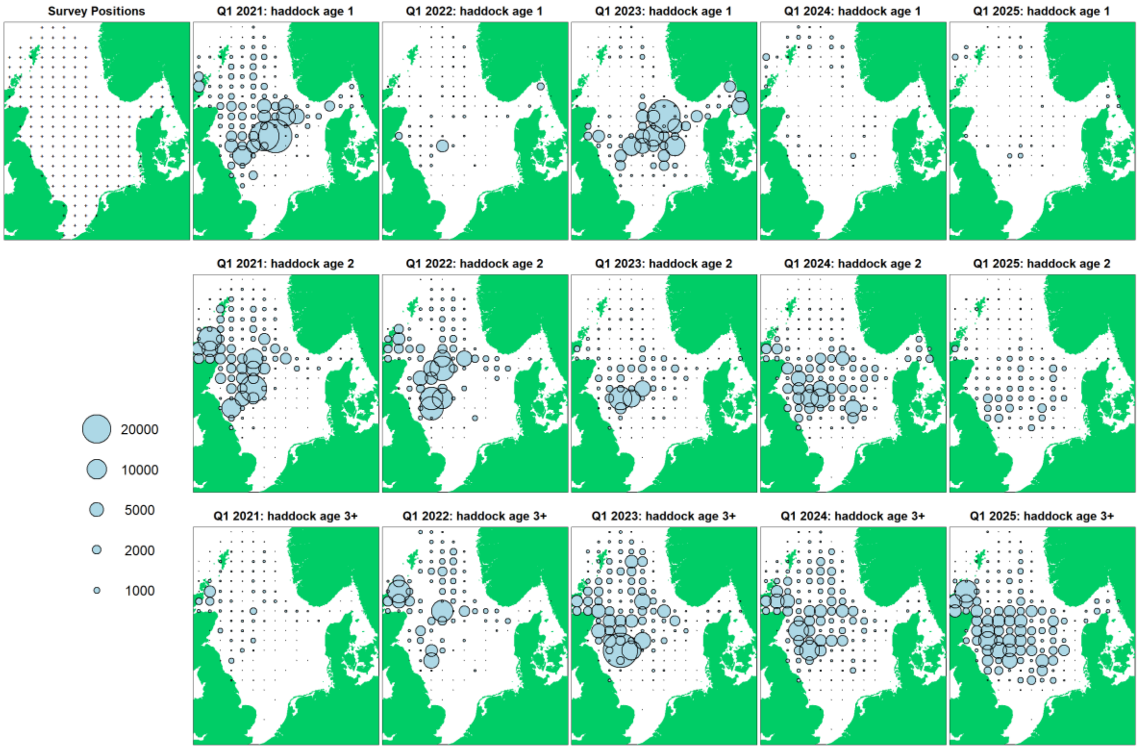


Figure 8.2.12. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Survey distributions by age for the international IBTS Q1 survey (North Sea).

Figure 8.2.13. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Survey distributions by age for the international IBTS Q3 survey (North Sea).

Figure 8.2.14. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Delta-GAM survey indices for NS-WC Q1 and NS-WC Q3+Q4.

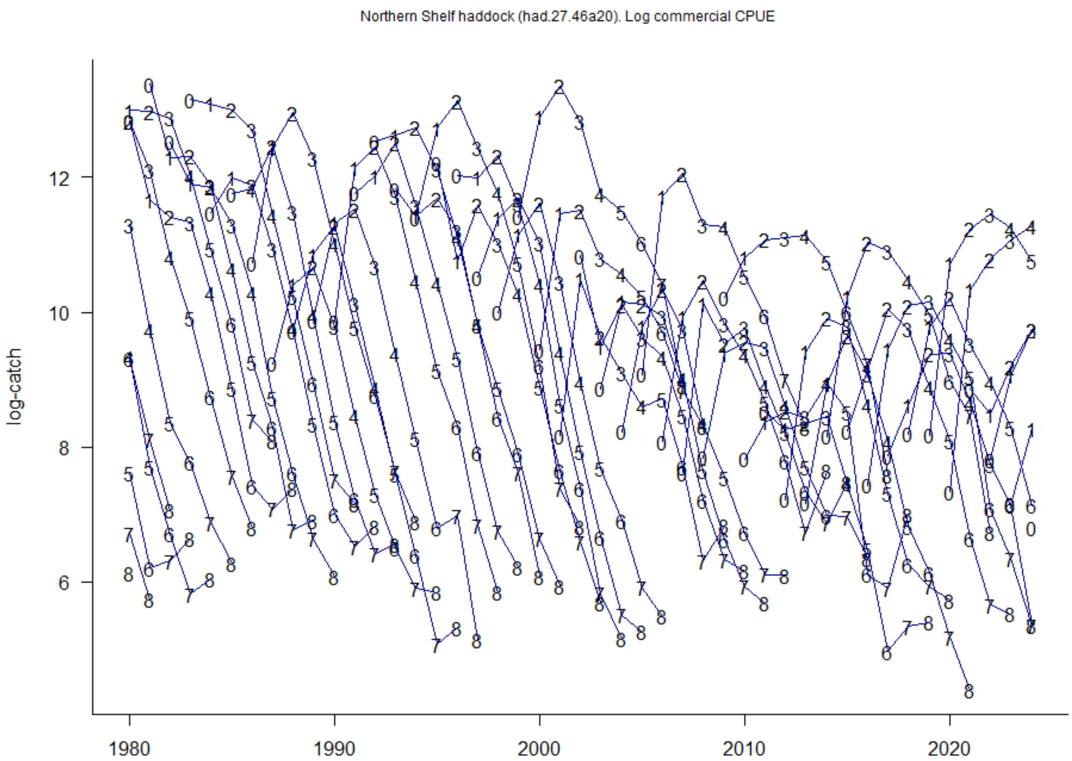


Figure 8.3.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Log catch curves by cohort for total catches.

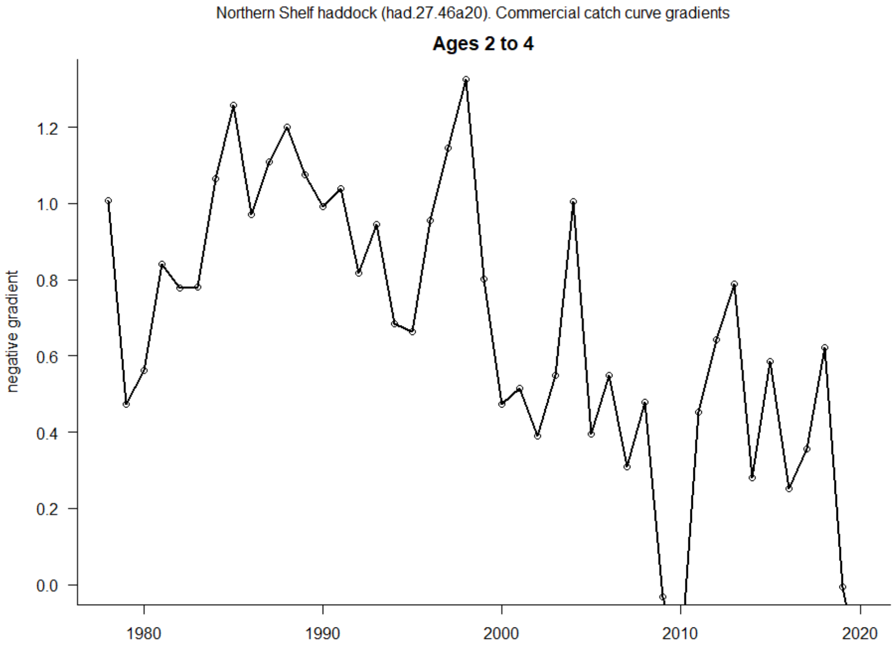


Figure 8.3.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Negative gradients of log catches per cohort, averaged over ages 2–4. The x-axis represents the spawning year of each cohort.

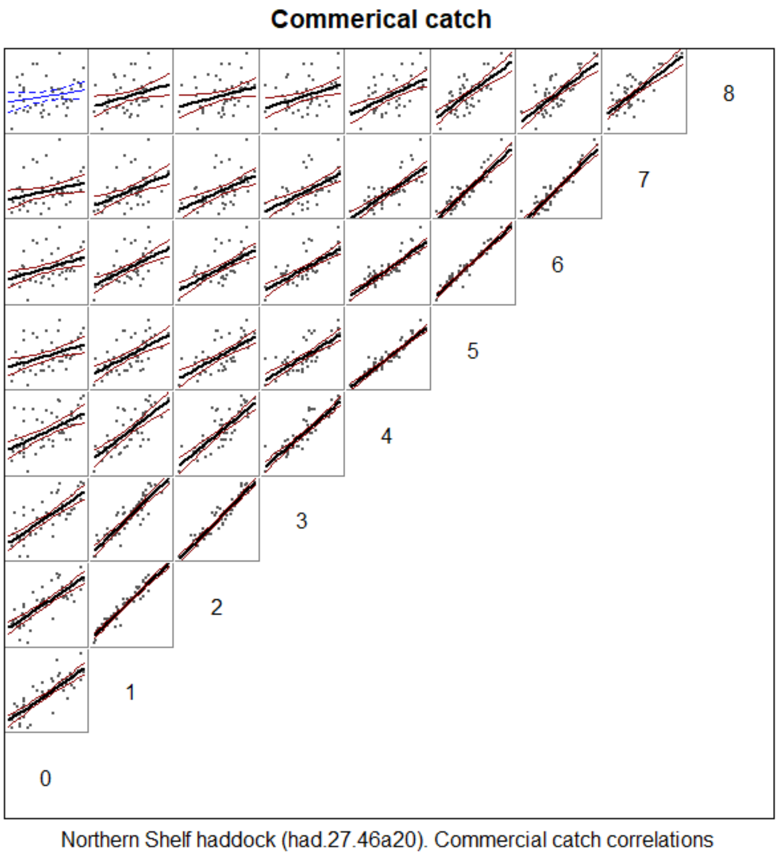


Figure 8.3.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Correlations in the catch-at-age matrix (including the plus-group for ages 8), comparing estimates at different ages for the same year classes (cohorts). In each plot, the straight line is a normal linear model fit: a thick line (and black points) represents a significant ($p < 0.05$) regression, while a thin line (and blue points) is not significant. Approximate 95% confidence intervals for each fit are also shown.

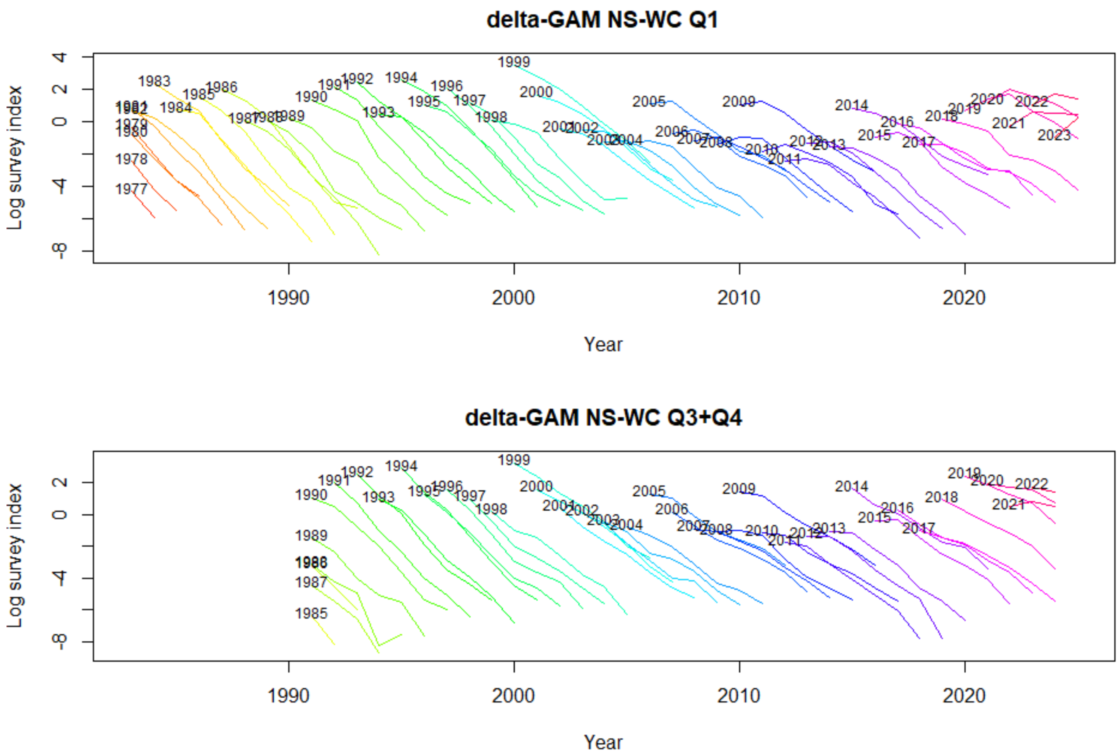


Figure 8.3.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Log abundance indices by cohort (survey “catch curves”) for each of the survey indices.

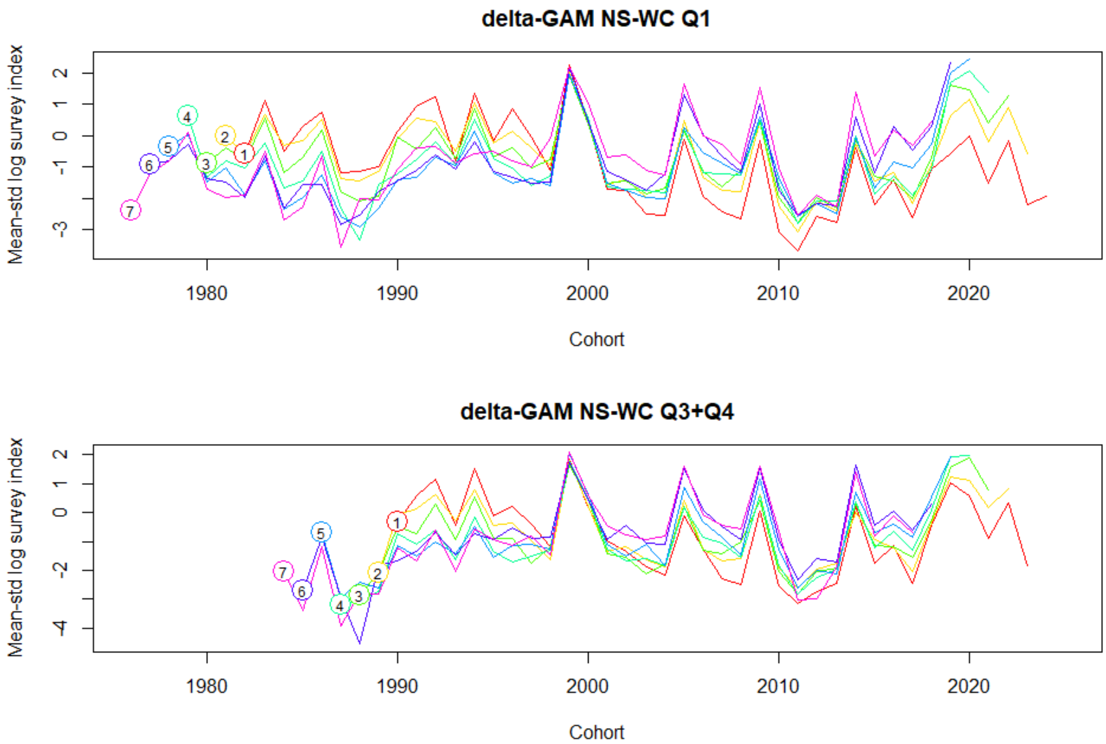
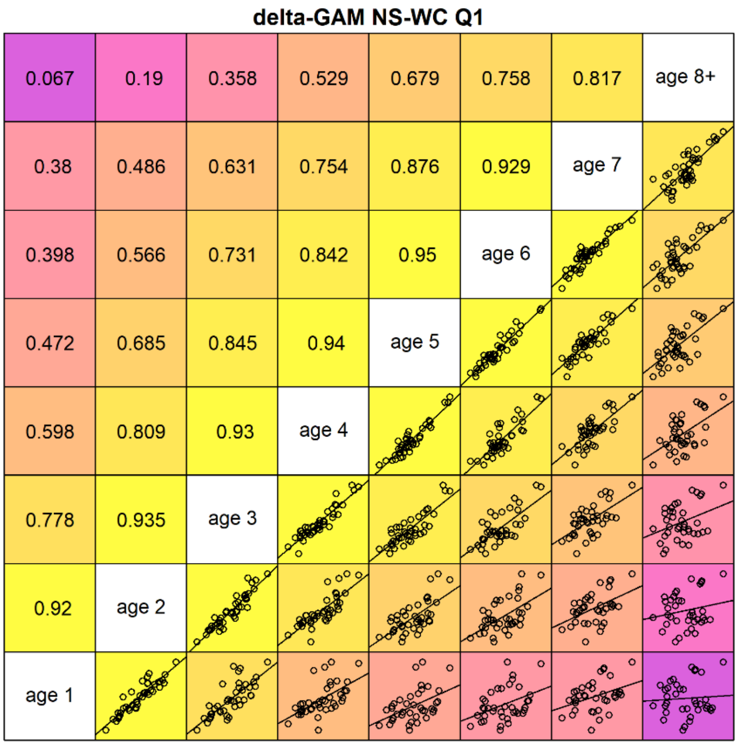


Figure 8.3.5. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean-standardized log abundance indices by age and cohort for each of the survey indices. The age represented by each line is indicated by a circled number at the start of the line.



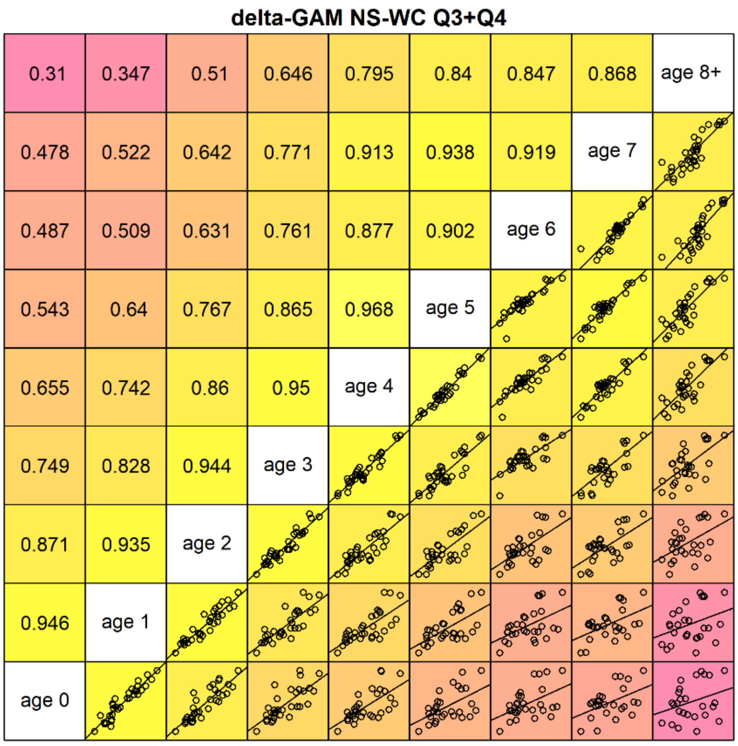


Figure 8.3.6. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Within-survey correlations for the delta-GAM NS-WC Q1 (upper) and Q3+Q4 (lower) survey series, comparing index values at different ages for the same year classes (cohorts). In each plot, the straight line is a normal linear model fit: a thick line (with black points) represents a significant ($p < 0.05$) regression, while a thin line (with blue points) is not significant. Approximate 95% confidence intervals for each fit are also shown.

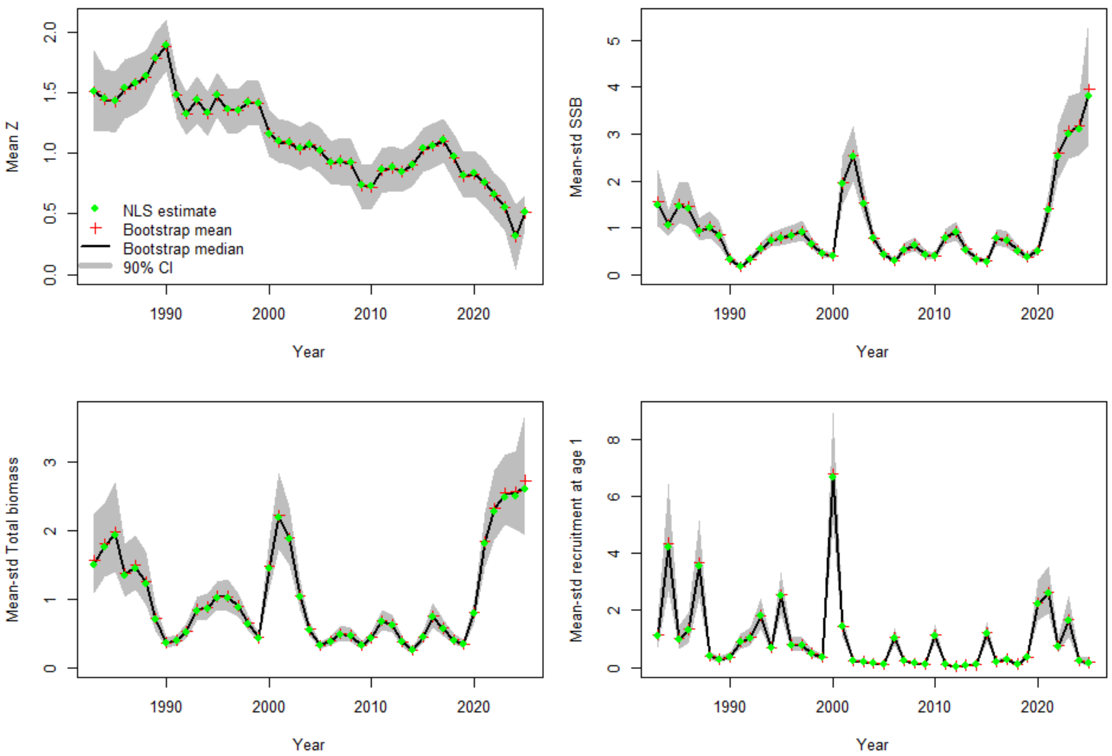


Figure 8.3.7. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Summary plots from an exploratory SURBAR assessment, using both available survey-derived indices (delta-GAM NS-WC Q1 and delta-GAM NS-WC Q3+Q4). Mean mortality Z (ages 2 to 4), relative spawning-stock biomass (SSB), relative total biomass (TSB), and relative recruitment. Shaded grey areas correspond to the 90% CI. Green points give the model estimates, while red crosses and black lines give (respectively) the mean and median values from the uncertainty estimation bootstrap.

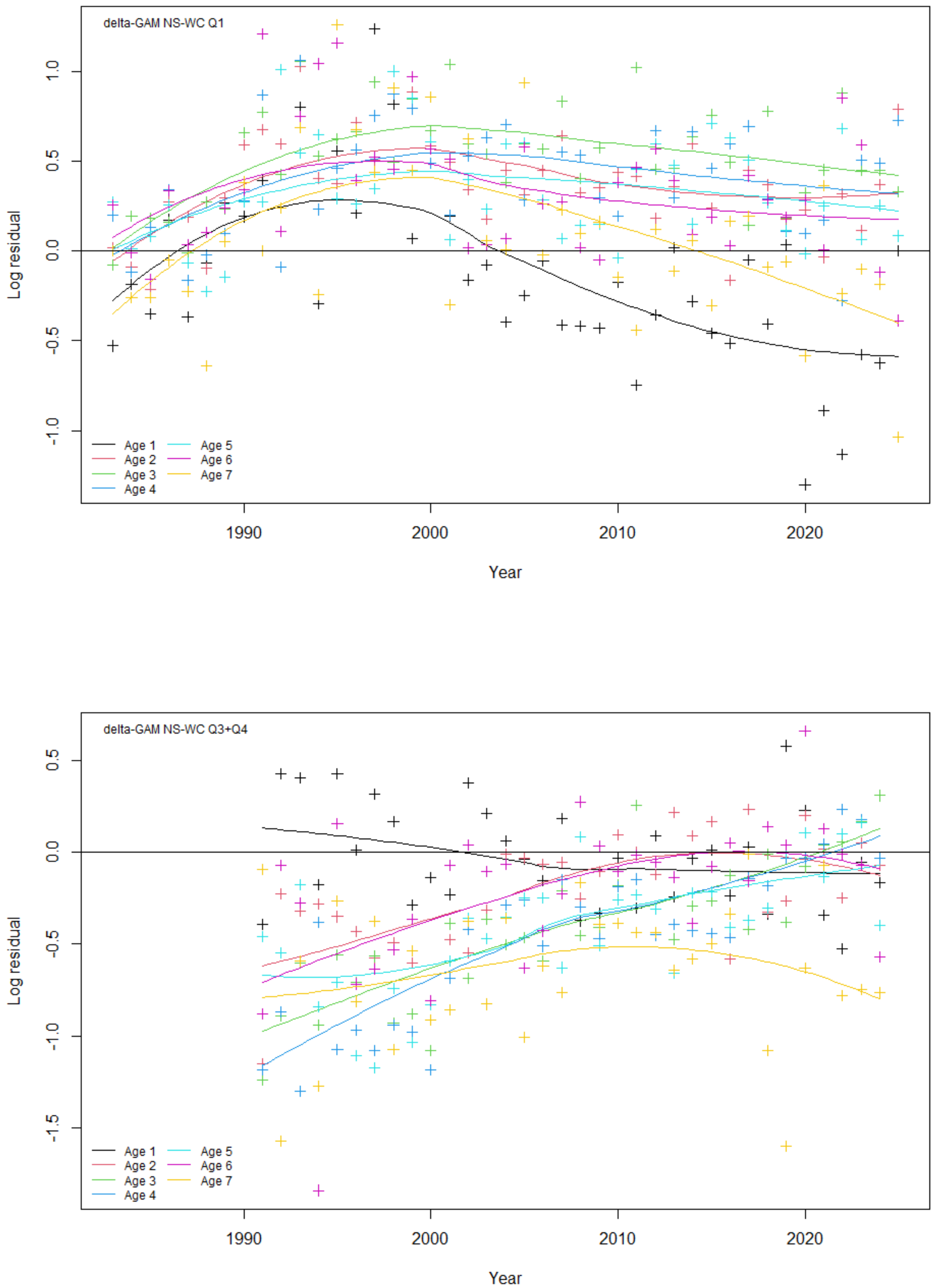


Figure 8.3.8. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Log residuals by age from an exploratory SURBAR assessment, using both available survey-derived indices (delta-GAM NS-WC Q1 and delta-GAM NS-WC Q3+Q4).

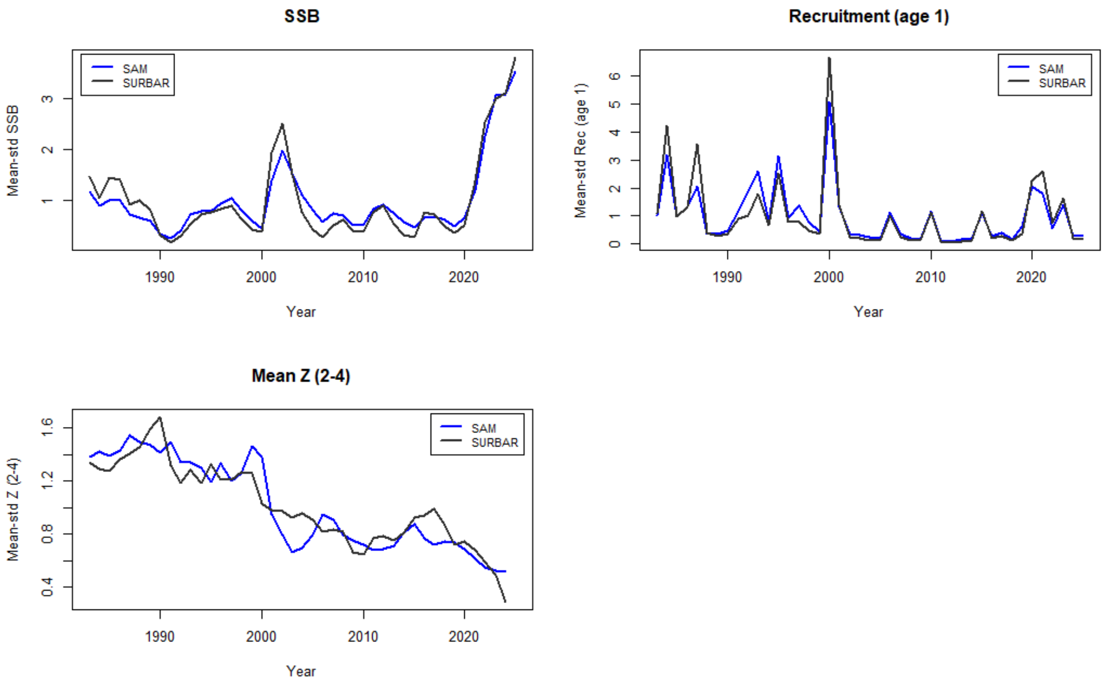


Figure 8.3.9. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparisons of mean-standardized stock summary estimates from SAM (blue) and SURBAR (black) models.

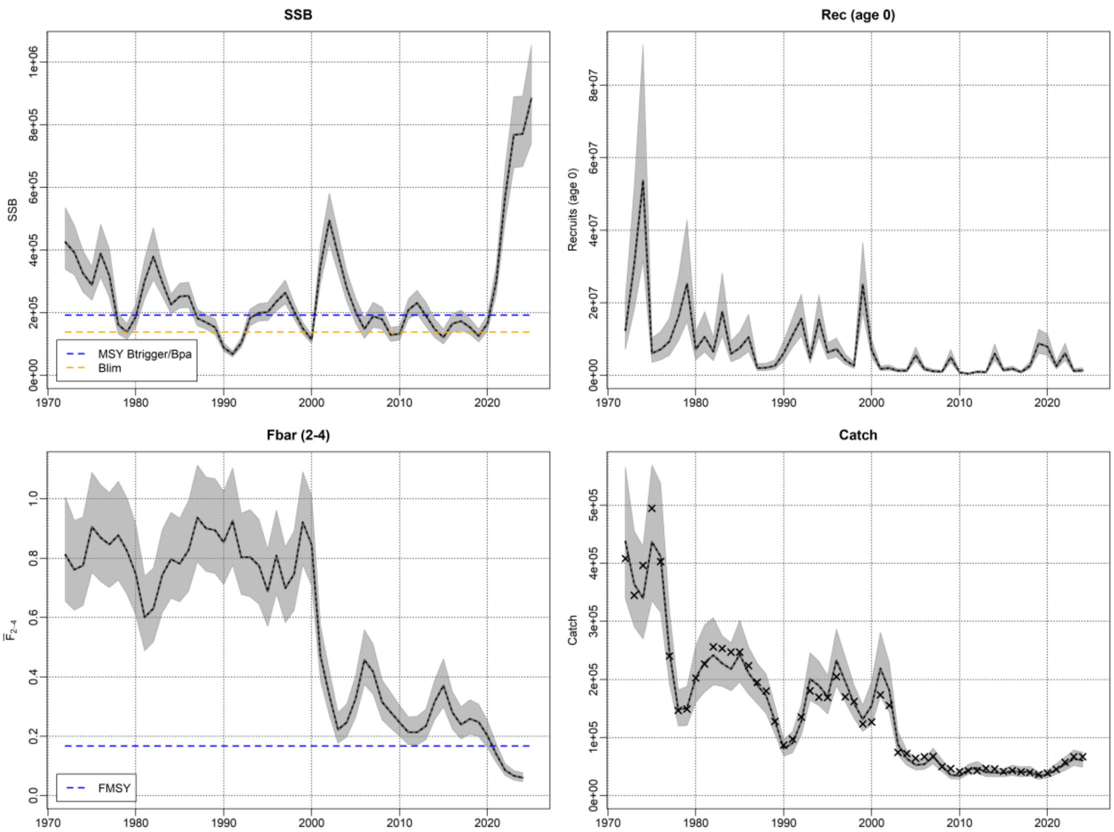


Figure 8.3.10. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Stock summary from final SAM assessment. Black lines give the assessment estimates for spawning-stock biomass (SSB; tonnes), fishing mortality (Fbar, ages 2–4), recruitment (age 0; thousands) and total catch (tonnes). Grey shading indicates the pointwise 95% confidence intervals, and black crosses give observed values for total catch. Horizontal dashed lines indicate various reference points in relation to SSB (B_{lim} in orange, MSY $B_{trigger}$ in blue) or fishing mortality (F_{MSY} in blue) as calculated at WGNSSK 2025 (see section 8.8).

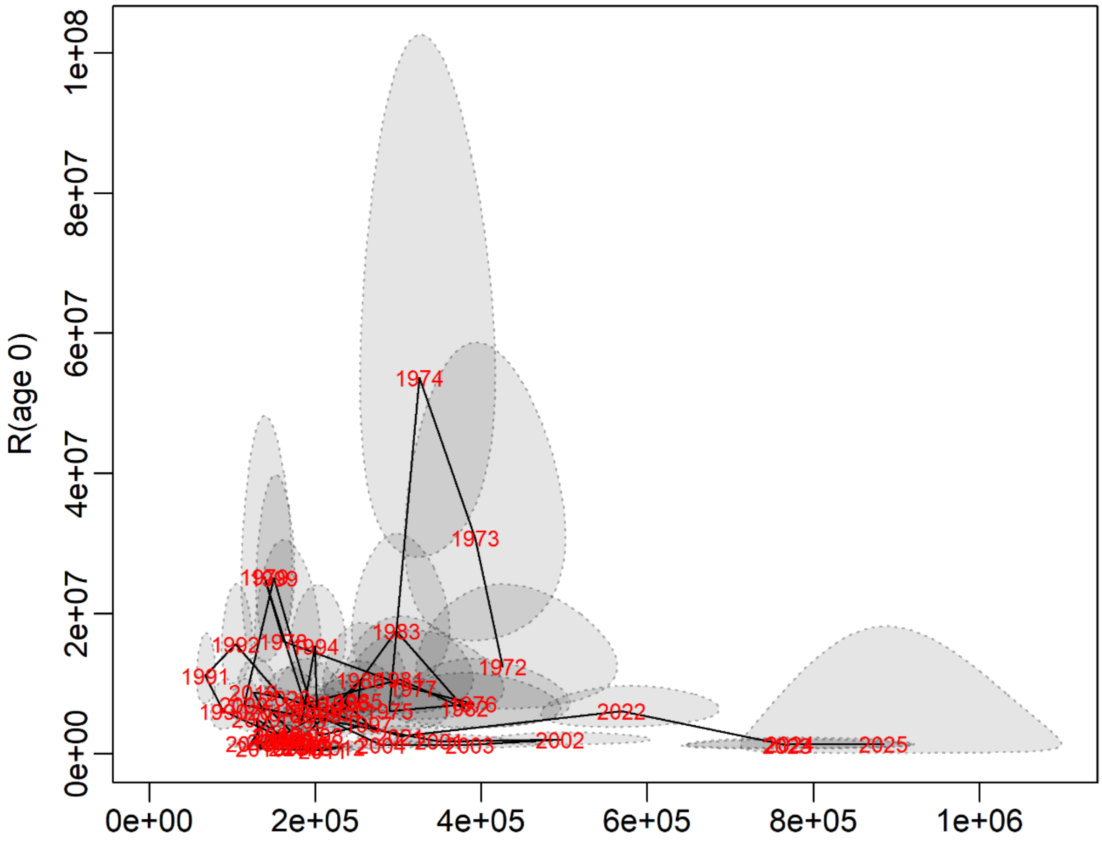


Figure 8.3.11. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Stock–recruitment estimates from the final SAM assessment. Point estimates are labelled by year class while the grey shaded areas indicate the 95% confidence intervals for each estimate.

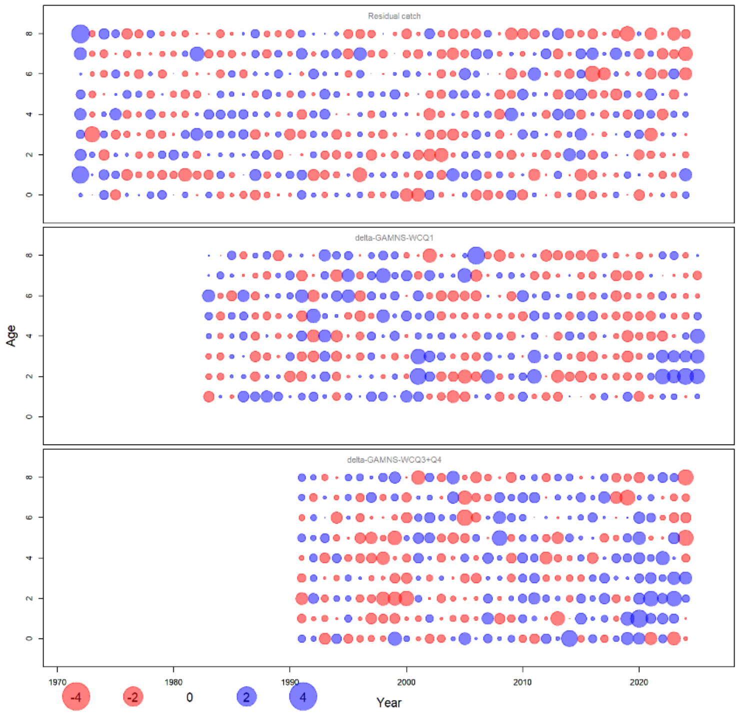


Figure 8.3.12. Haddock in Subarea 4, Division 6.a, and Subdivision 20. One-observation-ahead residuals for commercial catch, delta-GAM NS-WC Q1 and delta-GAM NS-CS Q3+Q4 from the final SAM assessment.

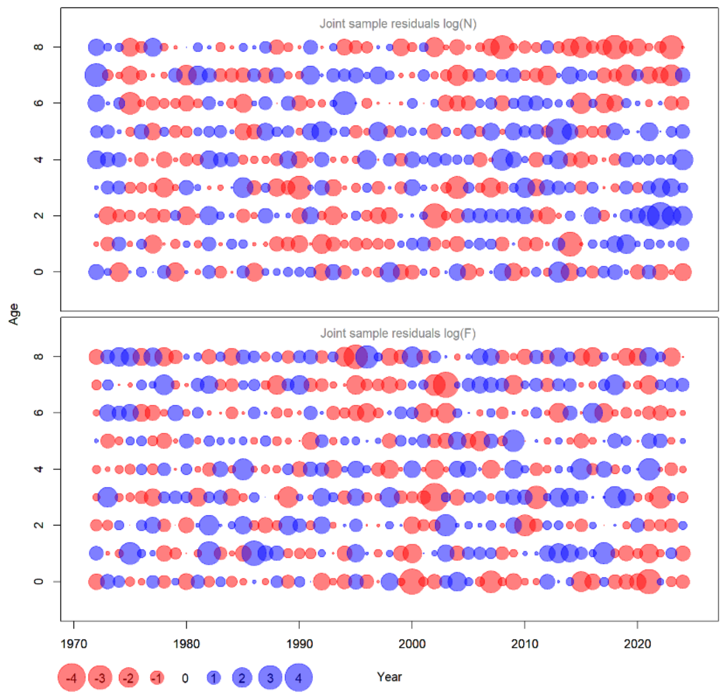


Figure 8.3.13. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Residuals for the N-process (survival) and F-process (fishing mortality) from the final SAM assessment.

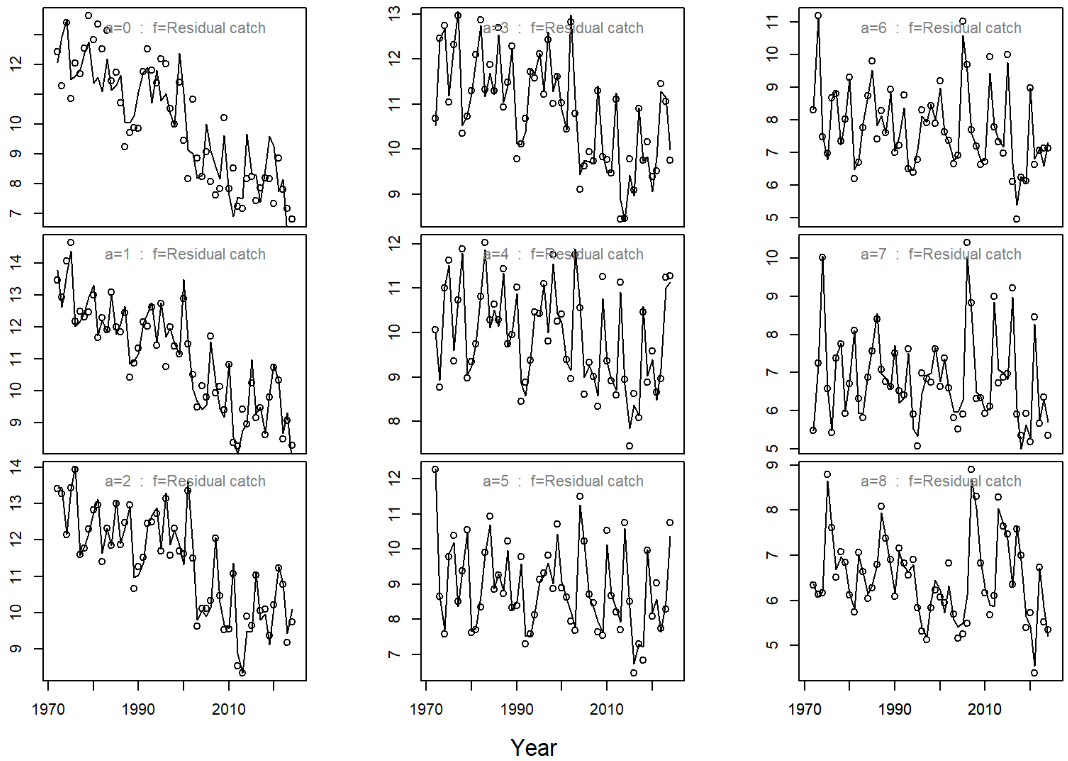


Figure 8.3.14. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Time-series of observed (points) and fitted (lines) values for total catch, by age.

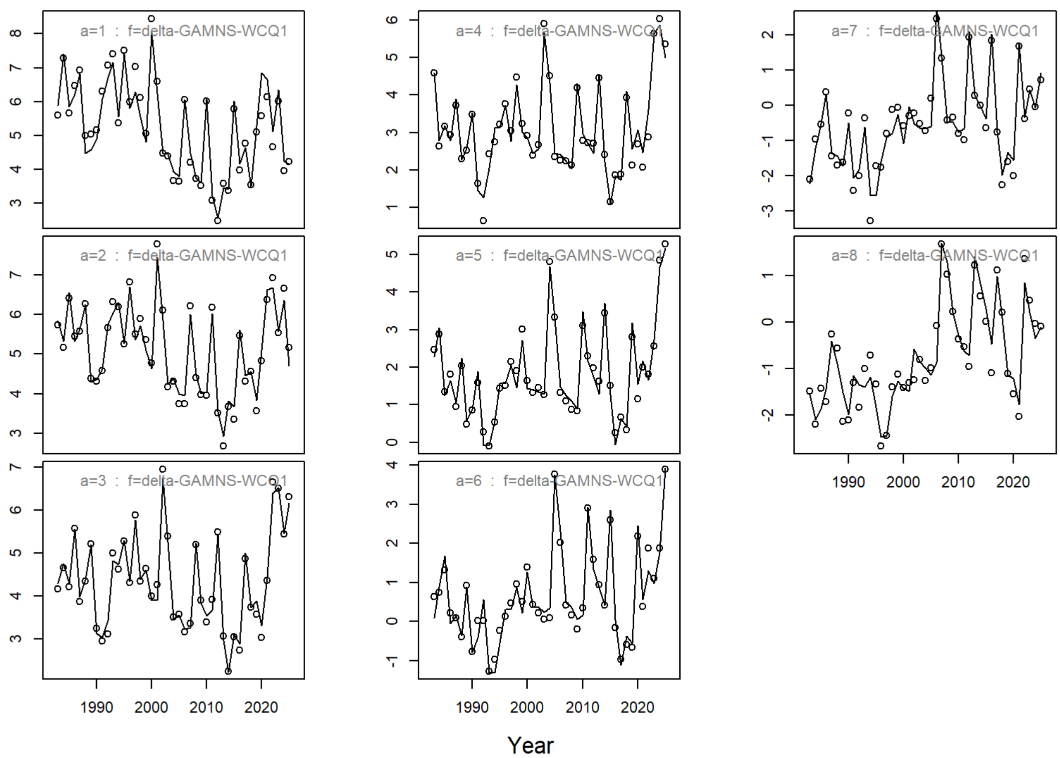


Figure 8.3.15. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Time-series of observed (points) and fitted (lines) values for the delta-GAM NS-WC Q1 survey index, by age.

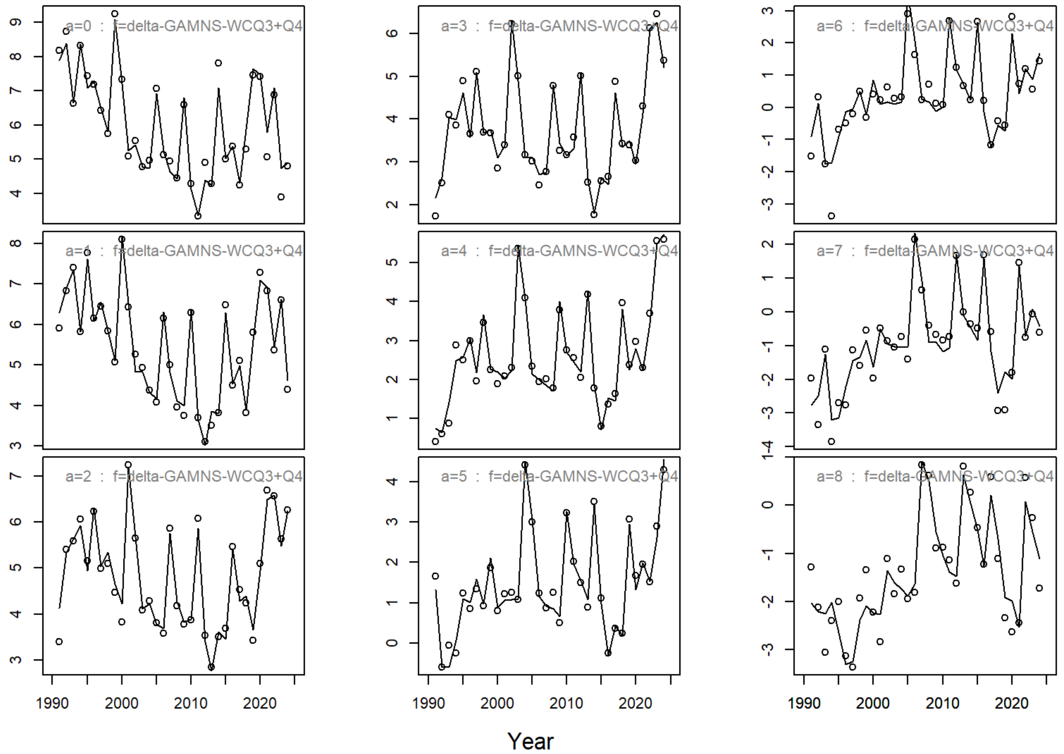


Figure 8.3.16. Haddock in Subarea 4, Division 6.a, and Subdivision 20 Time-series of observed (points) and fitted (lines) values for the delta-GAM NS-WC Q3+Q4 survey index, by age.

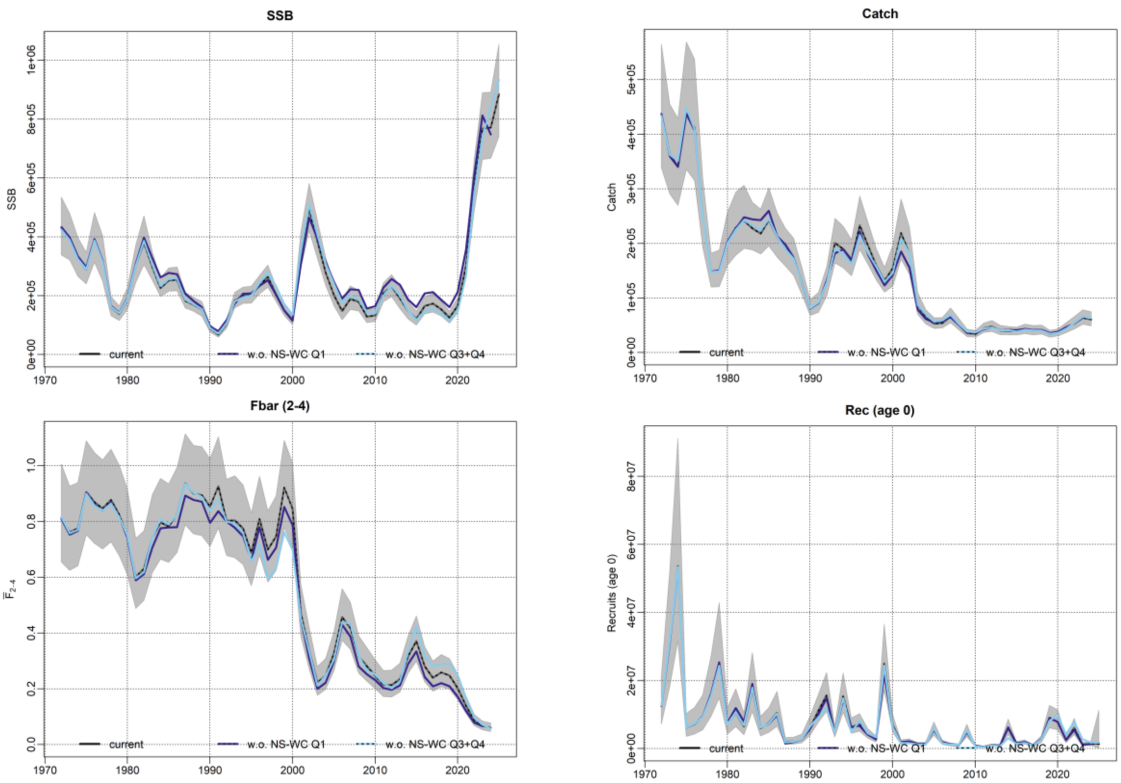


Figure 8.3.17. Haddock in Subarea 4, Division 6.a, and Subdivision 20 Leave one out analysis results for spawning-stock biomass (SSB; tonnes), fishing mortality (Fbar, ages 2–4), recruitment (age 0; thousands) and total catch (tonnes) from the final SAM assessment.

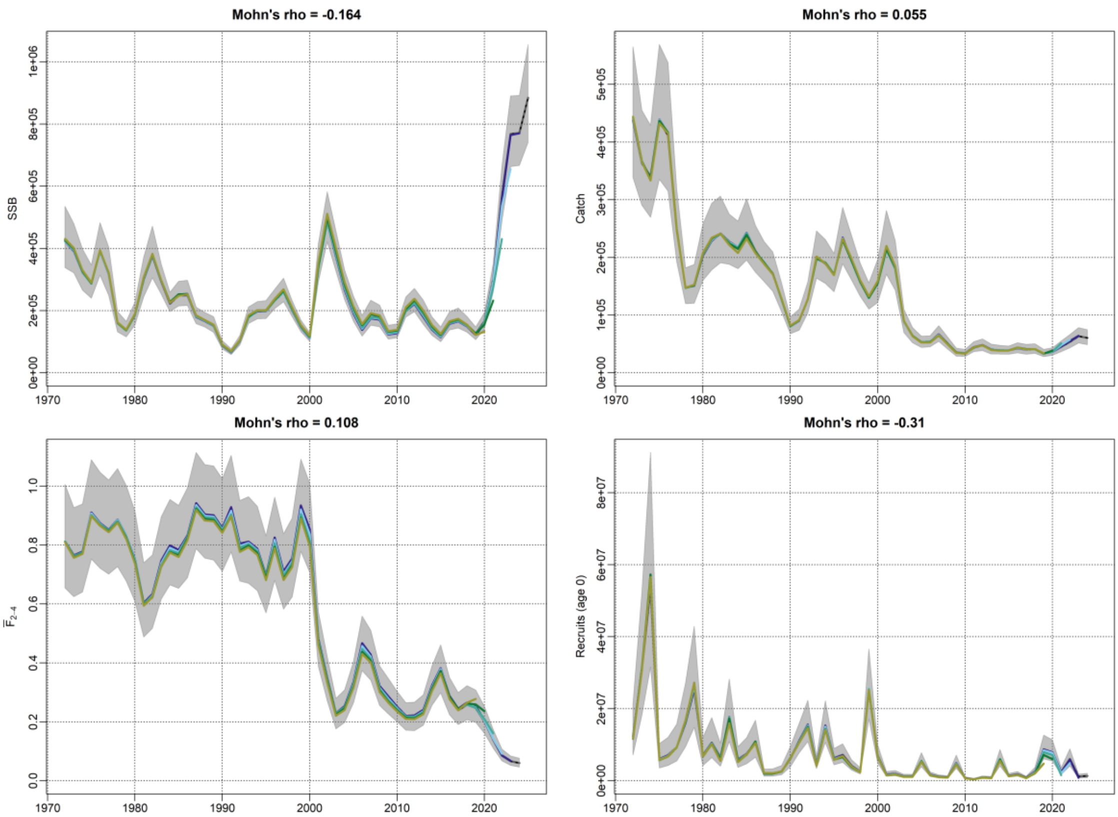


Figure 8.3.18. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Retrospective plots for the SAM assessment. The final-year run is shown in black with the pointwise 95% confidence interval indicated by the grey shading, while retrospective peels are shown with coloured lines. Mohn's rho estimates are given in the top right of each plot.

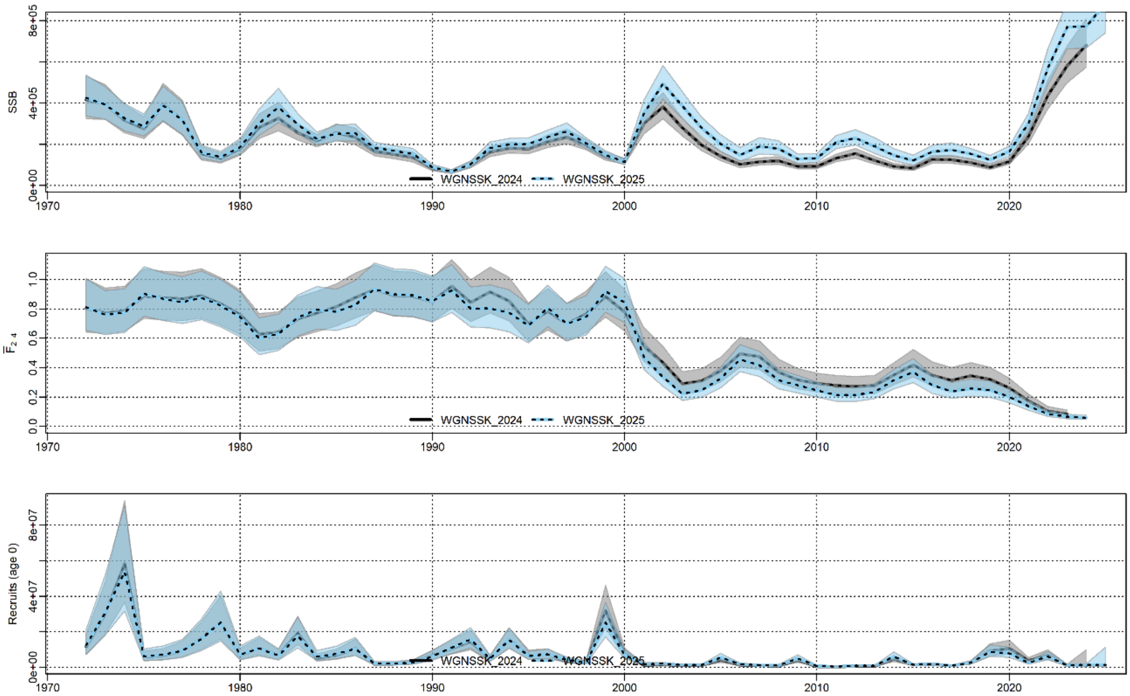


Figure 8.3.19. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparison of assessment results from previous stock assessment (WGNSSK 2024) and current stock assessment (WGNSSK 2025).

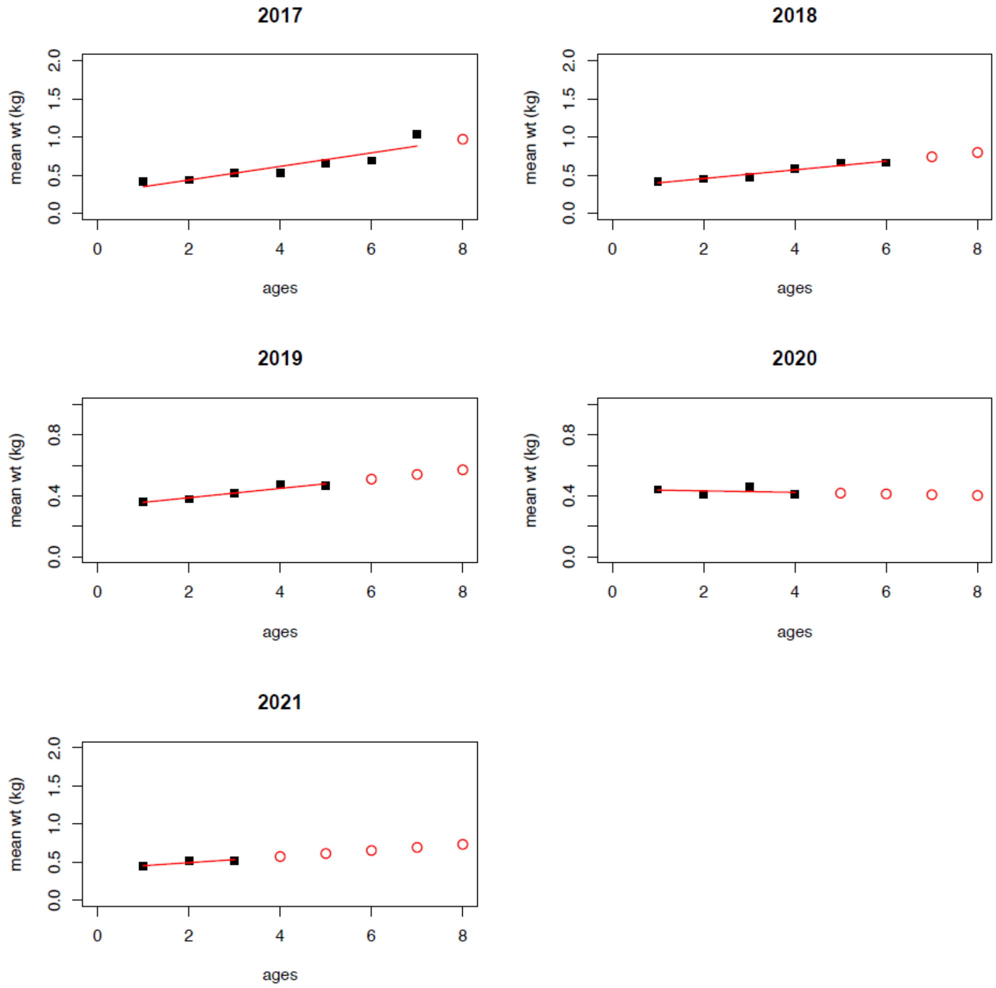


Figure 8.6.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Results of growth modelling for landings mean weights-at-age using cohort-based linear models (Jaworski, 2011). Cohorts 2017–2021 are shown here. Black points are available observations, red solid lines show linear fits to these points, and red points indicate projected weights for older ages.

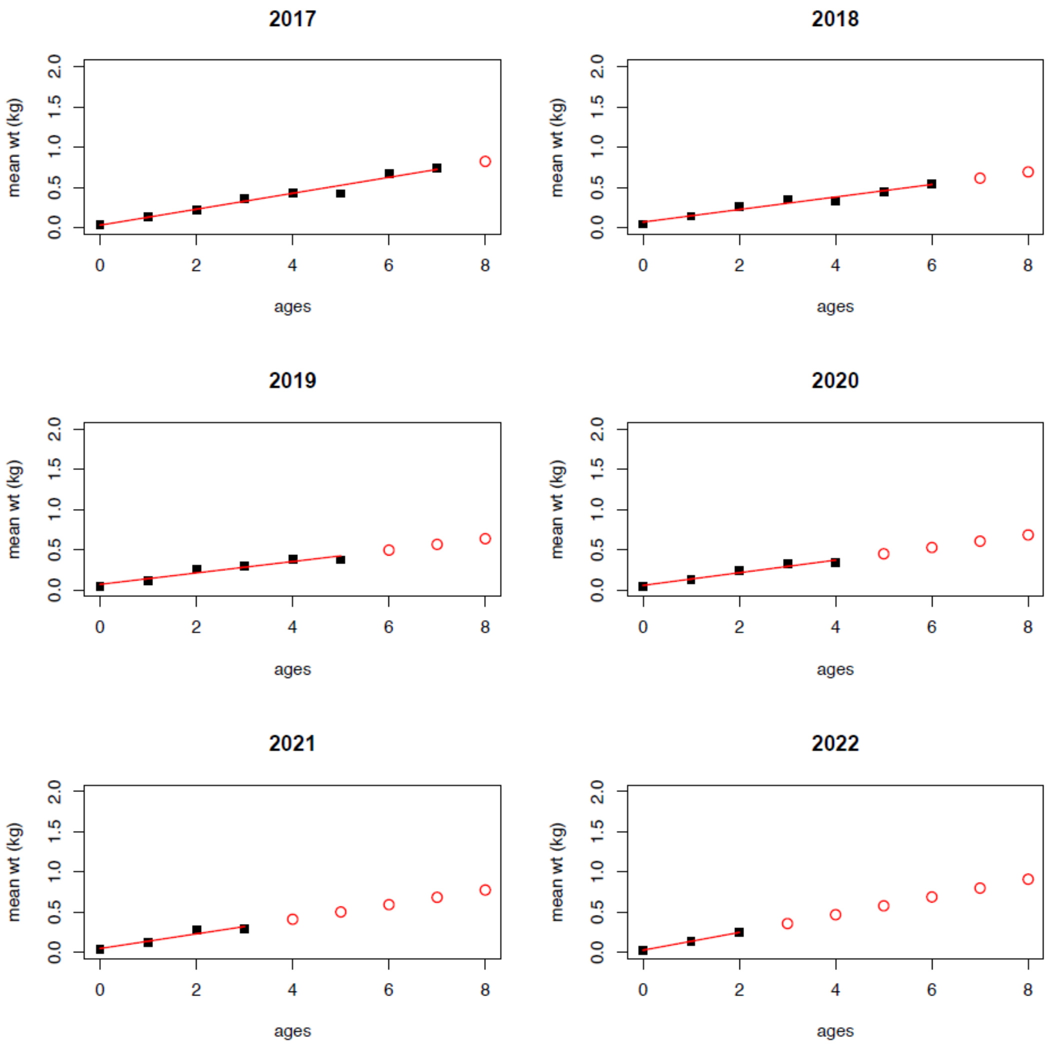


Figure 8.6.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Results of growth modelling for discards+IBC+BMS mean weights-at-age using cohort-based linear models (Jaworski, 2011). Cohorts 2017–2022 are shown here. Black points are available observations, red solid lines show linear fits to these points, and red points indicate projected weights for older ages.

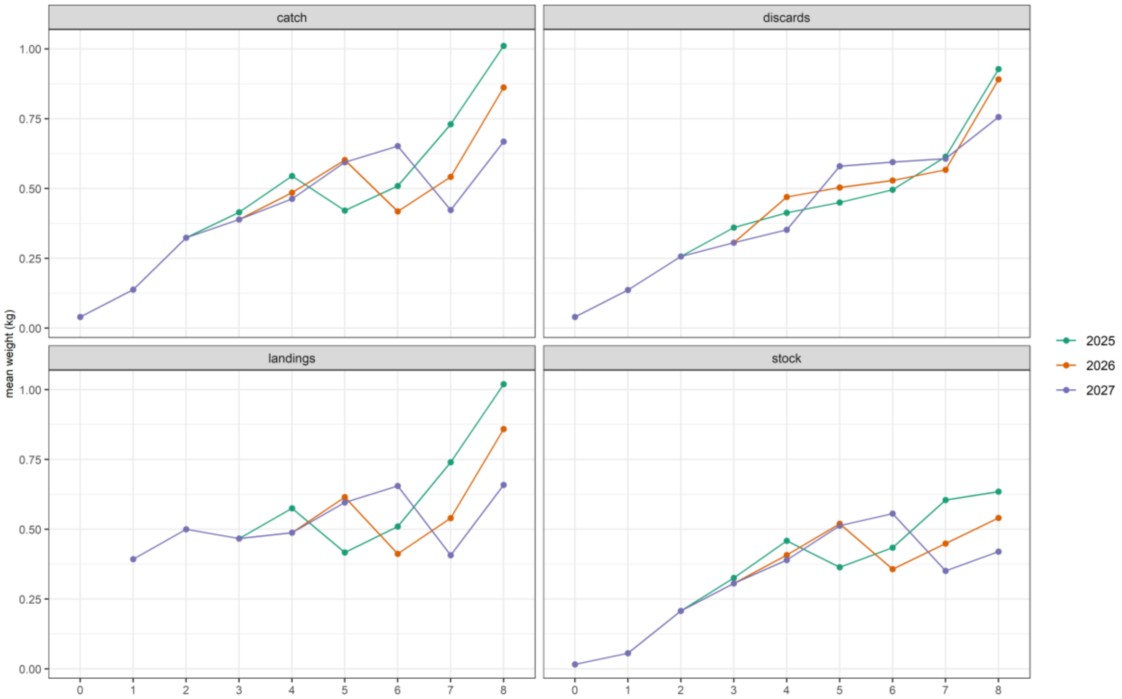


Figure 8.6.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean weights-at-age used in the forecast for total catch, landings, discards (including IBC and BMS) and the stock.

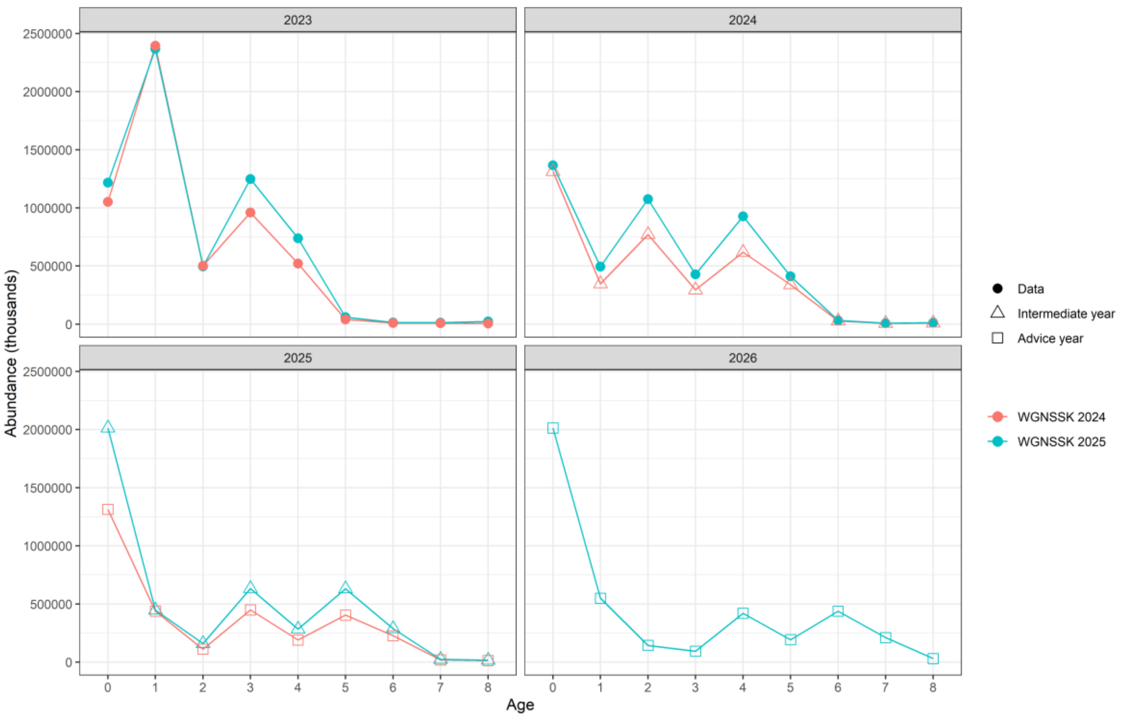


Figure 8.6.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparison of numbers-at-age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025).

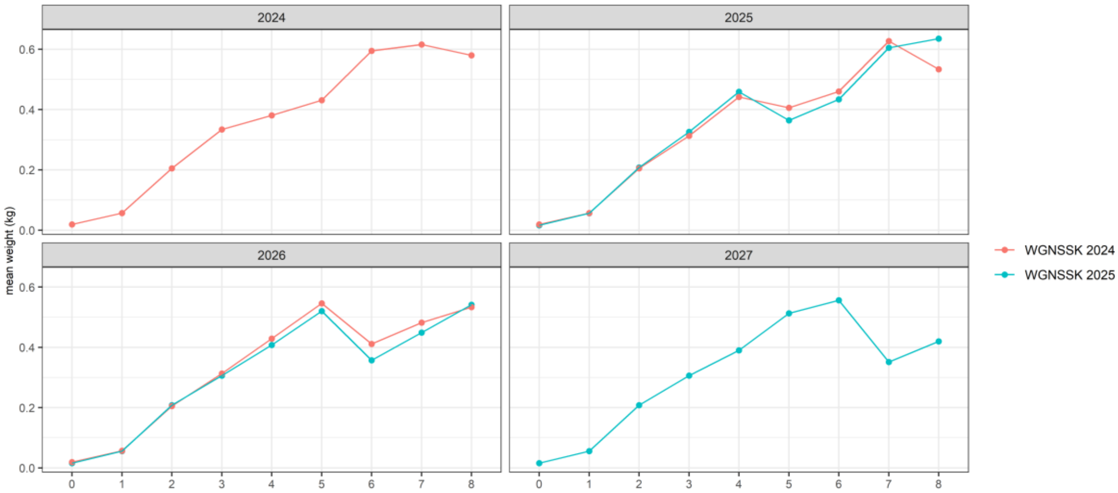


Figure 8.6.5. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparison of mean weights-at-age in the stock used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025).

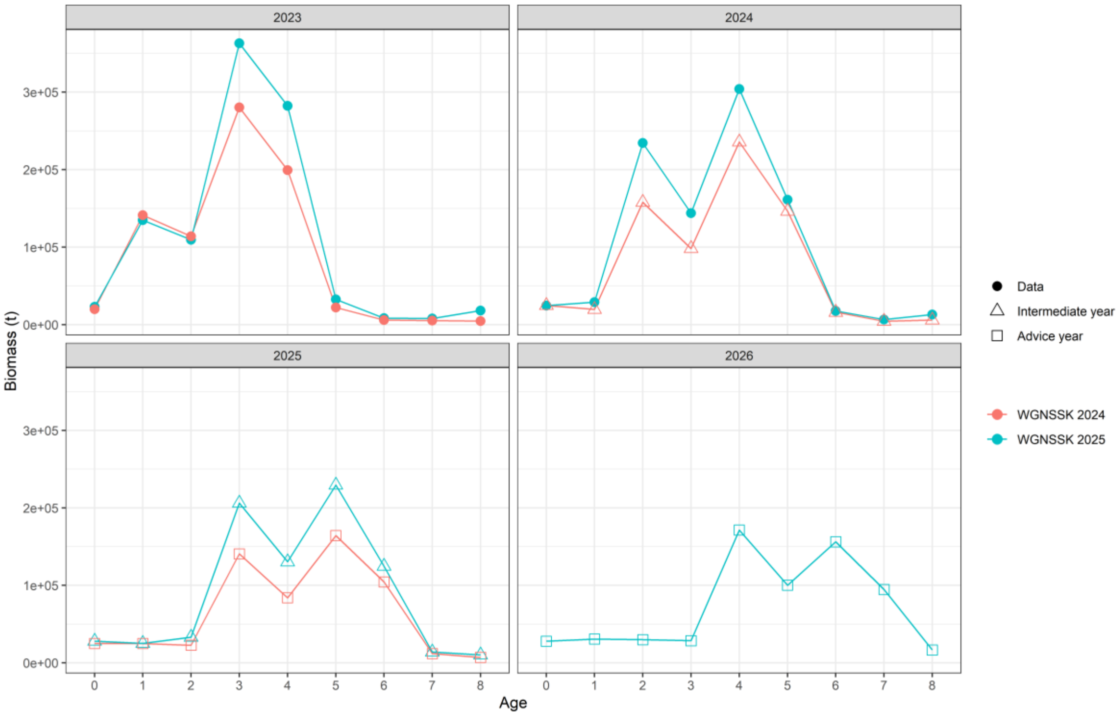


Figure 8.6.6. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparison of biomass at age from previous stock assessment and forecast results (WGNSSK 2024) and current stock assessment and forecast results (WGNSSK 2025).

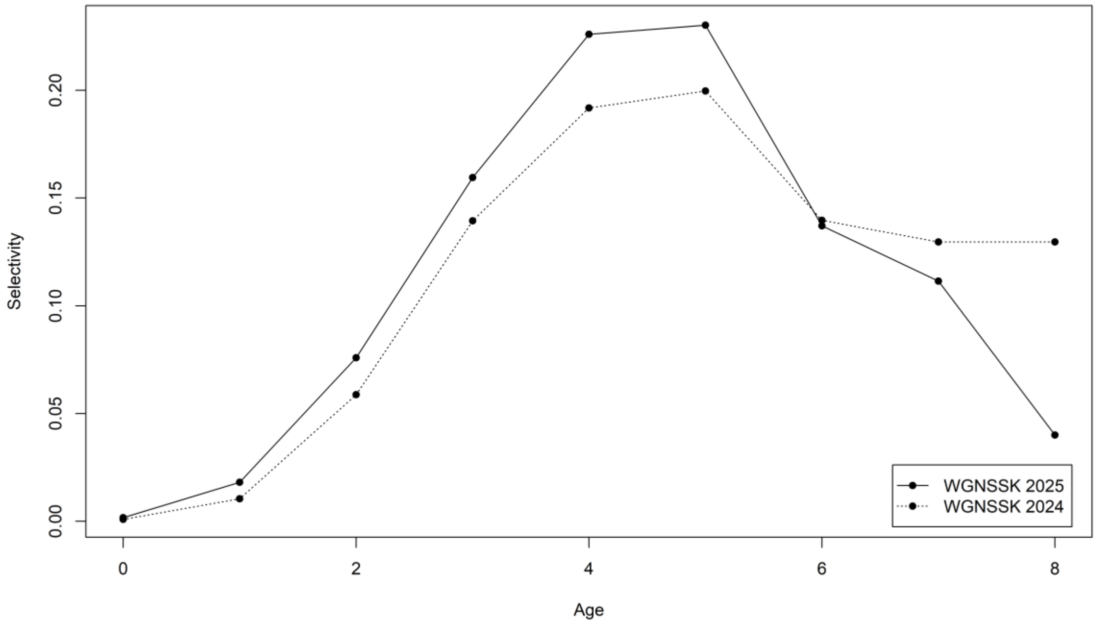


Figure 8.6.7. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparison of selectivity used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025).

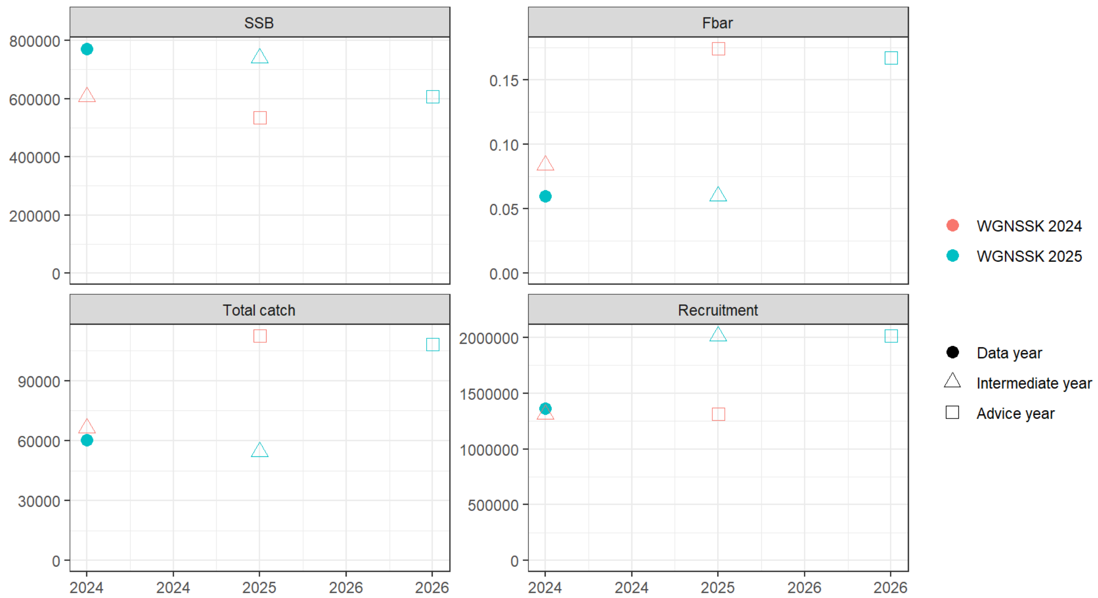


Figure 8.6.8. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Comparison of SSB, Fbar, total catch and recruitment assumptions used in the previous forecast (WGNSSK 2024) and current forecast (WGNSSK 2025).

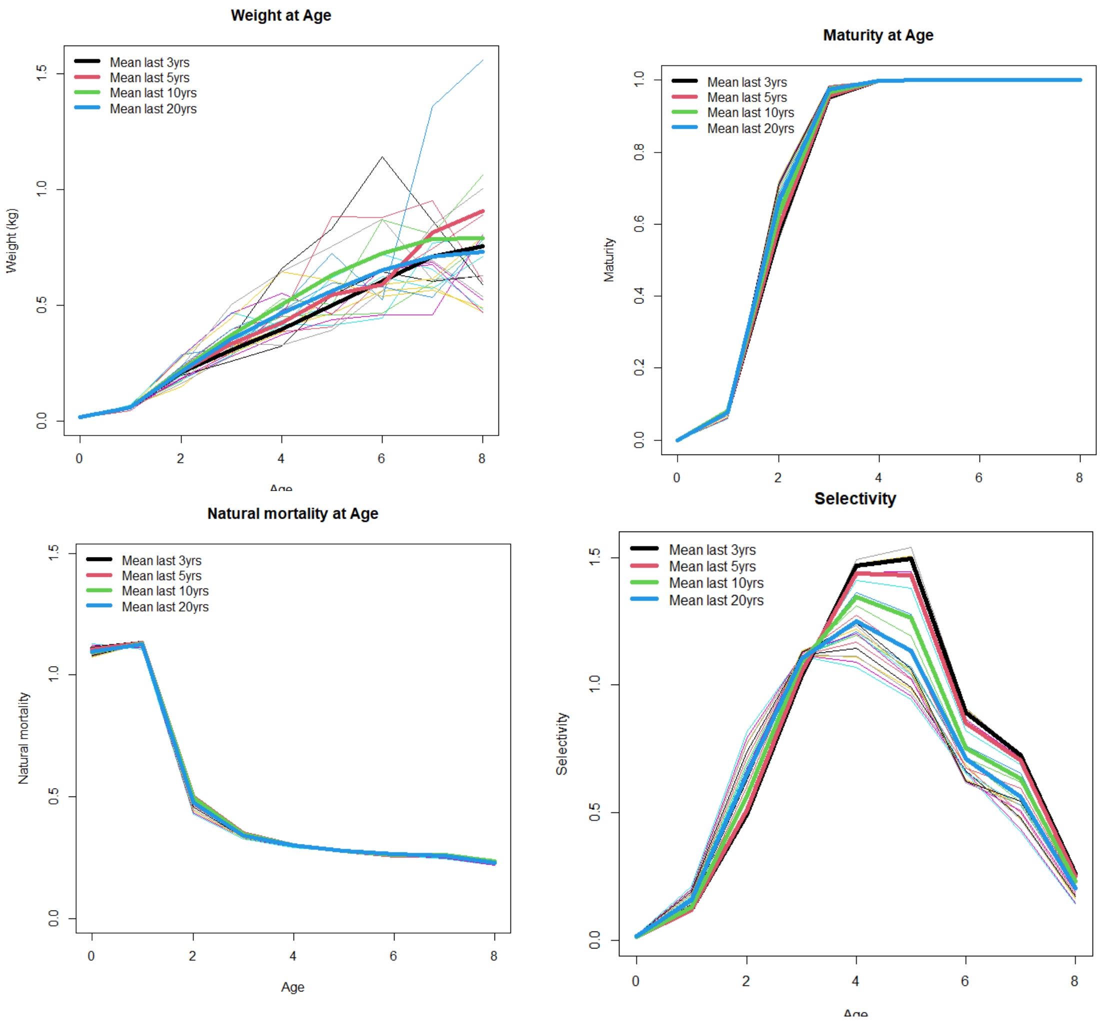


Figure 8.8.1. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Mean values of stock weights-at-age, maturity-at-age, natural mortality-at-age, and selectivity over the most recent 3, 5, 10 and 20 years.

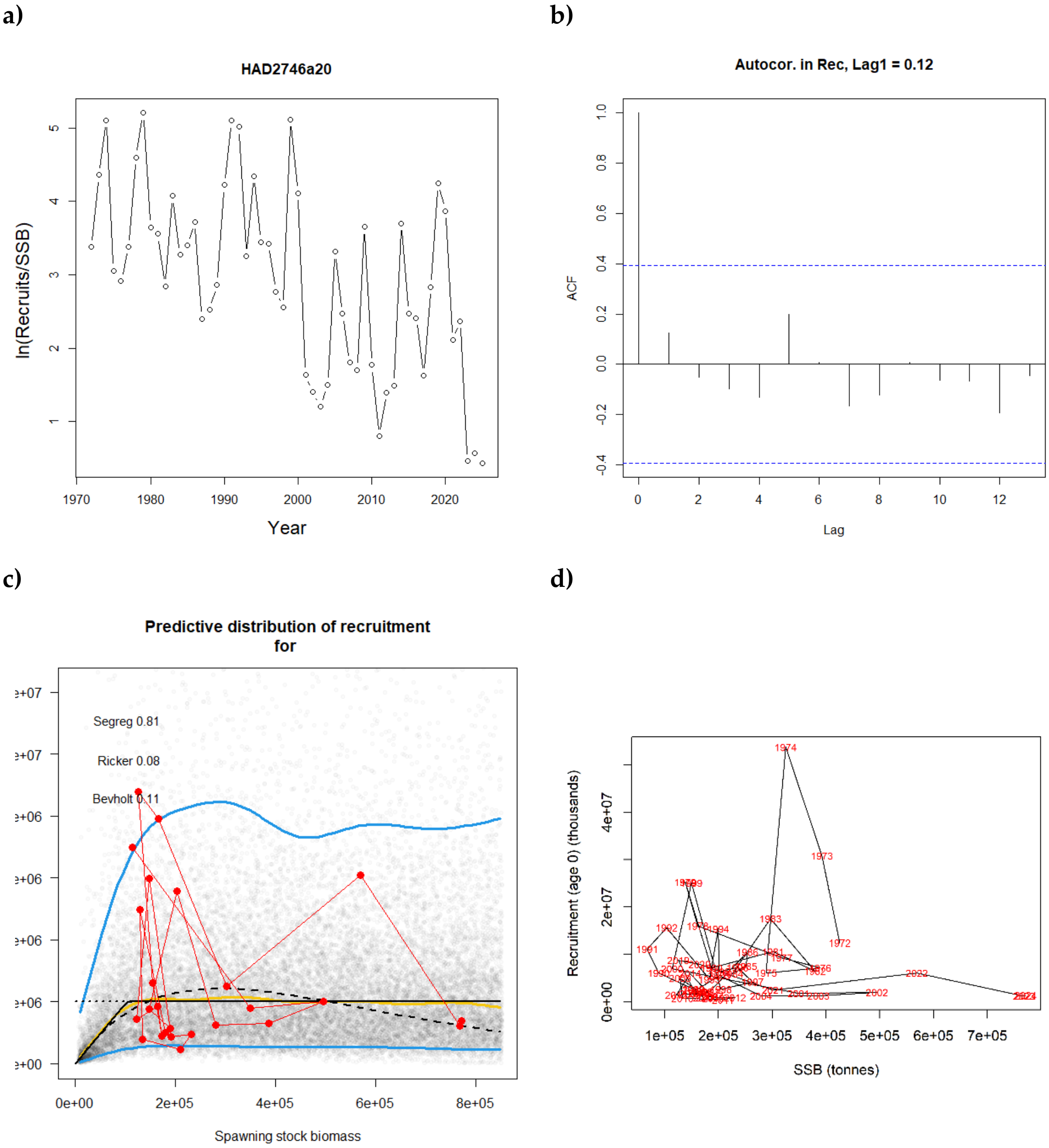


Figure 8.8.2. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Recruitment, stock–recruitment relationships and stock type: a) Recruits per SSB (log scale), b) estimation of autocorrelation in the recruitment time-series, c) fitting different stock–recruitment relationships, d) stock–recruitment relationship for determining stock type.

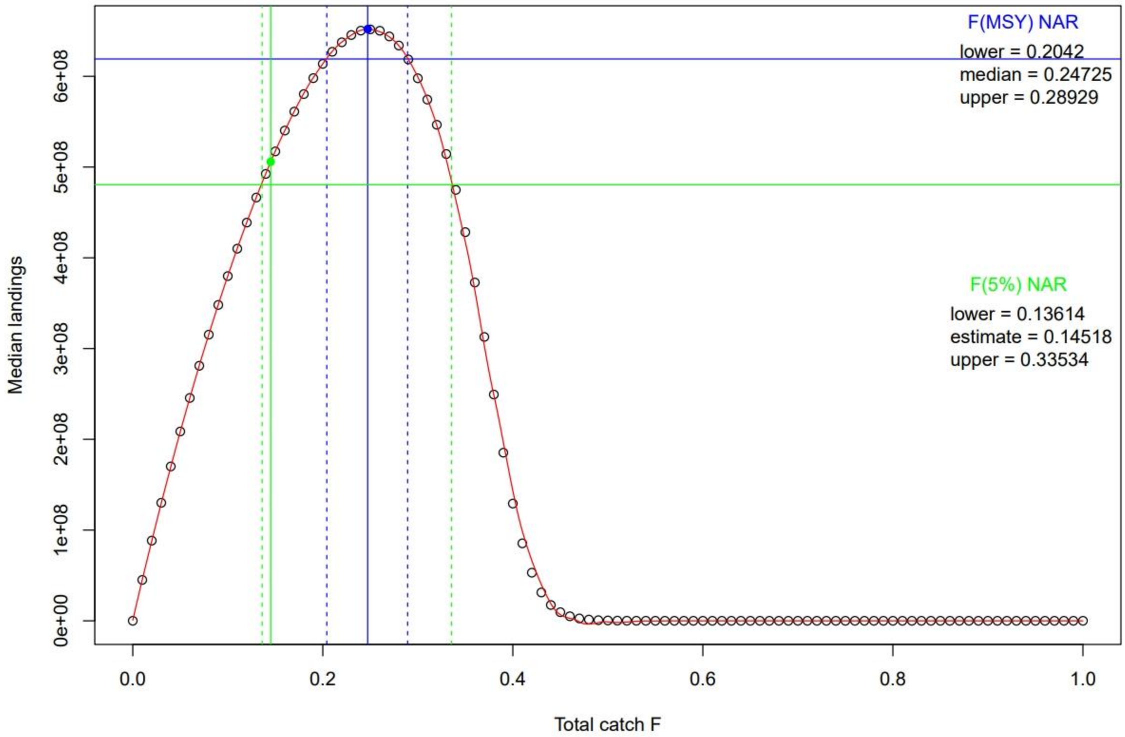


Figure 8.8.3. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Results of EqSim estimation of F_{MSY} and of $F_{P.05}$ with the assessment/advice error but no rule.

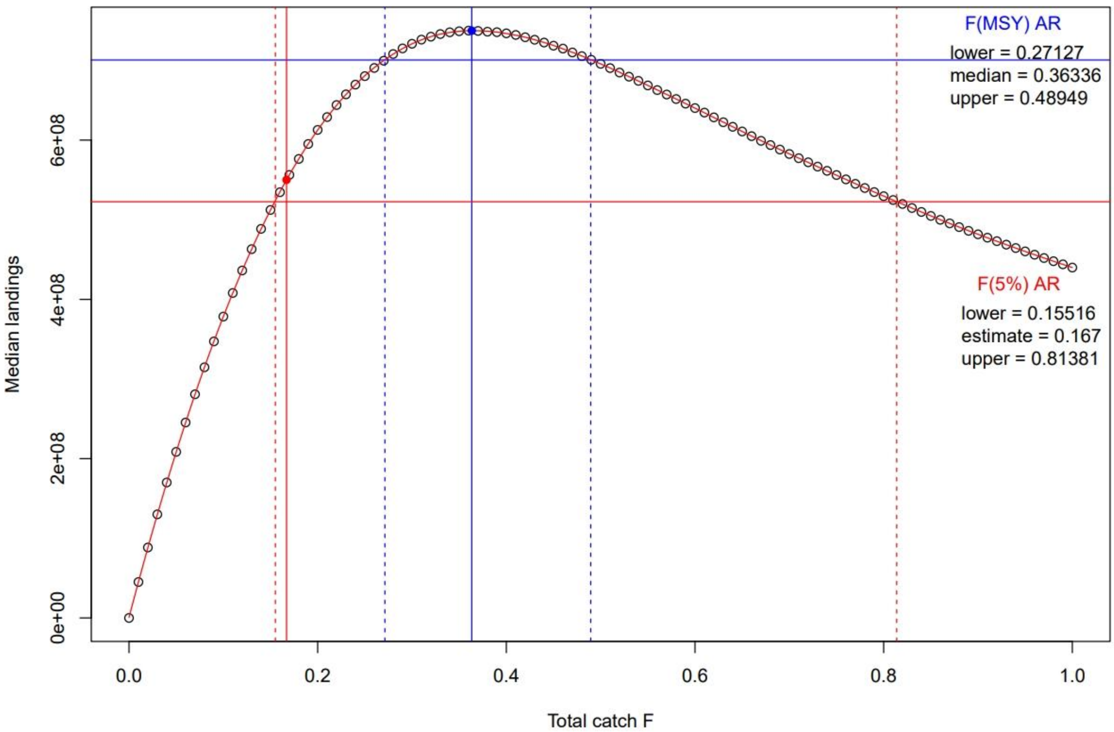


Figure 8.8.4. Haddock in Subarea 4, Division 6.a, and Subdivision 20. Results of EqSim estimation of F_{MSY} and of $F_{P.05}$ with both the assessment/advice error and rule.

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